

REPORT ON OPENAI’S COUNTEREXAMPLES TO THE LITTLE FINITISTIC DIMENSION CONJECTURE FOR FINITE-DIMENSIONAL ALGEBRAS

PREPARED AUTONOMOUSLY BY AI AGENTS (CLAUDE OPUS 5.5 AND GPT-6 ASTRA)

ABSTRACT. We reconstruct, with proofs written out in full (not verified by a human), the two constructions of finite-dimensional algebras of infinite little finitistic dimension released by OpenAI in September 2026: one realises a selection functor on modules over a finitely presented algebra by a bimodule complex and a trivial extension; the other obtains a counterexample to the Auslander–Reiten conjecture from a bimodule and a morphism in the stable module category of a symmetric algebra. We also prove criteria for infinite finitistic dimension and obstructions to simpler constructions.

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CAVEAT LECTOR

This report was produced as an aid for verifying and understanding the counterexamples to the little finitistic dimension conjecture in the preprints of OpenAI [Ope26b; Ope26c]. It was orchestrated by Gustavo Jasso and prepared by AI agents, as described in Appendix D. It has not been verified by a human, and it is provided as is.

1. INTRODUCTION

The little finitistic dimension of a finite-dimensional algebra A is the supremum of the projective dimensions of the finitely generated A -modules of finite projective dimension. Bass asked whether it is always finite [Bas60]; see [Zim95] for the history of the resulting finitistic dimension conjecture. In September 2026,

OpenAI released preprints that claim counterexamples. The first [Ope26b] constructs a finite-dimensional complex algebra of infinite little finitistic dimension. The second [Ope26c] constructs a counterexample to the Auslander–Reiten conjecture over the field $\mathbb{F}_2(q, H_1, H_2)$; by a theorem of Auslander and Reiten [AR75, Theorem 1.1(b)], together with Theorem 3.3, such a counterexample yields a further algebra of infinite little finitistic dimension.

This article reconstructs both constructions with proofs written out in full (not verified by a human), organised around general statements, some more general than the applications require. It also gives criteria and obstructions that explain the choices in the constructions. All proofs were written and checked by AI models, as described in Appendices B and D; no human has checked them, and the article has not been peer reviewed. The repository that accompanies the preprints poses the existence statements of both constructions as challenges in the proof assistant Lean and provides formal proofs of them; Appendix B compares these statements with the present article and records what has been checked.

The main construction. A *selection functor* $H = \Psi \otimes_R -$ on finite-dimensional modules over an algebra R , possibly infinite-dimensional, annihilates modules at arbitrarily late times (Definition 5.1). A bounded complex P of bimodules over a finite-dimensional algebra B of global dimension at most two realises H , in the sense that $P \otimes_B^{\mathbb{L}} M(Y) \cong M(HY) \oplus M(HY)[3]$ for a fully faithful embedding M (Theorem 6.18). The complex P is simulated by an ordinary bimodule X over an algebra Δ of finite global dimension, and the trivial extension $\Delta \ltimes X$ turns the extinction times of H into projective dimensions (Proposition 7.2 and corollary 4.5). The result is the following theorem, proved in Section 7; the main preprint treats the case of a central idempotent and an endomorphism of R .

Theorem (Theorem 7.3). *Let (R, Ψ) be selection data over a field, with R finitely presented and with unbounded extinction. Then there is a finite-dimensional algebra $A = \Delta \ltimes X$, with Δ of finite global dimension, such that $\text{findim } A = \infty$.*

Selection data with unbounded extinction exist: the complex group algebra of a finitely presented group with a family of independent central involutions, shifted by an automorphism, provides them (Theorem 5.17), which gives the counterexample of the main preprint (Corollary 7.4). The selection data must be infinite-dimensional in an essential way: their rank functions on finite-dimensional modules must span an infinite-dimensional space (Proposition 5.4), which excludes finite-dimensional algebras, commutative noetherian algebras and universal localisations of finite-dimensional hereditary algebras (Corollary 5.5). The realisation is the only step that is not explicit: it lifts a diagram from a Verdier quotient by the calculus of fractions, and the argument gives no bound on the length l of the complex P , hence on the number $3(l+2)$ of simple modules of the final algebra (Remarks 6.14 and 7.5). The argument works with an object $U \oplus U[3]$ instead of the summand U that encodes Ψ : the summand U need not lie in the Verdier quotient, while $U \oplus U[3]$ does (Proposition 6.7), and with the shift 3 the obstruction to splitting the realisation lies in an Ext^4 over an algebra of global dimension at most two (Remark 6.8).

The counterexample to the Auslander–Reiten conjecture. The second construction uses the following result over an arbitrary field (Theorem 8.4): for a symmetric algebra E , a bimodule F projective on both sides, a module S and a stable morphism $v: S \rightarrow FS$, if the maps $(g, j) \mapsto F(g)v - v[a]j$ between Tate cohomology groups are bijective in positive degrees and surjective with nonzero kernel in degree zero, then a module over the triangular algebra $\begin{pmatrix} E & 0 \\ F & E \end{pmatrix}$ is a counterexample to the Auslander–Reiten conjecture. The construction uses a ten-dimensional

algebra C , whose simple module has an aperiodic resolution, the symmetric algebra $T = C \ltimes DC$, over which this simple module has polynomial Ext algebra on a generator of degree three, and the tensor square $E = T \otimes T$, over which two cones and two twists produce the bimodule F (Section 9). With Theorem 3.7 this gives an algebra of infinite little finitistic dimension with nine simple modules (Corollary 9.18); the triangular algebra used to obtain it has eight simple modules and dimension of the order of 10^{35} (Remark 9.19).

Criteria and obstructions. The work that led to this article began with a search for a counterexample that could be verified by hand. The search did not succeed, and its results are the general statements of Sections 3, 4 and 10. Failure of the strong Nakayama conjecture produces explicit modules of every finite projective dimension (Theorem 3.3); simple modules witnessing this failure always come from a corner of the algebra and a counterexample to the Auslander–Reiten conjecture over that corner (Proposition 3.8); and finite projective dimensions, as well as finite extinction times, are bounded on modules of bounded dimension (Propositions 3.11 and 4.7), where the assertion about extinction times is for trivial extensions of algebras of finite global dimension. For the construction of the counterexample to the Auslander–Reiten conjecture, the analogous construction over T itself, with one cone instead of two, fails in degree zero whenever the simple module has polynomial Ext algebra on a generator of degree three (Proposition 10.2 and corollary 10.3); the proof uses Tate duality, which also forces the vanishing of the corresponding Toda brackets (Theorem 10.4). Whether a counterexample with a small explicit presentation exists remains open.

Structure. Section 2 fixes conventions and recalls Tate duality and Toda brackets. Section 3 contains the general criteria. The main construction occupies Sections 4 to 7: trivial extensions and extinction, selection data and the group, the realisation by a bimodule complex, and the simulation with the main theorem. The counterexample to the Auslander–Reiten conjecture is constructed in Sections 8 and 9, and Section 10 discusses obstructions to smaller constructions. Appendix A describes the computations used, Appendix B the verification and formalisation record, Appendix C the third preprint of the release, and Appendix D the use of AI in the preparation of this article.

2. CONVENTIONS AND PRELIMINARIES

Conventions. Throughout, $\mathbf{k}$ is a field and $D = \text{Hom}_{\mathbf{k}}(-, \mathbf{k})$ denotes the $\mathbf{k}$ -linear duality. Algebras are associative and unital; unless stated otherwise, they are finite-dimensional over $\mathbf{k}$, although the algebras of selection data in Section 5 need not be. Modules are finitely generated *left* modules unless they are called right modules, and right A -modules are identified with left A^{op} -modules. Composition of maps and products of paths in a quiver are written from right to left, so that an arrow $a: i \rightarrow j$ lies in $e_j A e_i$. Complexes are graded cohomologically, with $K[s]^n = K^{n+s}$ and $d_{K[s]} = (-1)^s d_K$. For a module M we write $M^* = \text{Hom}_A(M, A)$, which is a right module, ΩM for the kernel of a projective cover of M , and $\text{Tr } M$ for the transpose computed from a minimal projective presentation: if $P_1 \rightarrow P_0 \rightarrow M \rightarrow 0$ is minimal, then $\text{Tr } M = \text{coker}(P_0^* \rightarrow P_1^*)$.

The *little finitistic dimension* of an algebra A is

$$\text{findim } A := \sup \left\{ \text{pd}_A M \mid \begin{array}{l} M \text{ a finitely generated left } A\text{-module} \\ \text{pd}_A M < \infty \end{array} \right\}.$$

The *little finitistic dimension conjecture* asserts that $\text{findim } A < \infty$ for every finite-dimensional algebra A [Bas60]; see [Zim95] for its history. The finitistic dimension of A^{op} is the *right* finitistic dimension of A ; the two need not agree.

2.1. Stable categories of symmetric algebras. Let A be a symmetric algebra. Its stable module category $\underline{\text{mod}} A$ is triangulated, with suspension $[1]$ given by the cosyzygy Ω^{-1} , so that $[-1] = \Omega$, and with triangles induced by short exact sequences. For modules X and Y and an integer a , we write

$$\widehat{\text{Ext}}_A^a(X, Y) := \underline{\text{Hom}}_A(X, Y[a]),$$

so that $\widehat{\text{Ext}}_A^a(X, Y) = \text{Ext}_A^a(X, Y)$ for $a \geq 1$. Composition makes $\widehat{\text{Ext}}_A^*(X, X)$ a graded algebra.

Theorem 2.1 (Tate duality; [Lin13, Section 2, (2.1)]). *Let A be a symmetric algebra. For all modules X, Y and every integer a , there is an isomorphism*

$$D\widehat{\text{Ext}}_A^a(X, Y) \cong \widehat{\text{Ext}}_A^{1-a}(Y, X),$$

natural in X and Y .

Equivalently [Lin13, Section 2, (2.2)], there is a natural nondegenerate bilinear form

$$\langle -, - \rangle : \widehat{\text{Ext}}_A^a(Y, X) \times \widehat{\text{Ext}}_A^{-1-a}(X, Y) \longrightarrow \mathbf{k},$$

and it is compatible with composition: $\langle \zeta\eta, \tau \rangle = \langle \zeta, \eta\tau \rangle$ whenever the composites are defined [Lin13, Section 2, (2.8)].

2.2. Toda brackets. Let $\mathcal{T}$ be a triangulated category and let

$$X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} W$$

be morphisms with $gf = 0$ and $hg = 0$. Choose a triangle

$$Y \xrightarrow{g} Z \xrightarrow{i} C_g \xrightarrow{q} Y[1].$$

A *defining system* is a pair of morphisms $a: X[1] \rightarrow C_g$ and $b: C_g \rightarrow W$ with $qa = f[1]$ and $bi = h$; such pairs exist since $gf = 0$ and $hg = 0$. The *Toda bracket* $\langle h, g, f \rangle \subseteq \text{Hom}_{\mathcal{T}}(X[1], W)$ is the set of the composites ba over all defining systems. It is a coset of the *indeterminacy*

$$h \circ \text{Hom}_{\mathcal{T}}(X[1], Z) + \text{Hom}_{\mathcal{T}}(Y[1], W) \circ f[1].$$

In a graded endomorphism ring $\widehat{\text{Ext}}_A^*(X, X)$, the bracket of homogeneous classes of degrees p, q and r lies in degree $p + q + r - 1$.

Lemma 2.2. *In the situation above, let $u: X' \rightarrow X$ be a morphism. Then*

$$\langle h, g, f \rangle \circ u[1] \subseteq \langle h, g, fu \rangle.$$

Proof. If (a, b) is a defining system for $\langle h, g, f \rangle$, then $q(a \circ u[1]) = (fu)[1]$ and $bi = h$, so that $(a \circ u[1], b)$ is a defining system for $\langle h, g, fu \rangle$, with value $(ba) \circ u[1]$. $\square$

2.3. Trivial extensions. For an algebra Δ and a Δ -bimodule X , the *trivial extension* is the algebra $\Delta \ltimes X = \Delta \oplus X$ with multiplication

$$(d, x)(d', x') = (dd', dx' + xd').$$

Every Δ -module is a $(\Delta \ltimes X)$ -module through the projection $\Delta \ltimes X \rightarrow \Delta$; we call such modules *inflated*.

For a finite-dimensional algebra C , the trivial extension $C \ltimes DC$ by the bimodule DC , with actions $(a\phi)(c) = \phi(ca)$ and $(\phi a)(c) = \phi(ac)$, is symmetric: the linear form $\text{tr}(c, \phi) = \phi(1)$ satisfies $\text{tr}((a, \phi)(b, \psi)) = \psi(a) + \phi(b)$, which is symmetric in the two factors, and the associated bilinear form is nondegenerate, since for $a \neq 0$ there is ψ with $\psi(a) \neq 0$, and for $a = 0 \neq \phi$ there is b with $\phi(b) \neq 0$. Hence, $t \mapsto \text{tr}(-t)$ is an isomorphism of bimodules from $C \ltimes DC$ to its dual.

3. CRITERIA FOR INFINITE FINITISTIC DIMENSION

Both constructions use a module with prescribed homological behaviour. The following dimension-shifting argument produces modules of every finite projective dimension.

Proposition 3.1. *Let $\mathcal{A}$ be an abelian category with enough projective objects, and suppose given short exact sequences*

$$0 \rightarrow c_0 \rightarrow p_0 \rightarrow c_1 \rightarrow 0 \quad \text{and} \quad 0 \rightarrow c_n \rightarrow p_n \rightarrow c_{n+1} \rightarrow 0 \quad (n \geq 1)$$

in which every p_n is projective, c_0 is projective and c_1 is not. Then $\text{pd } c_n = n$ for every $n \geq 1$.

Proof. A short exact sequence $0 \rightarrow c_n \rightarrow p_n \rightarrow c_{n+1} \rightarrow 0$ with p_n projective gives $\text{pd } c_{n+1} \leq \text{pd } c_n + 1$, so $\text{pd } c_n \leq n$ by induction. Since c_1 is not projective, $\text{pd } c_1 = 1$. Let $n \geq 1$ and suppose that $\text{pd } c_n = n$. For every object V and every $j \geq 1$, the long exact sequence of $\text{Ext}(-, V)$ and the vanishing of $\text{Ext}^{\geq 1}(p_n, V)$ give an isomorphism $\text{Ext}^{j+1}(c_{n+1}, V) \cong \text{Ext}^j(c_n, V)$. Choosing V with $\text{Ext}^n(c_n, V) \neq 0$ and $j = n$ yields $\text{pd } c_{n+1} = n + 1$. $\square$

Remark 3.2. Proposition 3.1 is formally verified in Lean, against Mathlib, in an arbitrary abelian category; see Appendix B.

3.1. The strong Nakayama conjecture. The following argument shows that finiteness of the finitistic dimension of A^{op} implies the strong Nakayama conjecture for A [Cra, Section 3.2, Proposition 5]; it also gives the exact projective dimensions of the modules constructed.

Theorem 3.3. *Let A be a finite-dimensional algebra and $E \neq 0$ a right A -module such that*

$$\text{Ext}_{A^{\text{op}}}^i(E, A) = 0 \quad \text{for all } i \geq 0.$$

Let $\cdots \rightarrow P_1 \rightarrow P_0 \rightarrow E \rightarrow 0$ be a minimal projective resolution, and put $C_n = \text{coker}(P_{n-1}^ \rightarrow P_n^*)$ for $n \geq 1$. Then $C_n \cong \text{Tr } \Omega^{n-1} E$ and $\text{pd}_A C_n = n$ for every $n \geq 1$. In particular, $\text{findim } A = \infty$.*

Proof. The complex $0 \rightarrow P_0^* \rightarrow P_1^* \rightarrow P_2^* \rightarrow \cdots$ of projective left A -modules has cohomology $\text{Ext}_{A^{\text{op}}}^*(E, A) = 0$, so it is exact, and it splits into short exact sequences $0 \rightarrow C_0 \rightarrow P_1^* \rightarrow C_1 \rightarrow 0$ and $0 \rightarrow C_n \rightarrow P_{n+1}^* \rightarrow C_{n+1} \rightarrow 0$, where $C_0 = P_0^*$. If C_1 were projective, the map $P_0^* \rightarrow P_1^*$ would be a split monomorphism; since finitely generated projective modules are reflexive, its dual $P_1 \rightarrow P_0$ would then be a split epimorphism, and $E = \text{coker}(P_1 \rightarrow P_0)$ would vanish. Hence, Proposition 3.1 applies. Finally, $P_n \rightarrow P_{n-1} \rightarrow \Omega^{n-1} E \rightarrow 0$ is a minimal presentation, so that

$$\text{Tr } \Omega^{n-1} E = \text{coker}(P_{n-1}^* \rightarrow P_n^*) = C_n. \quad \square$$

Since $\text{Ext}_{A^{\text{op}}}^i(E, A) \cong D \text{Tor}_i^A(E, DA)$, the hypothesis says that the derived Nakayama functor $-\otimes_A^L DA$ annihilates E . It forces $\text{pd } E = \infty$: if $\text{pd } E = n < \infty$, then $\text{Ext}^n(E, A) \neq 0$.

Remark 3.4. Theorem 3.3 is formally verified in Lean over an arbitrary ring, including the equivalence between the vanishing of the groups $\text{Ext}_{A^{\text{op}}}^i(E, A)$ and the exactness of the dual complex; the identification of C_n with a transpose is proved for the presentation given by the resolution, without its minimality. See Appendix B.

The next proposition reformulates the hypothesis of Theorem 3.3 as a property of a module of projective dimension one, such as C_1 .

Proposition 3.5. *Let C be an A -module without nonzero projective summands and with $\text{pd } C = 1$. The following are equivalent:*

- (1) the right module $E = \text{Tr } C$ satisfies $\text{Ext}_{A^{\text{op}}}^i(E, A) = 0$ for all $i \geq 0$;
- (2) C is not projective and $\text{Ext}_{A^{\text{op}}}^i(\text{Tr } C, A) = 0$ for all $i \geq 1$;
- (3) C is not projective and admits an exact sequence $0 \rightarrow C \rightarrow Q_2 \rightarrow Q_3 \rightarrow \dots$ with each Q_m projective, in which every map from C or from a cokernel to the next Q_m is a left add A -approximation.

Moreover, $E \cong \text{Ext}_A^1(C, A)$.

Proof. Choose a minimal projective resolution $0 \rightarrow Q_0 \rightarrow Q_1 \rightarrow C \rightarrow 0$. Then $E = \text{coker}(Q_1^* \rightarrow Q_0^*) \cong \text{Ext}_A^1(C, A)$ and $E^* = \ker(Q_0 \rightarrow Q_1) = 0$; moreover $E \neq 0$, since otherwise $Q_1^* \rightarrow Q_0^*$ would split and C would be projective. This proves that (1) and (2) are equivalent. Assume (1). Dualising a minimal projective resolution of E gives an exact complex whose first cokernel is $\text{Tr } E = C$, since $\text{Tr } \text{Tr } C \cong C$ for a module C without nonzero projective summands, and whose tail is a projective coresolution of C ; the dual of each of its short exact sequences is a segment of the resolution of E , hence exact, which is the approximation property in (3). Conversely, dualising the sequence in (3) gives an exact sequence $\dots \rightarrow Q_3^* \rightarrow Q_2^* \rightarrow C^* \rightarrow 0$; splicing it with $0 \rightarrow C^* \rightarrow Q_1^* \rightarrow Q_0^* \rightarrow E \rightarrow 0$ gives a projective resolution of E whose dual $0 \rightarrow Q_0 \rightarrow Q_1 \rightarrow Q_2 \rightarrow \dots$ is exact, and (1) follows. $\square$

Corollary 3.6. *If a finite-dimensional algebra A admits a module E as in Theorem 3.3, then A has infinitely many isomorphism classes of indecomposable torsionless modules of finite projective dimension.*

Proof. The modules C_n of Theorem 3.3 embed into the projective modules P_{n+1}^* , by the short exact sequences in its proof, so their indecomposable summands are torsionless and of finite projective dimension. If there were only finitely many of them, their projective dimensions would bound $\text{pd } C_n = n$ for all n . $\square$

3.2. Counterexamples to the Auslander–Reiten conjecture. Let Λ be an Artin algebra. The *Auslander–Reiten conjecture* asserts that a Λ -module M with $\text{Ext}_\Lambda^i(M, M \oplus \Lambda) = 0$ for all $i \geq 1$ is projective. Auslander and Reiten showed that its failure produces a failure of the generalised Nakayama conjecture for an endomorphism algebra [AR75, Theorem 1.1(b)]. Combined with Theorem 3.3, this gives a counterexample to the finitistic dimension conjecture with one more simple module.

Theorem 3.7. *Let Λ be a finite-dimensional algebra and M a nonprojective Λ -module with $\text{Ext}_\Lambda^i(M, M \oplus \Lambda) = 0$ for all $i \geq 1$. Put $G = \Lambda \oplus M$ and $\Gamma = \text{End}_\Lambda(G)^{\text{op}}$, and let $\pi: P \rightarrow M$ be a projective cover. Then the Γ -module*

$$S := \text{coker}(\text{Hom}_\Lambda(G, \pi))$$

is nonzero and $\text{Ext}_\Gamma^i(S, \Gamma) = 0$ for all $i \geq 0$. Consequently, the algebra $\text{End}_\Lambda(G)$ has infinite little finitistic dimension.

Proof. Write $\mathbb{F} = \text{Hom}_\Lambda(G, -)$, which restricts to an equivalence $\text{add } G \simeq \text{proj } \Gamma$ with $\mathbb{F}G = \Gamma$. Put $K_0 = \ker \pi$ and choose right add G -approximations $g_j: G_j \rightarrow K_j$ with $K_{j+1} = \ker g_j$; they are surjective since $\Lambda \in \text{add } G$. The functor $\mathbb{F}$ is left exact and the maps $\mathbb{F}g_j$ are surjective, so

$$\dots \rightarrow \mathbb{F}G_1 \rightarrow \mathbb{F}G_0 \rightarrow \mathbb{F}P \rightarrow \mathbb{F}M \rightarrow S \rightarrow 0$$

is a projective resolution of S . If $\mathbb{F}\pi$ were surjective, the identity of the summand M of G would lift through π , and M would be projective; hence $S \neq 0$. For $Y \in \text{add } G$ there is a natural isomorphism $\text{Hom}_\Gamma(\mathbb{F}Y, \Gamma) \cong \text{Hom}_\Lambda(Y, G)$, so that $\text{Ext}_\Gamma^*(S, \Gamma)$ is the cohomology of

$$0 \rightarrow \text{Hom}_\Lambda(M, G) \rightarrow \text{Hom}_\Lambda(P, G) \rightarrow \text{Hom}_\Lambda(G_0, G) \rightarrow \text{Hom}_\Lambda(G_1, G) \rightarrow \dots,$$

with $\text{Hom}_\Lambda(M, G)$ in degree zero. Since $\text{Ext}_\Lambda^{\geq 1}(G, G) = 0$, the resolution $\cdots \rightarrow G_0 \rightarrow P \rightarrow M$ consists of $\text{Hom}_\Lambda(-, G)$ -acyclic modules and computes $\text{Ext}_\Lambda^*(M, G)$; as $\text{Ext}_\Lambda^{\geq 1}(M, G) = 0$ and the first map is injective, the complex is exact. The last assertion is Theorem 3.3 applied to $\Gamma^{\text{op}} = \text{End}_\Lambda(G)$. $\square$

Replacing M by an indecomposable nonprojective summand preserves the hypotheses of Theorem 3.7, since Ext is additive. If Λ is basic, the algebra $\text{End}_\Lambda(G)$ then has exactly one simple module more than Λ .

Conversely, simple modules witnessing failure of the strong Nakayama conjecture always arise in this way, on a corner of the algebra. In the next proposition, Af , M and their endomorphisms are taken as *right* B -modules.

Proposition 3.8. *Let S be a simple right A -module with $\text{Hom}_{A^{\text{op}}}(S, A) = 0 = \text{Ext}_{A^{\text{op}}}^1(S, A)$. Let e be the sum of the primitive idempotents of a complete decomposition of 1 corresponding to S , $f = 1 - e$, $B = fAf$ and $M = eAf$. Then left multiplication induces an algebra isomorphism*

$$A \xrightarrow{\sim} \text{End}_{B^{\text{op}}}(Af) = \text{End}_{B^{\text{op}}}(B \oplus M),$$

B has exactly one isomorphism class of simple modules fewer than A , and M is not a projective right B -module. If moreover $\text{Ext}_{A^{\text{op}}}^i(S, A) = 0$ for all $i \geq 0$, then $\text{Ext}_{B^{\text{op}}}^i(M, B \oplus M) = 0$ for all $i \geq 1$.

Proof. For a right A -module V , let $\eta_V: V \rightarrow \text{Hom}_{B^{\text{op}}}(Af, Vf)$ send v to $x \mapsto vx$. Multiplication by f identifies $\eta_V f$ with the identity of Vf , so the kernel and cokernel of η_V are modules over A/AfA , whose composition factors are copies of S . If $\text{Hom}_{A^{\text{op}}}(S, V) = 0$, then η_V is injective. If moreover $\text{Ext}_{A^{\text{op}}}^1(S, V) = 0$, then $\text{Ext}_{A^{\text{op}}}^1(T, V) = 0$ for every A/AfA -module T by induction on the length, so η_V splits; its cokernel is a summand of $\text{Hom}_{B^{\text{op}}}(Af, Vf)$ annihilated by f , hence zero, since $\text{Hom}_{A^{\text{op}}}(T, \text{Hom}_{B^{\text{op}}}(Af, Vf)) \cong \text{Hom}_{B^{\text{op}}}(Tf, Vf) = 0$. For $V = A$ this gives the isomorphism. Passing from A to B removes exactly the isomorphism class of S . If M were projective, $B \oplus M$ would be a progenerator of B and A would be Morita equivalent to B , which has fewer simple modules.

Assume now the vanishing in all degrees. A minimal injective resolution $A \rightarrow I^\bullet$ of the right regular module has no summand with socle S , since the number of such summands of I^j is the dimension of $\text{Ext}_{A^{\text{op}}}^j(S, A)$ over $\text{End}_{A^{\text{op}}}(S)$, so each I^j is a sum of modules $D(Ae_t)$ with $e_t \leq f$. Then $I^j f$ is a direct sum of modules $D(Be_t)$, hence an injective right B -module, and, by the first part of the proof applied to $V = I^j$, the complex $\text{Hom}_{B^{\text{op}}}(Af, I^\bullet f) \cong \text{Hom}_{A^{\text{op}}}(A, I^\bullet)$ computes $\text{Ext}_{B^{\text{op}}}^*(Af, Af)$, which therefore vanishes in positive degrees. $\square$

Example 3.9. The two vanishing conditions in the first part of Proposition 3.8 do not suffice for the last assertion. Let S be the right simple module at the vertex 3 of the algebra $A = \mathbf{k}(1 \xrightarrow{a} 2 \xrightarrow{b} 3)/(ba)$. Its minimal projective resolution is $0 \rightarrow e_1 A \rightarrow e_2 A \rightarrow e_3 A \rightarrow S \rightarrow 0$, and the dual complex shows that $\text{Hom}_{A^{\text{op}}}(S, A) = 0 = \text{Ext}_{A^{\text{op}}}^1(S, A)$, while $\text{Ext}_{A^{\text{op}}}^2(S, A) \cong Ae_1/\mathbf{k}a \neq 0$. The conclusion fails as well: here $B = \mathbf{k}(1 \xrightarrow{a} 2)$ and M is the simple right B -module at the vertex 2, with projective resolution $0 \rightarrow e_1 B \rightarrow e_2 B \rightarrow M \rightarrow 0$, so that $\text{Ext}_{B^{\text{op}}}^1(M, B) \cong Be_1/\mathbf{k}a \neq 0$.

3.3. Triangular matrix algebras and bounds on projective dimension. Triangular matrix algebras cannot produce a counterexample to the strong Nakayama conjecture when both diagonal algebras have finite little finitistic dimension.

Proposition 3.10. *Let $A = \begin{pmatrix} B & 0 \\ M & C \end{pmatrix}$ for a C - B -bimodule M , so that right A -modules are triples (X, Y, φ) with X a right B -module, Y a right C -module*

and $\varphi: Y \otimes_C M \rightarrow X$. Let $E = (X, Y, \varphi) \neq 0$ satisfy $\mathbb{R}\mathrm{Hom}_{A^{\mathrm{op}}}(E, A) = 0$, and put $Z = \mathrm{cone}(Y \otimes_C^{\mathbb{L}} M \rightarrow X)$ in $\mathrm{D}^-(\mathrm{mod} B^{\mathrm{op}})$. Then $\mathbb{R}\mathrm{Hom}_{B^{\mathrm{op}}}(Z, B) = 0$. If $Z \neq 0$, the little finitistic dimension of B is infinite; if $Z = 0$, then $Y \neq 0$, $\mathbb{R}\mathrm{Hom}_{C^{\mathrm{op}}}(Y, C) = 0$, and the little finitistic dimension of C is infinite.

Proof. Let $e = \mathrm{diag}(1, 0)$ and $f = \mathrm{diag}(0, 1)$, let i_* inflate right B -modules and put $j_!Y = Y \otimes_C^{\mathbb{L}} fA$. The derived counit gives a triangle $j_!Y \rightarrow E \rightarrow i_*Z \rightarrow (j_!Y)[1]$. Since $eA = i_*B$, adjunction gives $\mathbb{R}\mathrm{Hom}_{B^{\mathrm{op}}}(Z, B) \cong \mathbb{R}\mathrm{Hom}_{A^{\mathrm{op}}}(E, eA) = 0$. If $Z \neq 0$, represent it by a nonzero minimal bounded-above complex $P^\bullet$ of finitely generated projective right B -modules, with largest nonzero degree m . Its dual $0 \rightarrow (P^m)^* \rightarrow (P^{m-1})^* \rightarrow \cdots$ is exact, and its first cokernel is not projective, since otherwise the first map would split, against the minimality of $P^\bullet$; Proposition 3.1 applied to the successive cokernels gives unbounded finite projective dimensions. If $Z = 0$, then $E \simeq j_!Y$, so $Y \neq 0$, and $\mathbb{R}\mathrm{Hom}_{C^{\mathrm{op}}}(Y, C) \cong \mathbb{R}\mathrm{Hom}_{A^{\mathrm{op}}}(j_!Y, fA) = 0$ since $(fA)f = C$; now Theorem 3.3 applies to C . $\square$

Finite projective dimensions are bounded on modules of bounded dimension. The argument is due to Schofield, as explained by Happel in his survey [Hap90, Section 2.3]; the openness also follows from the semicontinuity of $\dim \mathrm{Ext}$ on module varieties [GLS24, Corollary 2.6]. Both sources work over an algebraically closed field; the argument below works over any field.

Proposition 3.11. *Let A be a finite-dimensional algebra and $d \geq 0$ an integer. The subset of the variety $\mathrm{Rep}_d(A)$ of d -dimensional modules of projective dimension at most n is open, and projective dimension is bounded on the modules of finite projective dimension in $\mathrm{Rep}_d(A)$. In particular, modules of unbounded finite projective dimension have unbounded dimension.*

Proof. Let J be the radical of A . A module M satisfies $\mathrm{pd} M \leq n$ if and only if $\mathrm{Tor}_{n+1}^A(A/J, M) = 0$. Choose a resolution $Q_\bullet$ of the right module A/J by finitely generated free modules. Over $\mathrm{Rep}_d(A)$, the complex

$$Q_{n+2} \otimes_A M \longrightarrow Q_{n+1} \otimes_A M \longrightarrow Q_n \otimes_A M$$

has terms of fixed dimension and differentials given by matrices whose entries are polynomial in the point M ; the vanishing of its middle homology is a lower bound on the sum of the ranks of the two differentials, which is an open condition. The open subsets $\{M \mid \mathrm{pd} M \leq n\}$ form an ascending chain in a noetherian topological space and are therefore stationary. $\square$

In Theorem 3.3, the dimension of $\mathrm{Tr} N$ is at most $(\dim A)^2 \dim N$. Hence, by Proposition 3.11, the syzygies $\Omega^n E$ of a module E as in Theorem 3.3 have unbounded dimension: families of modules of constant dimension whose syzygies move along a parameter, such as those of Schulz [Sch95, Example 7], cannot themselves satisfy the hypothesis of that theorem.

4. TRIVIAL EXTENSIONS AND EXTINCTION

Let Δ be a finite-dimensional algebra and X a finite-dimensional Δ -bimodule, and put

$$A = \Delta \ltimes X, \quad \Phi = X \otimes_{\Delta}^{\mathbb{L}} -.$$

For an inflated A -module N , $\Phi^r N \neq 0$ forces $\mathrm{pd}_A N \geq r$, and when Δ has finite global dimension, an inflated module N has finite projective dimension exactly when $\Phi^t N \simeq 0$ for some t . Hence, for such Δ , $\mathrm{findim} A = \infty$ as soon as finite extinction times of Φ are unbounded on inflated modules. Both statements are due to Minamoto and Yamaura [MY20]; we give a proof in the present conventions, without assuming that X is flat on either side.

4.1. The bar decomposition. Complexes are graded cohomologically, with the conventions of Section 2; the differential of a tensor product of complexes is $d(u \otimes v) = du \otimes v + (-1)^{|u|}u \otimes dv$. We write $D^-(\text{Mod } \Delta)$ for the derived category of bounded-above complexes of left Δ -modules.

Theorem 4.1 (Minamoto–Yamaura [MY20, Lemma 4.13(4) and proof of Theorem 4.17]). *Let N be a left Δ -module, regarded as an A -module through the projection $A \rightarrow \Delta$. There is an isomorphism, natural in N ,*

$$\Delta \otimes_A^{\mathbb{L}} N \cong \bigoplus_{r \geq 0} \Phi^r N[r] \quad \text{in } D^-(\text{Mod } \Delta),$$

where $\Phi^0 N = N$.

The source states the decomposition in terms of the internal grading of the graded algebra A , for right modules; the statement above is its translation to left modules through the opposite algebras. The proof below avoids the grading, and it does not assume that X is flat on either side. That the derived powers of X appear, and not the ordinary ones, matters already in small examples; see Example 4.2.

Proof. Choose a resolution $\tilde{X} \rightarrow X$ by finitely generated projective Δ -bimodules with $\tilde{X}^j = 0$ for $j > 0$, and a projective resolution $\tilde{N} \rightarrow N$ over Δ with $\tilde{N}^j = 0$ for $j > 0$. The complex $\tilde{A} = \Delta \oplus \tilde{X}$, with multiplication $(d, x)(d', x') = (dd', dx' + xd')$, is a dg algebra in which $\tilde{X}$ is an ideal of square zero, and the augmentation $\tilde{A} \rightarrow A$ is a quasi-isomorphism of dg algebras.

The bar module. For $r \geq 0$ put $F_r = \tilde{X}^{\otimes \Delta^r} \otimes_{\Delta} \tilde{N}$, with $F_0 = \tilde{N}$; each F_r is a bounded-above complex of projective left Δ -modules, since for a projective bimodule V and any left module L the module $V \otimes_{\Delta} L$ is a direct summand of a direct sum of modules $\Delta \otimes_{\mathbf{k}} L$. Let

$$\mathcal{B} = \bigoplus_{r \geq 0} (\tilde{A} \otimes_{\Delta} F_r)[r],$$

and write $s^r w$ for the element of the summand of index r corresponding to $w \in \tilde{A} \otimes_{\Delta} F_r$. Let $\tilde{A}$ act by $a \cdot s^r w = (-1)^{r|a|} s^r (aw)$, and define the differential by

$$D(s^r w) = (-1)^r s^r dw + b(s^r w),$$

$$b(s^r (a \otimes q_1 \otimes \cdots \otimes q_r \otimes v)) = s^{r-1} (aq_1 \otimes q_2 \otimes \cdots \otimes q_r \otimes v)$$

for $r > 0$, and $b = 0$ on the summand of index zero; here d is the differential of the tensor product $\tilde{A} \otimes_{\Delta} F_r$. The multiplication $\tilde{A} \otimes_{\Delta} F_r \rightarrow \tilde{A} \otimes_{\Delta} F_{r-1}$ is a chain map, so the mixed terms of D^2 cancel, with coefficients $(-1)^r + (-1)^{r-1}$; and $b^2 = 0$ because $q_1 q_2 = 0$ in $\tilde{A}$. Moreover, $b(a \cdot z) = (-1)^{|a|} a \cdot b(z)$, since $r|a| \equiv |a| + (r-1)|a|$ modulo two. Thus, $\mathcal{B}$ is a dg $\tilde{A}$ -module. The other faces of the usual bar construction vanish: they multiply two elements of $\tilde{X}$, or let $\tilde{X}$ act on $\tilde{N}$.

The bar module resolves N . Let $\epsilon: \mathcal{B} \rightarrow \tilde{N}$ be the map induced by $\tilde{A} \rightarrow \Delta$ on the summand of index zero and zero elsewhere, where $\tilde{N}$ is regarded as an $\tilde{A}$ -module through $\tilde{A} \rightarrow \Delta$. As a complex of left Δ -modules, $\tilde{A} \otimes_{\Delta} F_r = F_r \oplus F_{r+1}$, according to the decomposition $\tilde{A} = \Delta \oplus \tilde{X}$. Hence, $\mathcal{B}$ is the direct sum of $\tilde{N}$, in index zero, and of the complexes

$$F_r[r] \oplus F_r[r-1] \quad (r \geq 1),$$

in which the first summand is the part of index r with leading factor in Δ , the second is the part of index $r-1$ with leading factor in $\tilde{X}$, and the component of D from the first to the second is the identity of the underlying graded module, given by b . Each of these complexes is the mapping cone of an identity map, so the map of degree -1 sending the second summand identically to the first is a contraction.

These contractions give h with $Dh + hD = \text{id} - \iota\epsilon$, where ι is the inclusion of $\tilde{N}$. Hence, ϵ is a quasi-isomorphism.

Base change to A . Filter $\mathcal{B}$ by the subcomplexes of the summands of index at most r . The subquotients are $(\tilde{A} \otimes_{\Delta} F_r)[r]$, and the filtration is split as a filtration of graded $\tilde{A}$ -modules. For an acyclic right dg $\tilde{A}$ -module Z , the complex

$$Z \otimes_{\tilde{A}} (\tilde{A} \otimes_{\Delta} F_r)[r] \cong (Z \otimes_{\Delta} F_r)[r],$$

with the sign $(-1)^{r|z|}$ on $z \otimes s^r w$, is acyclic, because a bounded-above complex of projective modules is K-flat: its stupid truncations from below are finite extensions of projective modules, and it is their filtered colimit. By induction on r and passage to the colimit, $Z \otimes_{\tilde{A}} \mathcal{B}$ is acyclic, so $\mathcal{B}$ is K-flat over $\tilde{A}$. Since $\tilde{A} \rightarrow A$ is a quasi-isomorphism, so is $\mathcal{B} \rightarrow L := A \otimes_{\tilde{A}} \mathcal{B}$, and the augmentation $L \rightarrow N$ induced by ϵ and $\tilde{N} \rightarrow N$ is a quasi-isomorphism. The terms of L are direct sums of modules $(A \otimes_{\Delta} F_r^j)$, which are projective over A , and they vanish in positive degrees; in each degree only finitely many indices r contribute. Hence, L is a bounded-above projective resolution of N .

Tensoring with Δ . In $\Delta \otimes_A L$ the bar differential vanishes, since its image has a leading factor in the ideal X , which acts as zero on Δ . Hence,

$$\Delta \otimes_A L \cong \bigoplus_{r \geq 0} F_r[r],$$

with the differential $(-1)^r d_{F_r}$ on the summand of index r , which is the differential of $F_r[r]$. Finally, F_r represents $\Phi^r N$: by induction on r , F_{r-1} is a bounded-above complex of projective modules representing $\Phi^{r-1} N$, hence K-flat, and $F_r = \tilde{X} \otimes_{\Delta} F_{r-1} \rightarrow X \otimes_{\Delta} F_{r-1}$ is a quasi-isomorphism. Naturality follows by choosing $\tilde{N}$ functorially, for instance as the bar resolution of N over $\mathbf{k}$. $\square$

Example 4.2. Let $\Delta = \mathbf{k}(0 \xrightarrow{a} 1 \xrightarrow{b} 2)$, a path algebra of dimension six, and let $X = S_0 \otimes_{\mathbf{k}} S'_2$, where S_i and S'_i denote the simple left and right modules at the vertex i . Then A is the quotient of the path algebra of the cyclic quiver $0 \rightarrow 1 \rightarrow 2 \rightarrow 0$, with the new arrow $x: 2 \rightarrow 0$, by the relations $ax = 0 = xb$. For $N = S_1$ the ordinary tensor product $X \otimes_{\Delta} N$ is zero, whereas the projective resolution

$$0 \rightarrow e_1 \Delta \xrightarrow{b} e_2 \Delta \rightarrow S'_2 \rightarrow 0$$

gives $\Phi N \cong S_0[1]$ and $\Phi^2 N = 0$. The minimal projective resolution

$$0 \rightarrow Ae_1 \xrightarrow{a} Ae_0 \xrightarrow{x} Ae_2 \xrightarrow{b} Ae_1 \rightarrow S_1 \rightarrow 0,$$

by right multiplications, shows that $\text{pd}_A S_1 = 3$, in agreement with Proposition 4.4: the two finite terms of the formula are $\text{pd}_{\Delta} S_1 = 1$ and $\text{pd}_{\Delta}(S_0[1]) + 1 = 3$.

4.2. Projective dimension and extinction. Let $D^-(\text{mod } \Delta)$ be the derived category of bounded-above complexes with finitely generated cohomology. For a nonzero object C of it, let $\text{pd}_{\Delta} C$ be the least integer p such that C is represented by a bounded-above complex of projective modules vanishing in degrees below $-p$, or ∞ if there is none; for a module this is its projective dimension, and $\text{pd}(C[s]) = \text{pd } C + s$. We put $\text{pd } 0 = -\infty$.

Lemma 4.3. *Let Λ be a finite-dimensional algebra and C a nonzero object of $D^-(\text{mod } \Lambda)$. Then*

$$\text{pd}_{\Lambda} C = \sup\{n \in \mathbb{Z} \mid \text{Hom}(C, S[n]) \neq 0 \text{ for a simple module } S\}.$$

Proof. Represent C by a bounded-above complex J of finitely generated projective modules. If some differential $J^n \rightarrow J^{n+1}$ is nonzero on the tops of its source and target, it has an invertible component between isomorphic indecomposable

summands, and a change of bases splits off a contractible complex $P \xrightarrow{\text{id}} P$. Performing these cancellations from the highest degree downwards, finitely many in each degree, we may assume that J is minimal: every differential has image in the radical of its target. Then $\text{Hom}_\Lambda(J, S)$ has zero differential for every simple S , so $\text{Hom}(C, S[n]) = \text{Hom}_\Lambda(J^{-n}, S)$, which is nonzero for some S exactly when $J^{-n} \neq 0$. A representative vanishing below $-p$ forces these groups to vanish for $n > p$, and the minimal one realises the bound. $\square$

Proposition 4.4 (Minamoto–Yamaura [MY20, Corollary 4.11]). *Let N be a nonzero finitely generated left Δ -module, regarded as an A -module. Then*

$$\text{pd}_A N = \sup_{r \geq 0} (\text{pd}_\Delta(\Phi^r N) + r).$$

In particular, $\Phi^r N \not\simeq 0$ implies $\text{pd}_A N \geq r$.

Proof. Every simple A -module S is annihilated by X : the submodule XS is 0 or S , and $XS = S$ would give $S = X^2 S = 0$. Hence, the simple A -modules are the simple Δ -modules, inflated. For such S , adjunction and Theorem 4.1 give

$$\text{Ext}_A^n(N, S) \cong \text{Hom}_{D(\Delta)}(\Delta \otimes_A^{\mathbb{L}} N, S[n]) \cong \prod_{r \geq 0} \text{Hom}_{D(\Delta)}(\Phi^r N, S[n-r]),$$

where the factors with $r > n$ vanish since $\Phi^r N$ is represented by a complex concentrated in nonpositive degrees. Taking the supremum over n and S and applying Lemma 4.3 over A and over Δ gives the formula. If $\Phi^r N \not\simeq 0$, then $\text{pd}_\Delta \Phi^r N \geq 0$, again by Lemma 4.3. $\square$

Corollary 4.5. *Suppose that the left and the right global dimensions of Δ are at most d_L and d_R , respectively, and let N be a nonzero finitely generated left Δ -module. Then $\text{pd}_A N < \infty$ if and only if $\Phi^t N \simeq 0$ for some $t \geq 1$, and in that case*

$$\text{pd}_A N = \max_{0 \leq r < t} (\text{pd}_\Delta(\Phi^r N) + r) \leq d_L + (t-1)(d_R + 1).$$

In particular, if the finite extinction times $\min\{t \mid \Phi^t N \simeq 0\}$ of finitely generated Δ -modules N are unbounded, then $\text{findim } A = \infty$.

Proof. Truncating a projective bimodule resolution of X at its d_R -th syzygy, which is projective as a right module, gives a resolution $X' \rightarrow X$ in degrees $[-d_R, 0]$ by bi-modules that are projective as right modules. Its tensor powers compute the iterates of Φ , so $\Phi^r N$ has cohomology in degrees $[-rd_R, 0]$. A complex C with finitely generated cohomology in degrees $[u, 0]$ satisfies $\text{pd } C \leq d_L - u$, by Lemma 4.3 applied to the triangles of its cohomology truncations: in a triangle $C_1 \rightarrow C \rightarrow C_2 \rightarrow C_1[1]$, if $\text{Hom}(C_1, S[n]) = 0 = \text{Hom}(C_2, S[n])$, then $\text{Hom}(C, S[n]) = 0$, and $H^i(C)[-i]$ has projective dimension at most $d_L - i$. Hence, $\text{pd}_\Delta \Phi^r N \leq d_L + rd_R$. If $\Phi^t N \simeq 0$, then $\Phi^r N \simeq 0$ for all $r \geq t$, and Proposition 4.4 gives the displayed equality and bound. Conversely, if $\text{pd}_A N = p < \infty$, then $\Phi^{p+1} N \simeq 0$ by Proposition 4.4. For the last assertion, a module of extinction time $t \geq 1$ has $\Phi^{t-1} N \not\simeq 0$, so its projective dimension is finite and at least $t-1$. $\square$

4.3. Two constraints on extinction.

Remark 4.6. Suppose that Δ has finite global dimension, with n simple modules. The functor Φ preserves the bounded derived category of finitely generated modules, by the proof of Corollary 4.5, and induces an endomorphism $[\Phi]$ of $K_0(\Delta) \cong \mathbb{Z}^n$. If $\Phi^t N \simeq 0$, then $[\Phi]^t[N] = 0$, so $[N]$ lies in the generalised kernel of $[\Phi]$ on $\mathbb{Q}^n$, which is annihilated by $[\Phi]^n$. Hence, $[\Phi^r N] = [\Phi]^r[N] = 0$ for $r \geq n$, although $\Phi^r N$ may be nonzero. In the main construction, Proposition 7.2 and Theorem 6.18 give

$\Phi^2 M(Y) \simeq M(HY)[b] \oplus M(HY)[b+3]$, whose class is zero because the two shifts differ by an odd integer, as in Proposition 6.7.

For an integer $d \geq 0$, let $\text{Rep}_d(\Delta)$ be the set of $\mathbf{k}$ -algebra homomorphisms $\Delta \rightarrow M_d(\mathbf{k})$, that is, of left Δ -module structures on $\mathbf{k}^d$. It is a Zariski-closed subset of the affine space $\text{Hom}_{\mathbf{k}}(\Delta, M_d(\mathbf{k}))$, and therefore a noetherian topological space.

Proposition 4.7. *Suppose that Δ has finite right global dimension, and let $d \geq 0$. Then the finite extinction times $\min\{t \mid \Phi^t N \simeq 0\}$ of the modules $N \in \text{Rep}_d(\Delta)$ are bounded.*

Proof. Let $X' \rightarrow X$ be a bounded resolution by finite-dimensional bimodules that are projective as right modules, as in the proof of Corollary 4.5. For each t , the complex $X'^{\otimes_{\Delta} t}$ is bounded, with finite-dimensional terms projective as right modules, and computes Φ^t . Let $f_1, \dots, f_s$ be a complete set of orthogonal primitive idempotents of Δ . Every finitely generated projective right Δ -module is isomorphic to $\bigoplus_i f_i \Delta^{m_i}$ for some integers m_i , so that $\dim_{\mathbf{k}}(V \otimes_{\Delta} N) = \sum_i m_i \dim_{\mathbf{k}} f_i N$ for such a module V . The function $N \mapsto \dim_{\mathbf{k}} f_i N$ is the rank of an idempotent matrix depending polynomially on N , and it is locally constant: the conditions $\text{rank } f_i \geq c$ and $\text{rank}(1 - f_i) \geq d - c$ are open, and the two ranks add up to d . Hence, $\text{Rep}_d(\Delta)$ is the disjoint union of finitely many open and closed subsets on each of which all the numbers $\dim_{\mathbf{k}} f_i N$ are constant.

Fix such a subset Z and an integer $t \geq 0$, and write the terms of $X'^{\otimes_{\Delta} t}$ as direct summands $\varepsilon_j \Delta^{n_j}$ of free right modules, with idempotent matrices ε_j over Δ . For $N \in Z$, the complex $C = X'^{\otimes_{\Delta} t} \otimes_{\Delta} N$ is then identified with a complex of subspaces $C^j = \varepsilon_j(N^{n_j})$, whose dimensions do not depend on N , and whose differentials are the restrictions of linear maps $D^j: N^{n_j} \rightarrow N^{n_{j+1}}$ vanishing on the complements $(1 - \varepsilon_j)(N^{n_j})$ and given by matrices whose entries are polynomial functions of N . The rank of the differential $C^j \rightarrow C^{j+1}$ is the rank of D^j . The complex is exact at C^j if and only if $\text{rank } D^{j-1} + \text{rank } D^j \geq \dim C^j$, which is an open condition. Hence, the sets $U_t = \{N \in Z \mid \Phi^t N \simeq 0\}$ are open. They form an ascending chain, since Φ preserves zero objects, so the chain is stationary, say from T_Z on. Every $N \in Z$ of finite extinction time therefore has extinction time at most T_Z , and the maximum of the finitely many T_Z bounds all finite extinction times in $\text{Rep}_d(\Delta)$. $\square$

Compare Proposition 3.11, which bounds finite projective dimensions on modules of bounded dimension.

5. SELECTION DATA

The main construction uses an exact tensor endofunctor on finite-dimensional modules whose finite extinction times are unbounded. The example in [Ope26b, Section 2] is defined over the complex group algebra of a finitely presented group.

5.1. Selection data and extinction.

Definition 5.1. Let $\mathbf{k}$ be a field. *Selection data* over $\mathbf{k}$ consist of a $\mathbf{k}$ -algebra R and an R -bimodule Ψ on which the two actions of $\mathbf{k}$ agree and which is finitely generated and projective as a right R -module. The *selection functor* is

$$H = \Psi \otimes_R -: \text{mod}_{\text{fd}} R \longrightarrow \text{mod}_{\text{fd}} R.$$

The *extinction time* of a finite-dimensional left R -module Y is the least integer $t \geq 0$ with $H^t Y = 0$, or ∞ if there is none. The selection data have *unbounded extinction* if the finite extinction times of finite-dimensional modules are unbounded.

The functor H is well defined and exact: Ψ_R is a direct summand of some R^n , so that, as a vector space, HY is a direct summand of Y^n , naturally in Y , and exactness is detected on underlying vector spaces; see Lemma 6.9 for an explicit description. Iterates of H are written H^t , with H^0 the identity.

Example 5.2. Let $e \in R$ be a central idempotent and $\alpha: R \rightarrow R$ a homomorphism of unital $\mathbf{k}$ -algebras, and let $\Psi = {}_\alpha(eR)$ be the bimodule eR with its right regular action and the left action $r \cdot x = \alpha(r)x$. Then Ψ is projective as a right module, and $HY \cong {}_\alpha(eY)$ is the R -submodule eY of Y with r acting as $\alpha(r)$: the map $\Psi \otimes_R Y \rightarrow eY$, $x \otimes y \mapsto xy$, has the inverse $v \mapsto e \otimes v$, since $e \otimes xy = ex \otimes y = x \otimes y$. This is the case treated in [Ope26b, Section 2].

Selection data with unbounded extinction must detect infinitely many independent numerical invariants of finite-dimensional modules. To make this precise, let $K_0(R^{\text{op}})$ be the Grothendieck group of finitely generated projective right R -modules. For a finite-dimensional left R -module Y the *rank function* of Y is the additive map

$$\chi_Y: K_0(R^{\text{op}}) \longrightarrow \mathbb{Z}, \quad [P] \longmapsto \dim_{\mathbf{k}}(P \otimes_R Y),$$

and the bimodule Ψ induces the endomorphism $\mathsf{T}[P] = [P \otimes_R \Psi]$ of $K_0(R^{\text{op}})$; here $P \otimes_R \Psi$ is finitely generated and projective since P is a direct summand of some R^n . Associativity of the tensor product gives

$$(5.3) \quad \chi_Y(\mathsf{T}x) = \chi_{HY}(x) \quad \text{and} \quad \dim_{\mathbf{k}} H^t Y = \chi_Y(\mathsf{T}^t[R])$$

for every $x \in K_0(R^{\text{op}})$ and every $t \geq 0$. Let $\mathcal{F}$ be the $\mathbb{Q}$ -vector space of functions from the set of isomorphism classes of finite-dimensional left R -modules to $\mathbb{Q}$, and let

$$\chi: K_0(R^{\text{op}}) \otimes_{\mathbb{Z}} \mathbb{Q} \longrightarrow \mathcal{F}, \quad x \longmapsto (Y \mapsto \chi_Y(x)).$$

For the selection data, put

$$\mathcal{V} = \chi(\text{span}_{\mathbb{Q}}\{\mathsf{T}^t[R] \mid t \geq 0\}) \subseteq \mathcal{F}.$$

Proposition 5.4. *Let (R, Ψ) be selection data, and suppose that $r = \dim_{\mathbb{Q}} \mathcal{V} < \infty$. Then every finite-dimensional left R -module of finite extinction time has extinction time at most r .*

Proof. By (5.3), if $\chi(x) = 0$ then $\chi(\mathsf{T}x) = 0$, since HY is again finite-dimensional. Hence, T induces an endomorphism $\bar{\mathsf{T}}$ of $\mathcal{V}$, and $\mathcal{V}$ is spanned by the vectors $\bar{\mathsf{T}}^t v$ for $t \geq 0$, where $v = \chi([R])$. The Fitting decomposition of $\bar{\mathsf{T}}$ gives $\mathcal{V} = \mathcal{V}_0 \oplus \mathcal{V}_1$, where $\bar{\mathsf{T}}$ is nilpotent on $\mathcal{V}_0$, hence $\bar{\mathsf{T}}^r \mathcal{V}_0 = 0$, and invertible on $\mathcal{V}_1$. Write $v = v_0 + v_1$ accordingly, and let $\mathcal{V}_2 \subseteq \mathcal{V}_1$ be the span of the vectors $\bar{\mathsf{T}}^t v_1$ for $t \geq 0$. The restriction of $\bar{\mathsf{T}}$ to $\mathcal{V}_2$ is injective, hence bijective, so for every t_0 the vectors $\bar{\mathsf{T}}^t v_1$ with $t \geq t_0$ span $\bar{\mathsf{T}}^{t_0} \mathcal{V}_2 = \mathcal{V}_2$.

Let Y be a finite-dimensional module of finite extinction time t_0 , and let $\phi_Y: \mathcal{V} \rightarrow \mathbb{Q}$ be evaluation at Y . Then $\phi_Y(\bar{\mathsf{T}}^t v) = \dim_{\mathbf{k}} H^t Y = 0$ for $t \geq t_0$ by (5.3). For $t \geq \max(t_0, r)$ we have $\bar{\mathsf{T}}^t v = \bar{\mathsf{T}}^t v_1$, so ϕ_Y vanishes on $\mathcal{V}_2$. Hence,

$$\dim_{\mathbf{k}} H^r Y = \phi_Y(\bar{\mathsf{T}}^r v_0) + \phi_Y(\bar{\mathsf{T}}^r v_1) = 0. \quad \square$$

Corollary 5.5. *Selection data (R, Ψ) over $\mathbf{k}$ have bounded extinction in each of the following cases.*

- (1) *There are finitely many finitely generated projective right R -modules such that every finitely generated projective right R -module is a finite direct sum of copies of them; for example, R is finite-dimensional.*
- (2) *The algebra R is commutative and noetherian.*

- (3) *The algebra R is a universal localisation of a finite-dimensional hereditary algebra.*

Proof. By Proposition 5.4, it suffices to show that χ has image of finite dimension. In case (1) the classes of the finitely many modules span $K_0(R^{\text{op}}) \otimes_{\mathbb{Z}} \mathbb{Q}$; if R is finite-dimensional, the Krull–Schmidt theorem shows that every finitely generated projective module is a finite direct sum of indecomposable direct summands of R , of which there are finitely many up to isomorphism. In case (3), let $R = B_{\Sigma}$ be a universal localisation of a finite-dimensional hereditary algebra B . By a theorem of Schofield [Sch07, Theorem 2.3], every universal localisation of B is of the form studied in [Sch07], and for these the map from $K_0(B^{\text{op}})$ to $K_0(B_{\Sigma}^{\text{op}})$, for right modules as in [Sch07], is surjective [Sch07, Lemma 4.1]. Since $K_0(B^{\text{op}})$ is finitely generated by case (1), so is $K_0(B_{\Sigma}^{\text{op}})$.

In case (2), for a finitely generated projective R -module P , the function

$$\rho_P: \text{Spec } R \longrightarrow \mathbb{Z}, \quad \mathfrak{p} \longmapsto \dim_{\kappa(\mathfrak{p})} P \otimes_R \kappa(\mathfrak{p})$$

is locally constant [StacksProj, Tag 00NX, Lemma 10.78.2]. The space $\text{Spec } R$ has finitely many irreducible components [StacksProj, Tag 00FR, Lemma 10.31.6]; each of them is connected and contained in a connected component, so $\text{Spec } R$ has finitely many connected components $Z_1, \dots, Z_c$, and ρ_P takes a constant value $\rho_P(Z_i)$ on each of them. A finite-dimensional module Y has finite length, its composition factors are of the form $R/\mathfrak{m}$ with $\mathfrak{m}$ maximal and $R/\mathfrak{m}$ finite-dimensional, and

$$\dim_{\mathbf{k}}(P \otimes_R R/\mathfrak{m}) = \rho_P(\mathfrak{m}) \dim_{\mathbf{k}} R/\mathfrak{m}.$$

Since $P \otimes_R -$ is exact,

$$\chi_Y([P]) = \sum_{i=1}^c \rho_P(Z_i) d_i(Y),$$

where $d_i(Y)$ is the sum of the dimensions of the composition factors $R/\mathfrak{m}$ of Y with $\mathfrak{m} \in Z_i$. Hence, the image of χ lies in the span of the c functions d_i . $\square$

In case (2), the argument bounds the dimension of the image of χ by the number of connected components, without requiring $K_0(R)$ to have finite rank. In the example below, $\mathcal{V}$ is infinite-dimensional; see Remark 5.18.

5.2. The group. We write $[x, y] = xyx^{-1}y^{-1}$ for elements of a group, and put $I = \{1, 2, 3\}$. Let $\varepsilon_1, \varepsilon_2, \varepsilon_3$ be the standard basis of $\mathbb{Z}^3$, and for $n = (n_1, n_2, n_3) \in \mathbb{Z}^3$ and commuting elements T_1, T_2, T_3 write $T^n = T_1^{n_1} T_2^{n_2} T_3^{n_3}$.

Definition 5.6. Let G be the group with generators T_1, T_2, T_3 and $U_i(r), V_i(r), W_{ij}(r)$ for $i, j \in I$ with $i \neq j$ and $r \in \mathbb{Z}$, subject to the following relations, for all integers r and s and all indices satisfying the stated conditions: the T_ℓ commute pairwise;

$$(5.7) \quad T_\ell S(r) T_\ell^{-1} = S(r + \lambda_S(\varepsilon_\ell))$$

for every generator type $S \in \{U_i, V_i, W_{ij}\}$, where the *weights* are the homomorphisms $\lambda_S: \mathbb{Z}^3 \rightarrow \mathbb{Z}$ given by $\lambda_{U_i}(n) = -n_i$, $\lambda_{V_i}(n) = n_i$ and $\lambda_{W_{ij}}(n) = n_i - n_j$;

$$(5.8) \quad \begin{aligned} U_i(r)^2 &= 1, & [U_i(r), U_j(s)] &= [V_i(r), V_j(s)] = [U_i(r), V_j(s)] = 1 \quad (i \neq j), \\ [W_{ij}(s), U_h(r)] &= 1 \quad (h \neq i), & [W_{ij}(s), V_h(r)] &= 1 \quad (h \neq j), \\ [U_i(r), W_{ij}(s)] &= U_j(r+s), & [W_{ij}(s), V_j(r)] &= V_i(r+s). \end{aligned}$$

No relation between two generators of the same type with different parameters and no relation between two generators of types W_{ij} is imposed. The main preprint describes G as motivated by finitely presented matrix groups of Abels type, and its

commutator arguments parallel the proof of [San22, Proposition 4.9, relations (4.9) and (4.10)]; the proofs below are self-contained.

Proposition 5.9. *The group G is finitely presented. More precisely, put $u_i = U_i(0)$, $v_i = V_i(0)$ and $w_{ij} = W_{ij}(0)$. Then G is generated by the fifteen elements T_ℓ , u_i , v_i and w_{ij} , subject to the following finitely many relations: the T_ℓ commute pairwise; the relations (5.8) with $r = s = 0$; and $[T_h, u_i] = [T_h, v_i] = 1$ for $h \neq i$, together with $[T_i T_j, w_{ij}] = [T_k, w_{ij}] = 1$ for $\{i, j, k\} = I$.*

Proof. Let P be the group with the stated presentation. For a type S write s_S for the generator u_i , v_i or w_{ij} , and choose $q_{U_i} = -\varepsilon_i$, $q_{V_i} = \varepsilon_i$ and $q_{W_{ij}} = \varepsilon_i$, so that $\lambda_S(q_S) = 1$. The vectors b with $[T^b, s_S] = 1$ among the relations of P form a basis of the kernel K_S of λ_S : for U_i and V_i the kernel is $\{n \mid n_i = 0\}$, with basis ε_h for $h \neq i$, and for W_{ij} it is $\{n \mid n_i = n_j\}$, whose elements are $n_i(\varepsilon_i + \varepsilon_j) + n_k \varepsilon_k$. Hence, T^b commutes with s_S for every $b \in K_S$. In P put

$$\tilde{S}(r) = T^{r q_S} s_S T^{-r q_S} \quad (r \in \mathbb{Z}).$$

For $n \in \mathbb{Z}^3$ we have $n - \lambda_S(n) q_S \in K_S$, and the elements T_1, T_2, T_3 commute, so

$$(5.10) \quad T^n \tilde{S}(r) T^{-n} = \tilde{S}(r + \lambda_S(n)).$$

This gives the relations (5.7) for the elements $\tilde{S}(r)$, and conjugating $u_i^2 = 1$ by $T^{r q_{U_i}}$ gives $\tilde{U}_i(r)^2 = 1$. Each of the remaining relations in (5.8) involves two generators of types S and S' with prescribed parameters, and it is obtained by conjugating the corresponding relation with parameters zero by T^n , for an $n \in \mathbb{Z}^3$ with the prescribed values of $\lambda_S(n)$ and $\lambda_{S'}(n)$, and using (5.10). Such vectors n are, for distinct i, j, k :

$$\begin{array}{ll} [U_i(r), U_j(s)]: & -r\varepsilon_i - s\varepsilon_j, & [W_{ij}(s), V_i(r)]: & r\varepsilon_i + (r-s)\varepsilon_j, \\ [V_i(r), V_j(s)]: & r\varepsilon_i + s\varepsilon_j, & [W_{ij}(s), V_k(r)]: & s\varepsilon_i + r\varepsilon_k, \\ [U_i(r), V_j(s)]: & -r\varepsilon_i + s\varepsilon_j, & [U_i(r), W_{ij}(s)]: & -r\varepsilon_i - (r+s)\varepsilon_j, \\ [W_{ij}(s), U_j(r)]: & (s-r)\varepsilon_i - r\varepsilon_j, & [W_{ij}(s), V_j(r)]: & (r+s)\varepsilon_i + r\varepsilon_j. \\ [W_{ij}(s), U_k(r)]: & s\varepsilon_i - r\varepsilon_k, & \end{array}$$

In the last two cases the weight of the third generator is $\lambda_{U_j}(n) = r + s$ and $\lambda_{V_i}(n) = r + s$, as required. Hence, all defining relations of G hold for the elements T_ℓ and $\tilde{S}(r)$ of P .

Conversely, the relations (5.7) give $T^n S(r) T^{-n} = S(r + \lambda_S(n))$ in G for all $n \in \mathbb{Z}^3$, which implies the relations $[T^b, S(0)] = 1$ for $b \in K_S$; the other relations of P are instances of relations of G . Thus, there are homomorphisms $P \rightarrow G$ sending s_S to $S(0)$ and $G \rightarrow P$ sending $S(r)$ to $\tilde{S}(r)$, both fixing the T_ℓ . Their composites fix all generators, since $\tilde{S}(0) = s_S$ and $T^{r q_S} S(0) T^{-r q_S} = S(r)$. $\square$

Proposition 5.11. *For every $N \in \mathbb{Z}$ the element*

$$z_N = [U_i(a), V_i(b)] \quad (i \in I, a, b \in \mathbb{Z}, a + b = N)$$

of G does not depend on the choice of i , a and b . It is central and satisfies $z_N^2 = 1$.

Proof. Let $i \neq j$ and $a, s, b \in \mathbb{Z}$, and put $x = U_i(a)$, $w = W_{ij}(s)$, $v = V_j(b)$, $p = [x, w] = U_j(a + s)$ and $q = [w, v] = V_i(s + b)$. By the relations (5.8), the element x commutes with v , and q commutes with p and with v . Using $xwx^{-1} = pw$ and $wvw^{-1} = qv$, we compute

$$xqx^{-1} = x(wvw^{-1}v^{-1})x^{-1} = (pw)v(pw)^{-1}v^{-1} = p(qv)p^{-1}v^{-1} = q[p, v].$$

Since q commutes with p and v , it commutes with $[p, v]$, and multiplying by q^{-1} gives $[x, q] = [p, v]$, that is,

$$(5.12) \quad [U_i(a), V_i(s + b)] = [U_j(a + s), V_j(b)].$$

If $a + b = c + d$ and $i \neq j$, then (5.12) with $s = c - a$ and b replaced by d gives $[U_i(a), V_i(b)] = [U_j(c), V_j(d)]$. Two splittings at the same index i are both equal to $[U_j(N), V_j(0)]$ for some $j \neq i$. This proves the independence.

Let g be a generator of G . If g is $U_h(r)$ or $V_h(r)$, write z_N at an index $i \neq h$; then g commutes with $U_i(a)$ and $V_i(b)$ by (5.8). If $g = W_{ij}(r)$, write z_N at the index $k \notin \{i, j\}$, and use the relations $[W_{ij}, U_k] = [W_{ij}, V_k] = 1$. If $g = T_\ell$, then (5.7) gives

$$T_\ell z_N T_\ell^{-1} = [U_i(a - \delta_{\ell i}), V_i(b + \delta_{\ell i})] = z_N.$$

Hence, z_N is central. Finally, with $x = U_i(a)$ and $y = V_i(b)$, centrality of $z_N = [x, y]$ gives $xyx^{-1} = z_N y$ and then $x^2 y x^{-2} = z_N^2 y$; since $x^2 = 1$, we obtain $z_N^2 = 1$. $\square$

Proposition 5.13. *For every $c \in \mathbb{Z}$ there is an automorphism β_c of G fixing the T_ℓ , the $V_i(r)$ and the $W_{ij}(r)$, with $\beta_c(U_i(r)) = U_i(r + c)$. Moreover, $\beta_c \beta_d = \beta_{c+d}$, and $\alpha_G := \beta_1$ satisfies $\alpha_G(z_N) = z_{N+1}$ for every N .*

Proof. The assignment sends each defining relation to a defining relation:

- the relations (5.7) for U_i go to

$$T_\ell U_i(r + c) T_\ell^{-1} = U_i(r + c - \delta_{\ell i});$$

- the relations

$$\begin{aligned} U_i(r)^2 &= 1, & [U_i(r), U_j(s)] &= 1, \\ [U_i(r), V_j(s)] &= 1, & [W_{ij}(s), U_h(r)] &= 1 \end{aligned}$$

go to their instances with the parameter of each factor U replaced by its sum with c ;

- the relation $[U_i(r), W_{ij}(s)] = U_j(r + s)$ goes to its instance with first parameter $r + c$;
- all other relations are fixed.

Hence, β_c is an endomorphism, and the identities $\beta_c \beta_d = \beta_{c+d}$ and $\beta_0 = \text{id}$ hold on generators, so β_c is an automorphism with inverse β_{-c} . Applying β_1 to $z_N = [U_i(a), V_i(b)]$ gives $[U_i(a + 1), V_i(b)] = z_{N+1}$ by Proposition 5.11. $\square$

5.3. Finite quotients. For an integer $m \geq 1$ let $S_m = \mathbb{F}_2[t]/(t^m - 1)$. Division by the monic polynomial $t^m - 1$ shows that $1, t, \dots, t^{m-1}$ is a basis of S_m over $\mathbb{F}_2$, and t is a unit with inverse t^{m-1} ; the ring S_m need not be reduced. We write E_{ab} for the matrix units in $M_5(S_m)$ and $\mathbf{1}$ for the identity matrix.

Proposition 5.14. *For every $m \geq 1$ there is a homomorphism $\pi_m: G \rightarrow \text{GL}_5(S_m)$ with*

$$\begin{aligned} \pi_m(T_\ell) &= D_\ell, & \pi_m(U_i(r)) &= \mathbf{1} + t^r E_{1,i+1}, \\ \pi_m(V_i(r)) &= \mathbf{1} + t^r E_{i+1,5}, & \pi_m(W_{ij}(r)) &= \mathbf{1} + t^r E_{i+1,j+1}, \end{aligned}$$

where D_ℓ is the diagonal matrix with entry t in position $\ell + 1$ and entries 1 elsewhere. Its image F_m is finite, and $\pi_m(z_N) = \mathbf{1} + t^N E_{15}$ for every $N \in \mathbb{Z}$. The elements $\pi_m(z_0), \dots, \pi_m(z_{m-1})$ generate a central subgroup Z_m of F_m isomorphic to $(\mathbb{Z}/2)^m$.

Proof. Every matrix unit $A = E_{ab}$ with $a \neq b$ satisfies $A^2 = 0$, so $(\mathbf{1} + xA)^2 = \mathbf{1} + 2xA = \mathbf{1}$ for $x \in S_m$, since S_m has characteristic two; in particular, the matrices above are invertible and the relations $U_i(r)^2 = 1$ hold. The diagonal matrices commute, and $DE_{ab}D^{-1} = d_a d_b^{-1} E_{ab}$ for a diagonal matrix D with entries d_a ; for $D = D_\ell$ the factors for $E_{1,i+1}$, $E_{i+1,5}$ and $E_{i+1,j+1}$ are $t^{-\delta_{\ell i}}$, $t^{\delta_{\ell i}}$ and $t^{\delta_{\ell i} - \delta_{\ell j}}$, which gives the relations (5.7). If A and B are matrix units with $AB = BA = 0$, then $\mathbf{1} + xA$ and $\mathbf{1} + yB$ commute; since $E_{ab}E_{cd} = \delta_{bc}E_{ad}$, this applies to the pairs in the commutation relations of (5.8), by the restrictions on the indices: for instance, for W_{ij} and U_h with $h \neq i$ the products are $E_{i+1,j+1}E_{1,h+1} = 0$ and

$E_{1,h+1}E_{i+1,j+1} = \delta_{hi}E_{1,j+1} = 0$. For distinct a, b and c , the matrices $A = E_{ab}$ and $B = E_{bc}$ satisfy $BA = 0$ and $AB = E_{ac}$, and AB has zero product with A and B on both sides; hence

$$[\mathbf{1} + xA, \mathbf{1} + yB] = (\mathbf{1} + xA)(\mathbf{1} + yB)(\mathbf{1} + xA)(\mathbf{1} + yB) = \mathbf{1} + xyAB.$$

With $(a, b, c) = (1, i+1, j+1)$ and $(i+1, j+1, 5)$ this gives the last two relations of (5.8), and with $(1, i+1, 5)$ it gives $\pi_m(z_N) = \mathbf{1} + t^{a+b}E_{15} = \mathbf{1} + t^N E_{15}$. The image F_m is finite since $M_5(S_m)$ is finite.

The matrix unit E_{15} has zero product on both sides with every matrix unit E_{ab} , $a \neq b$, occurring above, and every D_ℓ has equal entries in positions 1 and 5, so $\pi_m(z_N)$ is central in F_m . Since $E_{15}^2 = 0$, we have

$$\prod_{q=0}^{m-1} \pi_m(z_q)^{c_q} = \mathbf{1} + \left(\sum_{q=0}^{m-1} c_q t^q \right) E_{15}$$

for $c_q \in \{0, 1\}$, which is the identity only if all c_q vanish. Together with $\pi_m(z_q)^2 = \mathbf{1}$ this gives $Z_m \cong (\mathbb{Z}/2)^m$. $\square$

Corollary 5.15. *The elements z_N , $N \in \mathbb{Z}$, generate a subgroup of the centre of G isomorphic to $\bigoplus_{N \in \mathbb{Z}} \mathbb{Z}/2$. In particular, the centre of G is not finitely generated.*

Proof. Let $N_1 < \dots < N_a$ be integers and choose $m > N_a - N_1$. The residues of the N_b modulo m are distinct, so Proposition 5.14 shows that every nonempty product of the z_{N_b} has nontrivial image in F_m . Since the z_N are central involutions by Proposition 5.11, they generate the stated subgroup. A subgroup of a finitely generated abelian group is finitely generated, which this subgroup is not. $\square$

The group G is finitely presented, has an automorphism shifting the independent central involutions, and has finite quotients F_m separating every finite subfamily of them.

5.4. Selection data with unbounded extinction. Let $R = \mathbb{C}G$ be the complex group algebra of the group G of Definition 5.6, let $\alpha: R \rightarrow R$ be the linear extension of α_G , and put

$$e = \frac{1}{2}(1 + z_0) \in R, \quad \Psi = \alpha(eR).$$

Lemma 5.16. *Let R be a $\mathbf{k}$ -algebra, $e \in R$ a central idempotent and α an automorphism of R , and let H be the selection functor of $\Psi = \alpha(eR)$. Put $e_q = \alpha^q(e)$ and $p_j = e_0 e_1 \dots e_{j-1}$, with $p_0 = 1$. Then for every finite-dimensional left R -module Y and every $j \geq 0$ there is an isomorphism*

$$H^j Y \cong_{\alpha^j} (p_j Y),$$

natural in Y , where $r \in R$ acts on the subspace $p_j Y$ of Y as $\alpha^j(r)$.

Proof. The elements e_q are central idempotents, since α is an automorphism. For $j = 0$ there is nothing to prove. If $H^j Y \cong_{\alpha^j} (p_j Y)$, then by Example 5.2 the module $H^{j+1} Y$ is the image of e acting on $_{\alpha^j} (p_j Y)$, that is, the subspace $\alpha^j(e)p_j Y = p_{j+1} Y$, with r acting as $\alpha^j(\alpha(r)) = \alpha^{j+1}(r)$. The identifications are restrictions of the identity of Y , hence natural. $\square$

Theorem 5.17. *The pair (R, Ψ) is selection data over $\mathbb{C}$ with R finitely presented. For every integer $m \geq 1$ there is a nonzero finite-dimensional left R -module Y_m of extinction time exactly m . In particular, (R, Ψ) has unbounded extinction.*

Proof. By Proposition 5.9, the group G has a presentation with fifteen generators $g_1, \dots, g_{15}$ and finitely many relators $w_1, \dots, w_b$. The algebra with generators $x_1, y_1, \dots, x_{15}, y_{15}$ and relations $x_i y_i = y_i x_i = 1$ and $w_c(x, y) = 1$, in which g_i^{-1} is replaced by y_i , is isomorphic to R : the homomorphism to R has an inverse given by

the linear extension of the group homomorphism from G to its group of units sending g_i to x_i . Hence, R is finitely presented. The map α is an automorphism with inverse the linear extension of α_G^{-1} , and e is a central idempotent by Proposition 5.11; so (R, Ψ) is selection data, by Example 5.2. Moreover, $e_q = \alpha^q(e) = \frac{1}{2}(1 + z_q)$ by Proposition 5.13.

Fix $m \geq 1$, and let F_m and Z_m be as in Proposition 5.14, with $\bar{z}_q = \pi_m(z_q)$. Since $Z_m \cong (\mathbb{Z}/2)^m$ with basis $\bar{z}_0, \dots, \bar{z}_{m-1}$, there is a character $\chi_m: Z_m \rightarrow \mathbb{C}^\times$ with $\chi_m(\bar{z}_q) = 1$ for $0 \leq q < m-1$ and $\chi_m(\bar{z}_{m-1}) = -1$. Put

$$p_{\chi_m} = \frac{1}{2^m} \sum_{z \in Z_m} \chi_m(z)^{-1} z \in \mathbb{C}F_m, \quad Y_m = (\mathbb{C}F_m)p_{\chi_m},$$

regarded as a left R -module through π_m . The coefficient of the identity in p_{χ_m} is 2^{-m} , so $Y_m \neq 0$, and Y_m is finite-dimensional. For $h \in Z_m$, the substitution $u = hz$ gives $hp_{\chi_m} = \chi_m(h)p_{\chi_m}$, and since Z_m is central in F_m , every $h \in Z_m$ acts on Y_m as the scalar $\chi_m(h)$. Hence, e_q acts on Y_m as the identity for $0 \leq q < m-1$ and as zero for $q = m-1$. By Lemma 5.16, $H^j Y_m$ is isomorphic to a twist of the subspace $p_j Y_m$, which is $Y_m \neq 0$ for $j < m$ and zero for $j \geq m$, since then p_j contains the factor e_{m-1} . $\square$

Remark 5.18. In the notation of Proposition 5.4, we have $\chi_Y(\mathbb{T}^j[R]) = \dim p_j Y$ by Lemma 5.16, and the values of these functions on the modules $Y_1, \dots, Y_s$ form a triangular matrix with nonzero diagonal, so $\mathcal{V}$ is infinite-dimensional. The module Y_m is the module induced from the character χ_m of Z_m , of dimension $[F_m : Z_m]$. The integer m enters only through the choice of the finite quotient F_m and of the values of χ_m on the central involutions; R , e and α are independent of m , as required in Section 7.

6. BIMODULE COMPLEXES FROM SELECTION DATA

This section turns selection data into a complex of bimodules over a finite-dimensional algebra of global dimension at most two. Let (R, Ψ) be selection data in the sense of Definition 5.1, with R finitely presented, and put $H = \Psi \otimes_R -$. Theorem 6.18 provides a finite-dimensional algebra B , an exact fully faithful functor M from finite-dimensional R -modules to B -modules, and a bounded complex P of B -bimodules such that

$$P \otimes_B^{\mathbb{L}} M(Y) \simeq M(HY) \oplus M(HY)[3]$$

for every finite-dimensional R -module Y . The argument has four steps.

- (1) The algebra B is constructed from a presentation of R by generators and relations of degree at most two, and R acts on an object E_0 of a Verdier quotient $\mathcal{Q}$ of the bounded homotopy category of projective right B -modules (Sections 6.2 and 6.3).
- (2) The bimodule Ψ determines a direct summand U of a finite sum of copies of E_0 in the idempotent completion of $\mathcal{Q}$. The object U need not lie in $\mathcal{Q}$, but $U \oplus U[3]$ does (Section 6.4).
- (3) A representation of the quiver with relations of B in $\mathcal{Q}$, with $U \oplus U[3]$ at every vertex, lifts to the homotopy category up to homotopies prescribed on the relations (Section 6.5).
- (4) Such a homotopy representation is replaced by a complex of bimodules (Section 6.6), and the extension between the two cohomology modules of $P \otimes_B^{\mathbb{L}} M(Y)$ vanishes because $\text{gldim } B \leq 2$ (Section 6.7).

The argument uses choices that are not made explicit in steps (2) and (3); see Remark 6.14.

Throughout this section $\mathbf{k}$ is a field, and all algebras, bimodules, categories and functors are $\mathbf{k}$ -linear; bimodules are $\mathbf{k}$ -central, that is, the two actions of $\mathbf{k}$ on them agree. For a finite-dimensional algebra B we write $\text{proj } B^{\text{op}}$ for the category of finitely generated projective right B -modules and $\text{mod } B$ for the category of finitely generated left B -modules. For a $\mathbf{k}$ -algebra R , not necessarily finite-dimensional, $\text{mod}_{\text{fd}} R$ denotes the category of finite-dimensional left R -modules.

6.1. Right fractions. Let $\mathcal{K}$ be an essentially small triangulated category and $\mathcal{S}$ a full triangulated subcategory. Let $\mathcal{W}$ be the class of morphisms u of $\mathcal{K}$ that occur in a triangle $X' \xrightarrow{u} X \rightarrow Z \rightarrow X'[1]$ with Z isomorphic to an object of $\mathcal{S}$. By [StacksProj, Tag 05RG, Lemma 13.6.6], $\mathcal{W}$ is a multiplicative system, and the Verdier quotient is defined as $\mathcal{K}/\mathcal{S} = \mathcal{W}^{-1}\mathcal{K}$, with quotient functor $q: \mathcal{K} \rightarrow \mathcal{K}/\mathcal{S}$ [StacksProj, Tag 05RI, Definition 13.6.7]. Since the categories of left and of right fractions of a multiplicative system are canonically isomorphic [StacksProj, Tag 04VL, Lemma 4.27.19], we may compute in $\mathcal{K}/\mathcal{S}$ with right fractions [StacksProj, Tag 04VH, Lemma 4.27.11]: a morphism $qX \rightarrow qY$ is the class of a pair (f, u) of morphisms $f: X' \rightarrow Y$ and $u: X' \rightarrow X$ with $u \in \mathcal{W}$. The composition rule gives $(f, u) \circ q(u) = q(f)$, so that the class of (f, u) is $q(f)q(u)^{-1}$, and $q(u)$ is invertible for every $u \in \mathcal{W}$ [StacksProj, Tag 04VK, Lemma 4.27.16].

Lemma 6.1. *In the situation above, let X be an object of $\mathcal{K}$.*

- (1) *For finitely many morphisms $\gamma_i: qX \rightarrow qY_i$ in $\mathcal{K}/\mathcal{S}$, there exist a morphism $u: X' \rightarrow X$ in $\mathcal{W}$ and morphisms $f_i: X' \rightarrow Y_i$ in $\mathcal{K}$ such that $\gamma_i = q(f_i)q(u)^{-1}$ for every i .*
- (2) *For finitely many morphisms $g_i: X \rightarrow Y_i$ in $\mathcal{K}$ with $q(g_i) = 0$, there exists a morphism $v: X' \rightarrow X$ in $\mathcal{W}$ such that $g_i v = 0$ for every i .*

Proof. (1) By induction on the number of morphisms, it suffices to give two roofs (f_1, u_1) and (f_2, u_2) from X , with $u_i: X_i \rightarrow X$, a common denominator. The Ore condition for the right multiplicative system $\mathcal{W}$ [StacksProj, Tag 04VC, Definition 4.27.1] gives morphisms $t: X' \rightarrow X_2$ in $\mathcal{W}$ and $g: X' \rightarrow X_1$ with $u_1 g = u_2 t$. The morphism $u := u_2 t$ lies in $\mathcal{W}$, and $q(g) = q(u_1)^{-1}q(u)$ is invertible. Hence,

$$q(f_1)q(u_1)^{-1} = q(f_1 g)q(u)^{-1} \quad \text{and} \quad q(f_2)q(u_2)^{-1} = q(f_2 t)q(u)^{-1}.$$

For the empty family take $u = \text{id}_X$.

(2) Replacing the family by the single morphism $g = (g_i)_i: X \rightarrow \bigoplus_i Y_i$, we may assume that there is one morphism. The roofs (g, id_X) and $(0, \text{id}_X)$ represent the same morphism of $\mathcal{K}/\mathcal{S}$, so that [StacksProj, Tag 04VJ, Lemma 4.27.14] gives $v: X' \rightarrow X$ in $\mathcal{W}$ with $gv = 0$. $\square$

6.2. The algebra B . Every finitely presented algebra has a presentation by relations of degree at most two. Indeed, given a finite presentation, introduce for each monomial $y_1 y_2 \cdots y_n$ of length $n \geq 3$ occurring in a relation new generators $w_2, \dots, w_{n-1}$ with the relations $w_2 = y_1 y_2$ and $w_{j+1} = w_j y_{j+1}$, and replace the monomial by $w_{n-1} y_n$. Sending w_j to $y_1 \cdots y_j$ and each old generator to its class defines mutually inverse homomorphisms between the two presented algebras.

Construction 6.2. Let R be a $\mathbf{k}$ -algebra with a presentation

$$R = \mathbf{k}\langle x_1, \dots, x_d \rangle / (p_1, \dots, p_t)$$

in which every p_j is a noncommutative polynomial of degree at most two; constant and linear terms are allowed. Let Q be the quiver with vertices 0, 1 and 2, arrows $s, a_1, \dots, a_d: 0 \rightarrow 1$ and $s', a'_1, \dots, a'_d: 1 \rightarrow 2$, and no other arrows. Write $\mathcal{A}_{01}$ and

$\mathcal{A}_{12}$ for the two sets of arrows. The *homogenisation* $\tilde{p}_j$ of p_j is the linear combination of paths of length two obtained by the substitutions

$$x_h x_l \mapsto a'_h a_l, \quad x_h \mapsto s' a_h, \quad 1 \mapsto s' s.$$

Let J_2 be the subspace of the span of the paths of length two spanned by the elements $a'_h s - s' a_h$ for $1 \leq h \leq d$ and $\tilde{p}_j$ for $1 \leq j \leq t$, and let $\mathcal{R}$ be a basis of J_2 . We write

$$\rho = \sum_{a \in \mathcal{A}_{01}, b \in \mathcal{A}_{12}} \rho_{ba} ba \quad (\rho \in \mathcal{R}, \rho_{ba} \in \mathbf{k}).$$

Put $B = \mathbf{k}Q/(J_2)$, with vertex idempotents e_0, e_1 and e_2 . For a finite-dimensional left R -module Y , let $M(Y)$ be the representation of Q with the space Y at each vertex, the identity of Y on s and s' , and the action of x_h on a_h and a'_h .

The algebra B depends on the presentation. Its defining relations span J_2 . In the quotient category of Section 6.3, s and s' become invertible and x_h acts as $s^{-1}a_h$.

Lemma 6.3. *Let Λ be a finite-dimensional algebra with a complete set of orthogonal idempotents $\varepsilon_0, \dots, \varepsilon_{n-1}$, where $n \geq 1$, such that $\varepsilon_i \Lambda \varepsilon_i = \mathbf{k} \varepsilon_i$ for every i and $\varepsilon_j \Lambda \varepsilon_i = 0$ for $j < i$. Then every left and every right Λ -module has projective dimension at most $n - 1$.*

Proof. Let N be a left Λ -module with $\varepsilon_i N = 0$ for $i < p$. The map

$$\bigoplus_{i=p}^{n-1} \Lambda \varepsilon_i \otimes_{\mathbf{k}} \varepsilon_i N \longrightarrow N, \quad c \otimes v \mapsto cv,$$

is surjective since $v = \sum_i \varepsilon_i v$, and its source is projective. The hypotheses give $\varepsilon_p \Lambda \varepsilon_i = 0$ for $i > p$ and $\varepsilon_p \Lambda \varepsilon_p = \mathbf{k} \varepsilon_p$, so the map is an isomorphism after multiplication by ε_p , and its kernel N' satisfies $\varepsilon_i N' = 0$ for $i < p + 1$. A module annihilated by $\varepsilon_0, \dots, \varepsilon_{n-2}$ is a direct sum of copies of $\Lambda \varepsilon_{n-1} = \mathbf{k} \varepsilon_{n-1}$, hence projective. Starting with $p = 0$ and repeating the construction $n - 1$ times gives a projective resolution of length at most $n - 1$. The opposite algebra, with the idempotents in the reverse order, satisfies the same hypotheses, which gives the statement for right modules. $\square$

Proposition 6.4. *In Construction 6.2, the algebra B is finite-dimensional,*

$$\dim_{\mathbf{k}} B = 3 + 2(d + 1) + (d + 1)^2 - \dim_{\mathbf{k}} J_2,$$

and every left and every right B -module has projective dimension at most two. Moreover, $M(Y)$ is a left B -module for every finite-dimensional left R -module Y , and M extends to an exact fully faithful functor $M: \text{mod}_{\text{fd}} R \rightarrow \text{mod } B$ with $e_i M(Y) = Y$ for every vertex i . In particular, $M(Y) = 0$ only if $Y = 0$.

Proof. The quiver has three paths of length zero, $2(d + 1)$ arrows, $(d + 1)^2$ paths of length two and no longer paths. No arrow ends at 0 and no arrow starts at 2, so the product of an element of J_2 with an arrow on either side is zero, and the ideal generated by J_2 is J_2 itself. This gives the dimension formula. The idempotents e_0, e_1, e_2 satisfy $e_i B e_i = \mathbf{k} e_i$ and $e_j B e_i = 0$ for $j < i$, since every arrow increases the vertex index; so Lemma 6.3 applies with $n = 3$.

The element $a'_h s - s' a_h$ acts on $M(Y)$ as $x_h - x_h = 0$. The element $\tilde{p}_j$ acts as $p_j(x_1, \dots, x_d)$, which is zero on Y . A homomorphism of R -modules $Y \rightarrow Y'$, placed at the three vertices, is a homomorphism $M(Y) \rightarrow M(Y')$. Conversely, a homomorphism $M(Y) \rightarrow M(Y')$ consists of three linear maps; compatibility with s and s' forces them to be equal, and compatibility with the arrows a_h forces the common map to commute with every x_h . Hence, M is fully faithful. It is exact because exactness of a sequence of B -modules is tested at each vertex. $\square$

6.3. The action on the quotient category. In the situation of Construction 6.2, put

$$\mathcal{K} := K^b(\text{proj } B^{\text{op}}), \quad E_i := e_i B \quad (i = 0, 1, 2).$$

An arrow $a: i \rightarrow j$ of Q lies in $e_j B e_i$, and left multiplication by a is a homomorphism of right modules $E_i \rightarrow E_j$, which we denote by a as well; composition of these homomorphisms is multiplication in B . Let $\mathcal{S}$ be the smallest thick subcategory of $\mathcal{K}$ containing the cones of $s: E_0 \rightarrow E_1$ and of $s': E_1 \rightarrow E_2$, let $\mathcal{W}$ be the class of morphisms with cone in $\mathcal{S}$, and let

$$q: \mathcal{K} \longrightarrow \mathcal{Q} := \mathcal{K}/\mathcal{S}$$

be the Verdier quotient. We write a morphism of $\mathcal{K}$ and its image under q by the same letter. In $\mathcal{Q}$ the morphisms s and s' are invertible.

For a finite-dimensional left R -module Y , the *evaluation* at Y is the functor

$$\text{ev}_Y: \mathcal{K} \longrightarrow D^b(\text{mod } \mathbf{k}), \quad T \longmapsto T \otimes_B M(Y).$$

Termwise tensor product preserves homotopies, mapping cones and shifts, so ev_Y is an exact functor. Since a bounded complex of projective right modules is K -flat, $\text{ev}_Y(T)$ computes $T \otimes_B^L M(Y)$. Under the isomorphisms $E_i \otimes_B M(Y) \cong e_i M(Y) = Y$, given by $c \otimes v \mapsto cv$, the morphism $\text{ev}_Y(a)$ is the action of the arrow a on $M(Y)$.

Proposition 6.5. *In the situation above, there is a unique homomorphism of unital algebras*

$$\theta: R \longrightarrow \text{End}_{\mathcal{Q}}(E_0) \quad \text{with} \quad \theta(x_h) = s^{-1} a_h \quad (1 \leq h \leq d).$$

For every finite-dimensional left R -module Y , the functor ev_Y factors uniquely through an exact functor $\text{ev}_Y: \mathcal{Q} \rightarrow D^b(\text{mod } \mathbf{k})$, and $\text{ev}_Y(\theta(r))$ is the action of r on $Y = \text{ev}_Y(E_0)$ for every $r \in R$.

Proof. In $\mathcal{Q}$ the relation $a'_h s = s' a_h$ gives $a'_h = s' a_h s^{-1}$, and therefore

$$\begin{aligned} (s's)^{-1} a'_h a_l &= s^{-1} (s')^{-1} s' a_h s^{-1} a_l = (s^{-1} a_h) (s^{-1} a_l), \\ (s's)^{-1} s' a_h &= s^{-1} a_h, \quad (s's)^{-1} s' s = \text{id}_{E_0}. \end{aligned}$$

Multiplying the relation $\tilde{p}_j = 0$ on the left by $(s's)^{-1}$ thus gives

$$p_j(s^{-1} a_1, \dots, s^{-1} a_d) = 0,$$

constant term included. Hence, the homomorphism from the free algebra sending x_h to $s^{-1} a_h$ factors through R ; it is unique since the x_h generate R .

The cones of s and s' are sent by ev_Y to cones of identity maps, which are zero. The full subcategory of objects sent to zero by the exact functor ev_Y is closed under shifts, cones and direct summands, so it contains $\mathcal{S}$, and the universal property of the Verdier quotient [StacksProj, Tag 05RJ, Lemma 13.6.8] gives the exact factorisation. Finally, $\text{ev}_Y(s^{-1} a_h)$ is $\text{id}^{-1} \circ x_h = x_h$, so $\text{ev}_Y \circ \theta$ and the action of R on Y agree on generators. $\square$

The argument uses the action θ and the evaluation functors; it does not require an identification of $\mathcal{Q}$ with a derived category of R .

6.4. A summand and its odd shifts. Let $\mathcal{T}$ be an additive category. Its *idempotent completion* $\text{Kar } \mathcal{T}$ has as objects the pairs (Z, p) of an object Z of $\mathcal{T}$ and an idempotent $p \in \text{End}(Z)$; a morphism $(Z, p) \rightarrow (Z', p')$ is a morphism $u: Z \rightarrow Z'$ with $u = p' u p$, and the identity of (Z, p) is p . The functor $Z \mapsto (Z, \text{id}_Z)$ is fully faithful, and we regard $\mathcal{T}$ as a full subcategory of $\text{Kar } \mathcal{T}$. If $\mathcal{T}$ is triangulated, the shift extends to $\text{Kar } \mathcal{T}$ by $(Z, p)[1] = (Z[1], p[1])$. We do not use a triangulated structure on $\text{Kar } \mathcal{T}$.

Lemma 6.6. *Let $\mathcal{T}$ be a triangulated category and*

$$X_1 \xrightarrow{f} X_2 \xrightarrow{j} X_3 \xrightarrow{v} X_1[1]$$

a triangle in $\mathcal{T}$. Suppose that there are decompositions $X_1 \cong I \oplus X'_1$ and $X_2 \cong I \oplus X'_2$ in $\text{Kar } \mathcal{T}$ in which f has the components id_I on I and zero elsewhere. Then $X_3 \cong X'_2 \oplus X'_1[1]$ in $\text{Kar } \mathcal{T}$.

Proof. Let $Z = (Z_0, p)$ be an object of $\text{Kar } \mathcal{T}$. The sequence obtained by applying $\text{Hom}(Z, -)$ to the triangle is the image of the idempotent endomorphism $\text{Hom}(p, -)$ of the long exact sequence obtained by applying $\text{Hom}(Z_0, -)$, and a direct summand of an exact sequence is exact. Hence, $\text{Hom}(Z, -)$ takes the triangle to a long exact sequence.

Let $\iota: X'_2 \rightarrow X_2$ and $\kappa: X'_1[1] \rightarrow X_1[1]$ be the inclusions, let $\pi_2: X_2 \rightarrow X'_2$ and $\pi: X_1[1] \rightarrow X'_1[1]$ be the projections, and put $j' = j\iota$ and $v' = \pi v$. Since $jf = 0$ and $f[1]v = 0$, the morphism j vanishes on I and the component of v in $I[1]$ is zero, so that $j = j'\pi_2$ and $v = \kappa v'$. We claim that

$$0 \longrightarrow \text{Hom}(Z, X'_2) \xrightarrow{j'_*} \text{Hom}(Z, X_3) \xrightarrow{v'_*} \text{Hom}(Z, X'_1[1]) \longrightarrow 0$$

is exact for every Z . If $j'g = 0$, then $\iota g = fh$ for some h ; the component of fh in X'_2 is zero, so $g = 0$. If $v'g = 0$, then $vg = 0$, so $g = jh = j'\pi_2 h$ for some h . If $g: Z \rightarrow X'_1[1]$, then $f[1]\kappa g = 0$, so $\kappa g = vh$ for some h and $g = \pi \kappa g = v'h$.

Taking $Z = X'_1[1]$ gives $\sigma: X'_1[1] \rightarrow X_3$ with $v'\sigma = \text{id}$. The morphism

$$(j', \sigma): X'_2 \oplus X'_1[1] \longrightarrow X_3$$

induces a bijection on $\text{Hom}(Z, -)$ for every Z : a morphism $g: Z \rightarrow X_3$ is the image of $(g', v'g)$, where $j'g' = g - \sigma v'g$, and the injectivity follows by applying v'_* , since $v'j' = \pi v j \iota = 0$, and then the injectivity of j'_* . By the Yoneda lemma, (j', σ) is an isomorphism. $\square$

Proposition 6.7. *Let $\mathcal{T}$ be an essentially small triangulated category, E an object of $\mathcal{T}$ and p an idempotent endomorphism of E , and put $U = (E, p) \in \text{Kar } \mathcal{T}$. Then there exist objects C and V of $\mathcal{T}$ such that*

$$C \cong U \oplus U[1] \quad \text{and} \quad V \cong U \oplus U[3] \quad \text{in } \text{Kar } \mathcal{T},$$

and $[C] = [V] = 0$ in $K_0(\mathcal{T})$.

Proof. Let C be a cone of $\text{id} - p: E \rightarrow E$. In the decomposition $E \cong (E, \text{id} - p) \oplus U$ the morphism $\text{id} - p$ is the identity on the first summand and zero on the second, so Lemma 6.6 gives $C \cong U \oplus U[1]$. Since $\mathcal{T}$ is a full subcategory of $\text{Kar } \mathcal{T}$, the morphism

$$g: C[1] \cong U[1] \oplus U[2] \xrightarrow{\begin{pmatrix} 0 & 0 \\ \text{id} & 0 \end{pmatrix}} U \oplus U[1] \cong C,$$

which identifies the two copies of $U[1]$, is a morphism of $\mathcal{T}$. Let V be a cone of g . By Lemma 6.6, with $I = U[1]$, we have $V \cong U \oplus U[3]$. The triangles defining C and V give $[C] = [E] - [E] = 0$ and $[V] = [C] - [C[1]] = 2[C] = 0$, where $[C[1]] = -[C]$ follows from the triangle $C \rightarrow 0 \rightarrow C[1] \xrightarrow{\text{id}} C[1]$. $\square$

Any object of $\mathcal{T}$ isomorphic to $U \oplus U[3]$ in $\text{Kar } \mathcal{T}$ is isomorphic to V in $\mathcal{T}$, since the inclusion of $\mathcal{T}$ is fully faithful, and so has class zero as well. The object U itself need not be isomorphic to an object of $\mathcal{T}$; for $p = 0$ or $p = \text{id}$ it is. The equality $[V] = 0$ is in $K_0(\mathcal{T})$. Cancellation between summands with shifts of opposite parity also occurs in the iterates of Corollary 6.20, whose classes vanish for $j \geq 1$, and in the second iterate of the functor of Section 7; see Remark 4.6.

Remark 6.8. The cones of Proposition 6.7 produce $U \oplus U[1]$ and $U \oplus U[3]$. If the diagram of Construction 6.10 were defined using $U \oplus U[c]$, the two cohomology modules of $P \otimes_B^{\mathbb{L}} M(Y)$ in the proof of Theorem 6.18 would be in degrees 0 and $-c$, and the extension class between them would lie in Ext_B^{c+1} , which vanishes when $c+1 > \text{gldim } B$. Since $\text{gldim } B \leq 2$, the shift 3 is the smallest positive odd integer c for which this argument applies; for $c = 1$ it would need the vanishing of an Ext_B^2 , which the bound on the global dimension does not provide.

We now apply Proposition 6.7 to selection data. Recall from Definition 5.1 that Ψ is an R -bimodule that is finitely generated and projective as a right R -module, and $H = \Psi \otimes_R -$. We write elements of R^n as columns, so that $M_n(R)$ acts on R^n from the left and $\text{End}_{R^{\text{op}}}(R^n) = M_n(R)$, with composition the product of matrices.

Lemma 6.9. *Let (R, Ψ) be selection data. Then there exist an integer $n \geq 1$, an idempotent $\varepsilon \in M_n(R)$ and a homomorphism of algebras $\rho: R \rightarrow \varepsilon M_n(R) \varepsilon$ with $\rho(1) = \varepsilon$ such that $\Psi \cong \varepsilon R^n$ as bimodules, where R acts on εR^n from the left through ρ . For every finite-dimensional left R -module Y there is an isomorphism, natural in Y ,*

$$HY \cong \varepsilon_Y(Y^n), \quad \text{on which } r \in R \text{ acts as } \rho(r)_Y,$$

where m_Y denotes the operator on Y^n given by a matrix $m \in M_n(R)$. In particular, H is exact and $\dim_{\mathbf{k}} HY \leq n \dim_{\mathbf{k}} Y$.

Proof. Choose a split surjection $\pi: R^n \rightarrow \Psi$ of right modules with section σ , and put $\varepsilon = \sigma\pi \in M_n(R)$. Then σ identifies Ψ with εR^n , and the endomorphisms of the right module εR^n , extended by zero on $(1 - \varepsilon)R^n$, form $\varepsilon M_n(R) \varepsilon$. The left action of R on Ψ thus becomes a homomorphism ρ with $\rho(1) = \varepsilon$; it is $\mathbf{k}$ -linear because Ψ is $\mathbf{k}$ -central. The isomorphism $R^n \otimes_R Y \cong Y^n$, $(r_i)_i \otimes y \mapsto (r_i y)_i$, restricts to $\varepsilon R^n \otimes_R Y \cong \varepsilon_Y(Y^n)$, and it carries the left action of r to $\rho(r)_Y$. After forgetting the left action of R , the functor H is a direct summand of the exact functor $R^n \otimes_R -$ with values in vector spaces; since exactness is detected on underlying vector spaces, H is exact. $\square$

Now suppose that R is finitely presented, fix a presentation as in Construction 6.2, and let B , $\mathcal{Q}$ and θ be as in Section 6.3. Choose n , ε and ρ as in Lemma 6.9. Applying θ to the entries of a matrix defines a homomorphism of algebras

$$\theta_n: M_n(R) \longrightarrow \text{End}_{\mathcal{Q}}(E_0^{\oplus n}).$$

Put

$$U := (E_0^{\oplus n}, \theta_n(\varepsilon)) \in \text{Kar } \mathcal{Q}.$$

For $r \in R$ the endomorphism $\theta_n(\rho(r))$ of $E_0^{\oplus n}$ satisfies

$$\theta_n(\rho(r)) = \theta_n(\varepsilon) \theta_n(\rho(r)) \theta_n(\varepsilon),$$

so it is an endomorphism of U , and $r \mapsto \theta_n(\rho(r))$ is a homomorphism of unital algebras $R \rightarrow \text{End}(U)$, since $\theta_n(\varepsilon) = \text{id}_U$.

By Proposition 6.7 there is an object V of $\mathcal{Q}$ with $V \cong U \oplus U[3]$ in $\text{Kar } \mathcal{Q}$; we fix such an isomorphism and transport to V the action of R on $U \oplus U[3]$ given by $\theta_n \circ \rho$ on both summands. Write $r_V \in \text{End}(V)$ for the action of $r \in R$; the map $r \mapsto r_V$ is a homomorphism of unital algebras.

Construction 6.10. A B -diagram in a $\mathbf{k}$ -linear category $\mathcal{C}$ consists of objects W_0 , W_1 and W_2 and a morphism $t_a: W_i \rightarrow W_j$ for every arrow $a: i \rightarrow j$ of $\mathcal{Q}$, such that $\sum_{b,a} \rho_{ba} t_b t_a = 0$ for every $\rho \in \mathcal{R}$. Let $\mathbb{V}$ be the B -diagram in $\mathcal{Q}$ with $W_0 = W_1 = W_2 = V$, with $t_s = t_{s'} = \text{id}_V$, and with $t_{a_h} = t_{a'_h} = (x_h)_V$ for $1 \leq h \leq d$.

The relations hold in $\mathbb{V}$ since they become $p_j((x_1)_V, \dots, (x_d)_V) = 0$ and $(x_h)_V - (x_h)_V = 0$. A bounded complex P of B -bimodules that is termwise finitely generated and projective as a right module defines a B -diagram $(e_i P, a \cdot -)$ in $\mathcal{K}$, and then in $\mathcal{Q}$. We lift $\mathbb{V}$ to $\mathcal{K}$ and replace the lift by a complex of bimodules.

Lemma 6.11. *For every finite-dimensional left R -module Y , the functor ev_Y extends to $\text{Kar } \mathcal{Q}$, and there is an isomorphism $\text{ev}_Y(V) \cong HY \oplus HY[3]$ in $D^b(\text{mod } \mathbf{k})$ under which $\text{ev}_Y(r_V)$ is the action of r on HY on both summands, for every $r \in R$.*

Proof. Idempotents split in $D^b(\text{mod } \mathbf{k})$: every bounded complex of vector spaces is isomorphic to its cohomology with zero differential, morphisms between such complexes have only components of degree zero, and an idempotent is then a family of idempotent linear maps, whose images are direct summands. Hence, ev_Y extends to $\text{Kar } \mathcal{Q}$, with (Z, p) sent to the image of $\text{ev}_Y(p)$. By Proposition 6.5, the functor ev_Y sends $E_0^{\oplus n}$ to Y^n and $\theta_n(m)$ to m_Y for $m \in M_n(R)$. Hence, it sends U to $\varepsilon_Y(Y^n) \cong HY$ and $\theta_n(\rho(r))$ to the action of r , by Lemma 6.9. Since ev_Y commutes with the shift, the statement follows. $\square$

6.5. Lifts of $\mathbb{V}$.

Proposition 6.12. *In the situation of Construction 6.10, there exist bounded complexes $\tilde{V}_0, \tilde{V}_1$ and $\tilde{V}_2$ of finitely generated projective right B -modules, chain maps $f_a: \tilde{V}_i \rightarrow \tilde{V}_j$ for the arrows $a: i \rightarrow j$ of Q , homogeneous maps $h_\rho: \tilde{V}_0 \rightarrow \tilde{V}_2$ of degree -1 of right modules for $\rho \in \mathcal{R}$, and isomorphisms $\phi_i: q\tilde{V}_i \rightarrow V$ in $\mathcal{Q}$ such that*

$$(6.13) \quad \begin{aligned} \phi_j \circ q(f_a) &= t_a \circ \phi_i \quad \text{for every arrow } a: i \rightarrow j, \\ dh_\rho + h_\rho d &= f_\rho := \sum_{b,a} \rho_{ba} f_b f_a \quad \text{for every } \rho \in \mathcal{R}. \end{aligned}$$

Proof. The objects of $\mathcal{Q}$ are those of $\mathcal{K}$, so V is an object of $\mathcal{K}$ and $qV = V$. We proceed from vertex 2 to vertex 0, so that the denominators introduced at a vertex do not affect the arrows already lifted.

Apply Lemma 6.1(1) to the morphisms $t_b: qV \rightarrow qV$ for $b \in \mathcal{A}_{12}$. This gives $u_1: \tilde{V}_1 \rightarrow V$ in $\mathcal{W}$ and morphisms $f_b: \tilde{V}_1 \rightarrow V$ with $t_b = q(f_b)q(u_1)^{-1}$. Put $\tilde{V}_2 = V$, $\phi_2 = \text{id}_V$ and $\phi_1 = q(u_1)$; then $\phi_2 q(f_b) = t_b \phi_1$. Next apply Lemma 6.1(1) to the morphisms $\phi_1^{-1} t_a: qV \rightarrow q\tilde{V}_1$ for $a \in \mathcal{A}_{01}$. This gives $u_0: V'_0 \rightarrow V$ in $\mathcal{W}$ and morphisms $f'_a: V'_0 \rightarrow \tilde{V}_1$, and with $\phi'_0 = q(u_0)$ the first identity of (6.13) holds for all arrows. For $\rho \in \mathcal{R}$ put

$$g_\rho = \sum_{b,a} \rho_{ba} f_b f'_a: V'_0 \longrightarrow \tilde{V}_2.$$

Then

$$\phi_2 q(g_\rho) = \left(\sum_{b,a} \rho_{ba} t_b t_a \right) \phi'_0 = 0,$$

so $q(g_\rho) = 0$. By Lemma 6.1(2) there is $v: \tilde{V}_0 \rightarrow V'_0$ in $\mathcal{W}$ with $g_\rho v = 0$ for every ρ . Put $f_a = f'_a v$ for $a \in \mathcal{A}_{01}$ and $\phi_0 = \phi'_0 q(v)$. The first identity of (6.13) still holds, and $\sum_{b,a} \rho_{ba} f_b f_a = g_\rho v = 0$ in $\mathcal{K}$; no further compatibility arises, since no arrow ends at 0.

Finally, choose chain maps representing the morphisms f_a of $\mathcal{K}$. For each $\rho \in \mathcal{R}$ the chain map f_ρ is then null-homotopic, and we choose a homotopy h_ρ with $dh_\rho + h_\rho d = f_\rho$. $\square$

All the data of Proposition 6.12 are chosen from $\mathbb{V}$, and do not depend on a module Y . Applying ev_Y to the first identity of (6.13) and using Lemma 6.11, we obtain for every Y isomorphisms $\text{ev}_Y(\tilde{V}_i) \cong HY \oplus HY[3]$ under which $\text{ev}_Y(f_s)$ and

$\text{ev}_Y(f_{s'})$ are identities and $\text{ev}_Y(f_{a_h})$ and $\text{ev}_Y(f_{a'_h})$ are the action of x_h on both summands.

Remark 6.14. The construction uses choices not given by explicit formulas at four points: the splitting morphisms in the proof of Lemma 6.6, the roofs representing the morphisms t_a in Lemma 6.1, the Ore squares producing common denominators, and the denominator v for which $g_\rho v = 0$ for every $\rho \in \mathcal{R}$. A proof that a cone lies in the thick subcategory $\mathcal{S}$ does not by itself provide a bounded number of cones, shifts and direct summands witnessing it, and none of these steps specifies the complexes $\tilde{V}_i$ or bounds their length. Once the $\tilde{V}_i$ and the f_a are given as complexes and chain maps, the homotopies h_ρ are solutions of finite systems of linear equations, and Section 6.6 is explicit: if all $\tilde{V}_i$ are concentrated in degrees $[a, b]$, the complex P of Proposition 6.15 is concentrated in degrees $[a - 2, b]$. The argument therefore does not determine the integer l used in Section 7. This is the absence of a bound in the argument, not a proof that no bound or algorithm exists; see Remark 7.5.

6.6. Rectification. The following construction applies to every algebra $B = \mathbf{k}Q/(J_2)$ for which Q has vertices 0, 1 and 2, finite sets of arrows $\mathcal{A}_{01}$ from 0 to 1 and $\mathcal{A}_{12}$ from 1 to 2 and no other arrows, and J_2 is a subspace of the span of the paths of length two, with basis $\mathcal{R}$.

Proposition 6.15. *Let B be as above, and suppose given bounded complexes $\tilde{V}_0, \tilde{V}_1$ and $\tilde{V}_2$ of finitely generated projective right B -modules, chain maps f_a and homogeneous maps h_ρ of degree -1 of right modules satisfying the second identity of (6.13). Then there are a bounded complex P of finite-dimensional B -bimodules whose terms are projective right modules, homotopy equivalences $\iota_i: \tilde{V}_i \rightarrow e_i P$ of complexes of right modules, and, for every arrow $a: i \rightarrow j$, a homotopy $\alpha \iota_i \simeq \iota_j f_a$, where a acts on P by left multiplication.*

Proof. All unadorned tensor products in this proof are over $\mathbf{k}$. On a tensor product $Be_j \otimes \tilde{V}_i$, the algebra B acts from the left on the first factor and from the right on the second, and the differential is $\text{id} \otimes d$. Consider the complexes of bimodules

$$L_0 = \bigoplus_i Be_i \otimes \tilde{V}_i, \quad L_1 = \bigoplus_{a: i \rightarrow j} Be_j \otimes \tilde{V}_i, \quad L_2 = \bigoplus_{\rho \in \mathcal{R}} Be_2 \otimes \tilde{V}_0,$$

with differentials d_0, d_1 and d_2 , and the bimodule maps $\partial_1: L_1 \rightarrow L_0$ and $\partial_2: L_2 \rightarrow L_1$ of degree zero and $\eta: L_2 \rightarrow L_0$ of degree -1 defined by

$$\begin{aligned} \partial_1(c \otimes v)_a &= (ca \otimes v)_i - (c \otimes f_a v)_j & \text{for } a: i \rightarrow j, \\ \partial_2(c \otimes v)_\rho &= \sum_{b,a} \rho_{ba} ((cb \otimes v)_a + (c \otimes f_a v)_b), \\ \eta(c \otimes v)_\rho &= (c \otimes h_\rho v)_2, \end{aligned} \tag{6.16}$$

where a subscript names the summand containing an element: in L_0 the summand of a vertex, in L_1 that of an arrow. These maps are bimodule homomorphisms because paths act on the first factor and f_a and h_ρ are homomorphisms of right modules on the second. Since the f_a are chain maps, $d_0 \partial_1 = \partial_1 d_1$ and $d_1 \partial_2 = \partial_2 d_2$. In the composite

$$\partial_1 \partial_2 (c \otimes v)_\rho = \sum_{b,a} \rho_{ba} ((cba \otimes v)_0 - (cb \otimes f_a v)_1 + (cb \otimes f_a v)_1 - (c \otimes f_b f_a v)_2) = -(c \otimes f_\rho v)_2$$

the two middle terms cancel, and the first term vanishes since $\sum \rho_{ba} ba = \rho = 0$ in B . Together with (6.13) this gives

$$d_0 \eta + \eta d_2 = -\partial_1 \partial_2. \tag{6.17}$$

Put

$$P := L_0 \oplus L_1[1] \oplus L_2[2], \quad d_P = \begin{pmatrix} d_0 & \partial_1 & \eta \\ 0 & -d_1 & \partial_2 \\ 0 & 0 & d_2 \end{pmatrix}.$$

In total degree m the three components lie in degrees $m, m+1$ and $m+2$ of L_0, L_1 and L_2 , so all entries of d_P have degree one; the diagonal signs are those of the shifts. The entries of d_P^2 are $d_0^2, d_1^2, d_2^2, d_0\partial_1 - \partial_1d_1, \partial_2d_2 - d_1\partial_2$ and $d_0\eta + \partial_1\partial_2 + \eta d_2$, and all of them vanish by the identities above and (6.17). As a right module, $Be_j \otimes \tilde{V}_i^m$ is a finite direct sum of copies of $\tilde{V}_i^m$, so the terms of P are finite-dimensional and projective on the right, and P is bounded.

For a vertex i , let $\iota_i: \tilde{V}_i \rightarrow e_iP$ send v to $(e_i \otimes v)_i \in e_iL_0$. It is a chain map and, in each degree, a split injection of right modules. We show that the quotient $C_i = e_iP/\iota_i(\tilde{V}_i)$ is acyclic. Filter e_iP by the subcomplexes F^r consisting of the summands whose second factor is $\tilde{V}_j$ with $j \geq r$; the differential preserves this filtration, since d and the path terms preserve j , the terms involving f_a increase it, and η increases it from 0 to 2. The induced filtration of C_i has the following subquotients. The subquotient for $j = i$ is zero, since the only summand of e_iP with second factor $\tilde{V}_i$ is $e_iBe_i \otimes \tilde{V}_i = \mathbf{k}e_i \otimes \tilde{V}_i$. The subquotients for $j > i$ are zero, since $e_iBe_j = 0$. For (j, i) equal to $(0, 1)$ or $(1, 2)$, the subquotient is the total complex of the double complex $K \otimes \tilde{V}_j$, where K is the complex

$$0 \longrightarrow \mathbf{k}\{a: j \rightarrow i\} \xrightarrow{a \mapsto a} e_iBe_j \longrightarrow 0$$

in degrees -1 and 0 , which is exact since all relations have length two. For $(j, i) = (0, 2)$ it is the total complex of $K \otimes \tilde{V}_0$ with

$$K: 0 \longrightarrow \mathbf{k}\mathcal{R} \xrightarrow{\rho \mapsto \sum \rho_{ba}(b, a)} \mathbf{k}\{(b, a)\} \xrightarrow{(b, a) \mapsto ba} e_2Be_0 \longrightarrow 0$$

in degrees $-2, -1$ and 0 , where (b, a) runs through $\mathcal{A}_{12} \times \mathcal{A}_{01}$. This complex is exact because $\mathcal{R}$ is linearly independent with span J_2 and e_2Be_0 is the quotient of the span of the paths of length two by J_2 ; here it matters that $\mathcal{R}$ is a basis and not merely a spanning set. An exact complex K of vector spaces has a contraction σ with $\partial_K\sigma + \sigma\partial_K = \text{id}$, and $\sigma \otimes \text{id}$ contracts the total complex of $K \otimes \tilde{V}_j$: in horizontal degree m its differential is

$$\partial_K \otimes \text{id} + (-1)^m \text{id} \otimes d,$$

and the terms of $(\sigma \otimes \text{id})(\text{id} \otimes d)$ and $(\text{id} \otimes d)(\sigma \otimes \text{id})$ appear with the opposite signs $(-1)^m$ and $(-1)^{m-1}$. These signs agree with those of d_P . Hence, all subquotients are acyclic, and so is C_i , by induction along the finite filtration and the long exact cohomology sequence.

The terms of C_i are projective right modules, since ι_i is split in each degree, and a bounded acyclic complex of projective modules is contractible. Choose a splitting $e_iP = \tilde{V}_i \oplus C_i$ of graded right modules, in which d_P has the form $\begin{pmatrix} d & t \\ 0 & d_C \end{pmatrix}$ with $dt + td_C = 0$, and a contraction σ of C_i . Then $r(x, c) = x - t\sigma c$ is a chain map with $r\iota_i = \text{id}$, and $K(x, c) = (0, \sigma c)$ satisfies $d_PK + Kd_P = \text{id} - \iota_i r$. Hence, ι_i is a homotopy equivalence.

For an arrow $a: i \rightarrow j$, let $H_a: \tilde{V}_i \rightarrow e_jP$ be the map of degree -1 sending v to $(e_j \otimes v)_a \in e_jL_1[1]$. The component of $d_PH_a(v)$ in $L_1[1]$ is $-(e_j \otimes dv)_a = -H_a(dv)$, and its component in L_0 is

$$\partial_1(e_j \otimes v)_a = (a \otimes v)_i - (e_j \otimes f_a v)_j.$$

Hence,

$$d_PH_a + H_ad = a\iota_i - \iota_j f_a.$$

□

The three columns of P are indexed by the vertices, the arrows and the relations of Q . Since every relation starts at 0 and ends at 2, no relation can be composed with a further arrow, and no homotopies beyond h_ρ are required.

6.7. The realisation theorem.

Theorem 6.18. *Let (R, Ψ) be selection data over $\mathbf{k}$ in which R is finitely presented, and put $H = \Psi \otimes_R -$. Let B and M be the algebra and the functor of Construction 6.2 for a presentation of R by relations of degree at most two. Then there exists a bounded complex P of finite-dimensional B -bimodules, whose terms are projective as right modules, such that for every finite-dimensional left R -module Y*

$$(6.19) \quad P \otimes_B^{\mathbb{L}} M(Y) \cong M(HY) \oplus M(HY)[3] \quad \text{in } D^b(\text{mod } B).$$

The complex P does not depend on Y .

Proof. Let P be the complex obtained by applying Proposition 6.15 to the data of Proposition 6.12; it depends only on (R, Ψ) , the presentation and the choices made there. Fix Y and put $T_Y = P \otimes_B M(Y)$, which computes $P \otimes_B^{\mathbb{L}} M(Y)$ since P is a bounded complex of projective right modules. For each vertex i we have $e_i T_Y = \text{ev}_Y(e_i P)$, and the action of an arrow a on T_Y is ev_Y of left multiplication by a .

Applying ev_Y to the homotopy equivalences ι_i and the homotopies of Proposition 6.15 identifies $\text{ev}_Y(e_i P)$ with $\text{ev}_Y(\tilde{V}_i)$, compatibly with the arrows, and the paragraph after Proposition 6.12 identifies the latter with $HY \oplus HY[3]$. Taking cohomology, we obtain isomorphisms of vector spaces at each vertex that commute with the actions of all arrows, hence isomorphisms of left B -modules

$$H^m(T_Y) \cong \begin{cases} M(HY) & \text{if } m \in \{0, -3\}, \\ 0 & \text{otherwise.} \end{cases}$$

Put $N_0 = H^0(T_Y)$ and $N_{-3} = H^{-3}(T_Y)$. The good truncation T' of T_Y to the degrees $[-3, 0]$ is quasi-isomorphic to T_Y , and its cohomology in degree -3 is its degree -3 cycle module, which is a subcomplex isomorphic to $N_{-3}[3]$. The quotient complex has cohomology N_0 in degree zero only and terms in nonpositive degrees, so it is quasi-isomorphic to N_0 . The short exact sequence of complexes gives a triangle

$$N_{-3}[3] \longrightarrow T_Y \longrightarrow N_0 \xrightarrow{\delta} N_{-3}[4]$$

in $D^b(\text{mod } B)$. By Proposition 6.4, N_0 has a projective resolution of length at most two, so that $\delta \in \text{Hom}(N_0, N_{-3}[4]) = \text{Ext}_B^4(N_0, N_{-3}) = 0$. Hence, the identity of N_0 lifts to a morphism $N_0 \rightarrow T_Y$, and together with the first morphism of the triangle it gives a morphism $N_{-3}[3] \oplus N_0 \rightarrow T_Y$, which is a quasi-isomorphism by the long exact cohomology sequence. This proves (6.19). $\square$

Corollary 6.20. *In the situation of Theorem 6.18, put $\mathcal{F} = P \otimes_B^{\mathbb{L}} -$. For every finite-dimensional left R -module Y and every integer $j \geq 0$,*

$$\mathcal{F}^j M(Y) \cong \bigoplus_{r=0}^j (M(H^j Y)[3r])^{\oplus \binom{j}{r}}.$$

In particular, $\mathcal{F}^j M(Y) = 0$ if and only if $H^j Y = 0$.

Proof. For $j = 0$ there is nothing to prove. Suppose the formula holds for j . The module $H^j Y$ is finite-dimensional by Lemma 6.9, and $\mathcal{F}$ commutes with finite direct sums and shifts, so Theorem 6.18 applied to $H^j Y$ gives

$$\mathcal{F}^{j+1} M(Y) \cong \bigoplus_{r=0}^j (M(H^{j+1} Y)[3r] \oplus M(H^{j+1} Y)[3r+3])^{\oplus \binom{j}{r}},$$

in which the shift $3r$ occurs with multiplicity $\binom{j}{r} + \binom{j}{r-1} = \binom{j+1}{r}$. The last assertion follows from Proposition 6.4. $\square$

Remark 6.21. The main preprint [Ope26b, Sections 3 and 4] treats the case of a central idempotent $e \in R$ and an algebra endomorphism α of R , with $\Psi = {}_\alpha(eR)$ and $HY = {}_\alpha(eY)$. This is the case $n = 1$, $\varepsilon = e$ and $\rho(r) = e\alpha(r)e$ of Lemma 6.9; the centrality of e makes ρ multiplicative. The proof above uses neither the centrality of ε nor any property of ρ beyond its being a homomorphism of unital $\mathbf{k}$ -algebras into $\varepsilon M_n(R)\varepsilon$, and it does not use that Ψ is projective as a left module.

7. TRIVIAL EXTENSIONS FROM BIMODULE COMPLEXES

Theorem 6.18 provides a complex P of bimodules whose derived tensor functor iterates the selection functor, while Corollary 4.5 requires a single bimodule over an algebra of finite global dimension. A bounded complex of bimodules is simulated, up to a shift, by the square of the derived tensor functor of an ordinary bimodule over a larger algebra.

7.1. An ordinary bimodule associated with a complex. For an integer $l \geq 0$, let K_l be the path algebra of the quiver

$$0 \xrightarrow{c_1} 1 \xrightarrow{c_2} \dots \xrightarrow{c_l} l$$

modulo all paths of length two, with vertex idempotents $1_0, \dots, 1_l$, so that $c_i \in 1_i K_l 1_{i-1}$. Let W be the simple right K_l -module at the vertex l . For $i > 0$ the right module $1_i K_l$ has basis $1_i, c_i$ and radical spanned by c_i , and $1_0 K_l = \mathbf{k}1_0$. Hence, left multiplication by c_i maps $1_{i-1} K_l$ onto the radical of $1_i K_l$ with kernel the radical of $1_{i-1} K_l$, and

$$(7.1) \quad 0 \longrightarrow 1_0 K_l \xrightarrow{c_1} 1_1 K_l \xrightarrow{c_2} \dots \xrightarrow{c_l} 1_l K_l \longrightarrow W \longrightarrow 0$$

is a projective resolution of W ; we write R_W for the complex with $R_W^n = 1_{l+n} K_l$ for $-l \leq n \leq 0$.

Proposition 7.2. *Let B be a finite-dimensional algebra whose left and right global dimensions are at most d_L and d_R , and let P be a bounded complex of finite-dimensional B -bimodules, projective as right modules, with $P^j = 0$ for $j \notin [a, b]$; put $l = b - a \geq 0$. Put $B_1 = B \otimes_{\mathbf{k}} K_l$ and $\Delta = B \times B_1$, and let π_0 and π_1 be the central idempotents of Δ corresponding to the two factors. Define*

- the (B_1, B) -bimodule O by $1_i O = P^{a+i}$ for $0 \leq i \leq l$, with $c_i: 1_{i-1} O \rightarrow 1_i O$ acting as the differential d_P^{a+i-1} and with the two actions of B on the terms of P ;
- the (B, B_1) -bimodule $Y = B \otimes_{\mathbf{k}} W$, with $b'(b \otimes w)(b'' \otimes c) = b'bb'' \otimes wc$;

and regard $X = O \oplus Y$ as a Δ -bimodule with $O = \pi_1 O \pi_0$ and $Y = \pi_0 Y \pi_1$. Let $\Phi = X \otimes_{\Delta}^{\mathbb{L}} -$. Then for every complex N of left B -modules, regarded as a complex of Δ -modules through the first factor,

$$\Phi^2 N \cong (P \otimes_B^{\mathbb{L}} N)[b] \quad \text{in } \mathbf{D}(\text{Mod } \Delta).$$

Moreover, the left and right global dimensions of Δ are at most $d_L + l$ and $d_R + l$.

Proof. The bimodule O is well defined: the relations $c_{i+1}c_i = 0$ hold because $d_P^2 = 0$, and the actions of B commute with those of the c_i because the differential of P is a bimodule map. As a right B -module, O is the direct sum of the terms of P , hence projective. The complex $R_Y = B \otimes_{\mathbf{k}} R_W$ is a resolution of Y by (B, B_1) -bimodules whose terms $B \otimes_{\mathbf{k}} 1_{l+n} K_l = (1 \otimes 1_{l+n})B_1$ are projective right B_1 -modules. In degree $n \in [-l, 0]$ there is an isomorphism of B -bimodules

$$(B \otimes_{\mathbf{k}} 1_{l+n} K_l) \otimes_{B_1} O \longrightarrow 1_{l+n} O = P^{b+n}, \quad (\beta \otimes u) \otimes o \longmapsto \beta u o,$$

with inverse $o \mapsto (1 \otimes 1_{l+n}) \otimes o$, and under these isomorphisms the differential of $R_Y \otimes_{B_1} O$ becomes d_P , since the differential of R_W is left multiplication by the arrows. The complex $P[b]$ has the same terms in the same degrees and the differential $(-1)^b d_P$; the maps $(-1)^{bn} \text{id}$ on the terms of degree n form an isomorphism of complexes of bimodules $R_Y \otimes_{B_1} O \cong P[b]$, since $(-1)^{b(n+1)} = (-1)^b (-1)^{bn}$.

The complex $R_X = O \oplus R_Y$ is a bounded resolution of X by Δ -bimodules projective as right modules, so it computes Φ on all complexes. For a complex N of left B -modules, one application gives $O \otimes_B N$, a complex of B_1 -modules, and a second gives

$$R_Y \otimes_{B_1} (O \otimes_B N) \cong (R_Y \otimes_{B_1} O) \otimes_B N \cong P[b] \otimes_B N,$$

a complex of B -modules; the associativity isomorphism involves no sign since the order of the factors is unchanged. Since $P[b]$ is a bounded complex of projective right modules, $P[b] \otimes_B N$ computes $(P \otimes_B^{\mathbb{L}} N)[b]$.

For the global dimensions, the resolution (7.1), ending at any vertex, shows that the right simple K_l -modules have projective dimension at most l , and similarly for the left simple modules, whose successive syzygies are simple at the following vertices. The ideal $I = J_B \otimes K_l + B \otimes J_K$ of B_1 , where J_B and J_K denote the radicals, is nilpotent, being the sum of two commuting nilpotent ideals, and $B_1/I \cong (B/J_B)^{l+1}$ is semisimple. Hence, I is the radical of B_1 and the simple left B_1 -modules are the modules $S \otimes_{\mathbf{k}} L$ for simple modules S over B and L over K_l . The tensor product of projective resolutions of S and L is a projective resolution of $S \otimes_{\mathbf{k}} L$ of length at most $d_L + l$. Since every module has a finite filtration by semisimple modules, the left global dimension of B_1 is at most $d_L + l$; the same argument applies on the right, and the bounds pass to the product $\Delta = B \times B_1$. $\square$

The vertices of K_l correspond to the degrees of P , the arrows act on O by the differential of P , and $Y \otimes_{B_1}^{\mathbb{L}} O \cong P[b]$. The construction uses $l + 2$ copies of B ; the resulting number of simple modules and the bounds on global dimension grow linearly with l .

7.2. The main theorem.

Theorem 7.3. *Let (R, Ψ) be selection data over a field $\mathbf{k}$ in which R is finitely presented, and let $H = \Psi \otimes_R -$. Then there is a finite-dimensional algebra $A = \Delta \ltimes X$ over $\mathbf{k}$, with Δ of finite global dimension, and an integer $l \geq 0$ with the following property: for every finite-dimensional left R -module Y of finite extinction time $t \geq 1$, there is a finite-dimensional left A -module N with*

$$2t - 2 \leq \text{pd}_A N \leq (l + 2) + (2t - 1)(l + 3).$$

In particular, if (R, Ψ) has unbounded extinction, then $\text{findim } A = \infty$.

Proof. Fix a presentation of R by relations of degree at most two, the algebra B and the functor M of Construction 6.2, and the complex P of Theorem 6.18. Choose integers $a \leq b$ with $P^j = 0$ for $j \notin [a, b]$, put $l = b - a$, and let Δ , X and Φ be as in Proposition 7.2; put $A = \Delta \ltimes X$. By Proposition 6.4, B has left and right global dimension at most two, so Δ has left and right global dimension at most $l + 2$ by Proposition 7.2. All these choices are made before the module Y is chosen.

Let Y be a finite-dimensional left R -module of extinction time $t \geq 1$, and let N be $M(Y)$, regarded as a Δ -module through the first factor and then as an A -module. By Proposition 7.2, $\Phi^{2r} N \cong (\mathcal{F}^r M(Y))[rb + 3j]$ for every $r \geq 0$, where $\mathcal{F} = P \otimes_B^{\mathbb{L}} -$, and Corollary 6.20 gives

$$\Phi^{2r} N \cong \bigoplus_{j=0}^r (M(H^r Y)[rb + 3j])^{\oplus \binom{r}{j}}.$$

Hence, $\Phi^{2t}N \simeq 0$, since $H^tY = 0$, and $\Phi^{2t-2}N \not\simeq 0$, since $H^{t-1}Y \neq 0$ and hence $M(H^{t-1}Y) \neq 0$ by Proposition 6.4. By Corollary 4.5, with $d_L = d_R = l + 2$ and with $2t$ in place of t , the module N has finite projective dimension at most $(l + 2) + (2t - 1)(l + 3)$, and by Proposition 4.4 at least $2t - 2$. The last assertion follows since every finite-dimensional A -module is finitely generated. $\square$

Corollary 7.4 ([Ope26b, Theorem 1.1]). *There is a finite-dimensional complex algebra A with $\text{findim } A = \infty$. More precisely, there are an integer $l \geq 0$ and a finite-dimensional complex algebra $A = \Delta \ltimes X$ with $3(l + 2)$ simple modules, where Δ has global dimension at most $l + 2$, such that for every $m \geq 1$ some finite-dimensional A -module N_m satisfies $2m - 2 \leq \text{pd}_A N_m < \infty$.*

Proof. Apply Theorem 7.3 to the selection data of Theorem 5.17 and to the modules Y_m of extinction time m . The simple A -modules are the simple Δ -modules, since X is an ideal of square zero in A . The algebra B has three simple modules, one for each vertex, and the simple modules of $B_1 = B \otimes K_l$ are the $3(l + 1)$ modules $S \otimes L$ of the proof of Proposition 7.2. $\square$

Remark 7.5. The group G and its presentation (Section 5.2), the selection data (Section 5.4), the algebra B with its dimension (Proposition 6.4), the rectification (Proposition 6.15) once its input is given, and the simulation (Proposition 7.2) are given by formulas. The exception is the input of the rectification: the complexes $\tilde{V}_i$, the chain maps f_a and the homotopies h_ρ of Proposition 6.12 exist by the calculus of fractions, but the proof gives no bound on the lengths of the $\tilde{V}_i$ (Remark 6.14). Consequently, the integer l , the number $3(l + 2)$ of simple modules of A and the dimension of A are not determined by the argument. Whether the lifting can be made effective, with a bound on l , is not settled here. Applying the procedure of Section 6.2 to the chosen presentation of $\mathbb{C}G$ introduces auxiliary generators for the relators of degree larger than two, in addition to the thirty symbols for the fifteen generators of G and their inverses. For a resulting presentation with d generators, B has $2(d + 1)$ arrows.

Remark 7.6. The main preprint bounds the global dimensions of Δ by $3l + 2$, using the ordering of the vertices [Ope26b, Proposition 5.1]; the bound $l + 2$ of Proposition 7.2 comes from the tensor product description of the radical of B_1 . Neither bound is needed beyond finiteness, except for the explicit upper bound in Theorem 7.3.

8. TATE COHOMOLOGY AND THE AUSLANDER–REITEN CONJECTURE

By Theorem 3.7, a counterexample to the Auslander–Reiten conjecture [AR75] yields an algebra of infinite little finitistic dimension. The preprint [Ope26c] constructs a counterexample from data in the stable category of a symmetric algebra E : a bimodule F and a stable morphism $v: S \rightarrow F \otimes_E S$. Theorem 8.4 gives this construction over an arbitrary field; the preprint treats characteristic two, in which the signs disappear. The theorem reduces the vanishing of the self-extensions of a module over the triangular algebra $\begin{pmatrix} E & 0 \\ F & E \end{pmatrix}$ to the bijectivity of a map between Tate cohomology groups over E .

8.1. Complete resolutions. Let A be a finite-dimensional algebra. For complexes $\mathbb{P}$ and $\mathbb{W}$ of A -modules, the Hom complex has the terms and differential

$$\text{Hom}_A(\mathbb{P}, \mathbb{W})^a = \prod_{n \in \mathbb{Z}} \text{Hom}_A(\mathbb{P}^n, \mathbb{W}^{n+a}), \quad \partial f = d_{\mathbb{W}}f - (-1)^a f d_{\mathbb{P}};$$

its cocycles of degree a are the chain maps $\mathbb{P} \rightarrow \mathbb{W}[a]$, up to the signs of the shift, and its coboundaries are the null-homotopic ones. For an exact complex $\mathbb{P}$ put

$C^n(\mathbb{P}) = \text{coker}(\mathbb{P}^{n-1} \rightarrow \mathbb{P}^n)$; the differential induces injections $C^n(\mathbb{P}) \rightarrow \mathbb{P}^{n+1}$ and short exact sequences

$$(8.1) \quad 0 \longrightarrow C^n(\mathbb{P}) \longrightarrow \mathbb{P}^{n+1} \longrightarrow C^{n+1}(\mathbb{P}) \longrightarrow 0.$$

An exact complex $\mathbb{P}$ of finitely generated projective modules is *totally acyclic* if $\text{Hom}_A(\mathbb{P}, A)$ is exact; then $\text{Hom}_A(\mathbb{P}, L)$ is exact for every projective module L , since $\text{Hom}_A(\mathbb{P}, A^{(I)}) = \text{Hom}_A(\mathbb{P}, A)^{(I)}$ by finite generation. A finitely generated module is *Gorenstein-projective* if it is isomorphic to $C^0(\mathbb{P})$ for a totally acyclic complex $\mathbb{P}$, which is then called a *complete resolution* of it.

Lemma 8.2. *Let $\mathbb{P}$ be a totally acyclic complex and $\mathbb{W}$ an exact complex of finitely generated projective A -modules, and put $M = C^0(\mathbb{P})$ and $N = C^0(\mathbb{W})$. Then for every integer a there are isomorphisms*

$$H^a \text{Hom}_A(\mathbb{P}, \mathbb{W}) \cong \underline{\text{Hom}}_A(C^{-a}(\mathbb{P}), N) \cong \underline{\text{Hom}}_A(M, C^a(\mathbb{W})),$$

induced by $f \mapsto C^0(f)$ and compatible with composition of chain maps and with exact additive functors that send $\mathbb{P}$ and $\mathbb{W}$ to complexes of the same kind, and for $a > 0$ these groups are isomorphic to $\text{Ext}_A^a(M, N)$. Moreover, $\text{Ext}_A^a(M, L) = 0$ for every $a > 0$ and every projective module L .

Proof. Let $\epsilon_n: \mathbb{P}^n \rightarrow C^n(\mathbb{P})$ and $j_n: C^n(\mathbb{P}) \rightarrow \mathbb{P}^{n+1}$ be the induced maps. If L is projective, every map $u: C^n(\mathbb{P}) \rightarrow L$ extends across j_n : the map $u\epsilon_n$ is a cocycle of $\text{Hom}_A(\mathbb{P}, L)$, hence a coboundary, so $u\epsilon_n = bd_{\mathbb{P}}^n$ for some $b: \mathbb{P}^{n+1} \rightarrow L$, and $bj_n = u$.

A chain map $f: \mathbb{P} \rightarrow \mathbb{W}$ induces a map $M \rightarrow N$. Conversely, given $u: M \rightarrow N$, lift $u\epsilon_0$ to $f^0: \mathbb{P}^0 \rightarrow \mathbb{W}^0$, and construct f^n for $n < 0$ by lifting through the surjections $\mathbb{W}^n \rightarrow \text{im } d_{\mathbb{W}}^n$, using projectivity of the $\mathbb{P}^n$. For $n \geq 0$, once f^n is chosen, $d_{\mathbb{W}}^n f^n$ vanishes on the image of $d_{\mathbb{P}}^{n-1}$ and so factors through $C^n(\mathbb{P})$; the extension property gives f^{n+1} with $f^{n+1}d_{\mathbb{P}}^n = d_{\mathbb{W}}^n f^n$.

If $f = d_{\mathbb{W}}h + hd_{\mathbb{P}}$, then the induced map is induced by $h^1d_{\mathbb{P}}^0$ and factors through $\mathbb{P}^1$. Conversely, suppose the induced map factors as $M \xrightarrow{\alpha} L \xrightarrow{\beta} N$ with L projective. Extending α to $\mathbb{P}^1$ and lifting β to $\mathbb{W}^0$ gives $h^1: \mathbb{P}^1 \rightarrow \mathbb{W}^0$ whose boundary induces $\beta\alpha$; subtracting it, we may assume that f induces zero. Then f^0 takes values in the image of $d_{\mathbb{W}}^{-1}$, and we construct h with $f = d_{\mathbb{W}}h + hd_{\mathbb{P}}$ by two recursions, starting with $h^1 = 0$ and a lift h^0 of f^0 through $d_{\mathbb{W}}^{-1}$. For $n < 0$, the map $f^n - h^{n+1}d_{\mathbb{P}}^n$ takes values in $\ker d_{\mathbb{W}}^n = \text{im } d_{\mathbb{W}}^{n-1}$ and lifts to h^n by projectivity of $\mathbb{P}^n$. For $n \geq 1$, the map $f^n - d_{\mathbb{W}}^{n-1}h^n$ vanishes on the image of $d_{\mathbb{P}}^{n-1}$, hence factors through $C^n(\mathbb{P})$, and the extension property gives h^{n+1} . Since the Hom complex is a product, no finiteness condition arises. This proves the case $a = 0$; the compatibility with composition and with additive functors follows from the construction.

The case of general a follows by applying the case $a = 0$ to the pairs $(\mathbb{P}[-a], \mathbb{W})$ and $(\mathbb{P}, \mathbb{W}[a])$, since $C^0(\mathbb{P}[-a]) = C^{-a}(\mathbb{P})$ and $C^0(\mathbb{W}[a]) = C^a(\mathbb{W})$. For $a > 0$, the truncation $\cdots \rightarrow \mathbb{P}^{-1} \rightarrow \mathbb{P}^0$ is a projective resolution of M , and it computes $\text{Ext}_A^a(M, N)$ as the maps $C^{-a}(\mathbb{P}) \rightarrow N$ modulo those that extend to $\mathbb{P}^{-a+1}$; by the extension property, these are exactly the maps factoring through a projective module. Applying the same description with N replaced by a projective module L gives $\text{Ext}_A^a(M, L) = 0$, since every map $C^{-a}(\mathbb{P}) \rightarrow L$ extends. $\square$

Now let E be a finite-dimensional symmetric algebra. Then $E \cong DE$ as bimodules, so finitely generated projective E -modules are injective, and every finitely generated module M embeds into a finitely generated projective module, by dualising a projective cover of the right module DM . Splicing a projective resolution of M with a coresolution by projective modules gives an exact complex $\mathbb{P}$ of finitely generated projective modules with $C^0(\mathbb{P}) = M$, and $\mathbb{P}$ is totally acyclic, since $\text{Hom}_E(-, L)$ is exact for projective L . By Lemma 8.2, two such complexes for M are homotopy

equivalent, the functor C^0 identifies the homotopy category of these complexes with the stable category $\underline{\text{mod}} E$, the shift $[1]$ corresponds to the cosyzygy by (8.1), and

$$\widehat{\text{Ext}}_E^a(M, N) = H^a \text{Hom}_E(\mathbb{P}_M, \mathbb{P}_N)$$

for complete resolutions $\mathbb{P}_M$ and $\mathbb{P}_N$. Let F be a finitely generated E -bimodule that is projective as a left and as a right module, and write $FM = F \otimes_E M$. The functor $F \otimes_E -$ is exact, since F_E is projective, and preserves projective modules, since ${}_E F$ is projective; hence $F\mathbb{P}$ is a complete resolution of FM . In particular, $F \otimes_E -$ induces an exact functor on $\underline{\text{mod}} E$ with $F(M[a]) \cong (FM)[a]$, and for a stable morphism $g: M \rightarrow N[a]$ we write $F(g): FM \rightarrow (FN)[a]$.

8.2. Triangular algebras. For E and F as above, let

$$\Lambda = \begin{pmatrix} E & 0 \\ F & E \end{pmatrix}, \quad (a, f, b)(a', f', b') = (aa', fa' + bf', bb'),$$

where (a, f, b) denotes the matrix with diagonal entries a , b and lower left entry f . The two diagonal idempotents split a left Λ -module into components M and N , and the action of the lower left corner gives an E -linear map $\eta: FM \rightarrow N$; conversely, every such triple (M, N, η) is a Λ -module, with $(a, f, b)(m, n) = (am, \eta(f \otimes m) + bn)$. A morphism $(M, N, \eta) \rightarrow (M', N', \eta')$ is a pair of E -linear maps (α, β) with $\beta\eta = \eta'F(\alpha)$, and a sequence of triples is exact if and only if both component sequences are exact. Define exact functors

$$\mathbf{L}_1(M) = (M, FM, \text{id}), \quad \mathbf{L}_2(N) = (0, N, 0).$$

Lemma 8.3. *In the situation above, the following hold.*

- (1) *There are natural isomorphisms $\text{Hom}_\Lambda(\mathbf{L}_i M, \mathbf{L}_i N) \cong \text{Hom}_E(M, N)$ for $i = 1, 2$, $\text{Hom}_\Lambda(\mathbf{L}_2 M, \mathbf{L}_1 N) \cong \text{Hom}_E(M, FN)$ and $\text{Hom}_\Lambda(\mathbf{L}_1 M, \mathbf{L}_2 N) = 0$.*
- (2) *We have $\Lambda = \mathbf{L}_1(E) \oplus \mathbf{L}_2(E)$, so $\mathbf{L}_1$ and $\mathbf{L}_2$ preserve finitely generated projective modules, and the first component of every finitely generated projective Λ -module is a projective E -module.*
- (3) *If $\mathbb{P}$ is a totally acyclic complex of E -modules, then $\mathbf{L}_1 \mathbb{P}$ and $\mathbf{L}_2 \mathbb{P}$ are totally acyclic complexes of Λ -modules.*

Proof. (1) A morphism $\mathbf{L}_1 M \rightarrow (M', N', \eta')$ is determined by its first component α , the second being $\eta'F(\alpha)$, and a morphism $\mathbf{L}_2 M \rightarrow (M', N', \eta')$ is an arbitrary map $M \rightarrow N'$. This gives the first three isomorphisms. For the last, the compatibility condition forces the second component to vanish, since the structure map of $\mathbf{L}_1 M$ is the identity.

(2) The first column of Λ is $(E, F, \text{id}) = \mathbf{L}_1(E)$ and the second is $(0, E, 0) = \mathbf{L}_2(E)$. The first component of a direct summand of Λ^r is a direct summand of E^r .

(3) The terms are finitely generated projective by (2), and both complexes are exact. By (1) and (2),

$$\begin{aligned} \text{Hom}_\Lambda(\mathbf{L}_1 \mathbb{P}, \Lambda) &\cong \text{Hom}_E(\mathbb{P}, E), \\ \text{Hom}_\Lambda(\mathbf{L}_2 \mathbb{P}, \Lambda) &\cong \text{Hom}_E(\mathbb{P}, F) \oplus \text{Hom}_E(\mathbb{P}, E), \end{aligned}$$

which are exact since $\mathbb{P}$ is totally acyclic and ${}_E F$ is projective. $\square$

8.3. A module with no positive self-extensions.

Theorem 8.4. *Let E be a finite-dimensional symmetric algebra over a field $\mathbf{k}$, let F be a finite-dimensional E -bimodule that is projective as a left and as a right module, let S be a finite-dimensional left E -module, and let $v: S \rightarrow FS$ be a morphism in $\underline{\text{mod}} E$. For $a \in \mathbb{Z}$ consider the linear map*

$$\delta^a: \widehat{\text{Ext}}_E^a(S, S)^{\oplus 2} \longrightarrow \widehat{\text{Ext}}_E^a(S, FS), \quad (g, j) \longmapsto F(g)v - v[a]j.$$

Suppose that δ^0 is surjective with nonzero kernel and that δ^a is bijective for every $a > 0$. Choose a homomorphism $v_0: S \rightarrow FS$ representing v and an injective homomorphism $i: S \rightarrow Q$ into a finitely generated projective module Q , and put

$$Z_2 = (FS \oplus Q)/\{(v_0(s), i(s)) \mid s \in S\},$$

$$\iota: FS \rightarrow Z_2, \quad t \mapsto [t, 0], \quad Z = (S, Z_2, \iota).$$

Then Z is a finite-dimensional, nonprojective and Gorenstein-projective Λ -module, and

$$\text{Ext}_\Lambda^a(Z, Z) = 0 = \text{Ext}_\Lambda^a(Z, \Lambda) \quad \text{for all } a > 0.$$

In particular, Z is a counterexample to the Auslander–Reiten conjecture for Λ .

Proof. The complete resolution. Choose a complete resolution $\mathbb{P}$ of S over E with $\mathbb{P}^1 = Q$ and $d_{\mathbb{P}}^0 = i\epsilon_0$, where $\epsilon_0: \mathbb{P}^0 \rightarrow S$ is the projection; this is possible by starting the coresolution of S with i . By Lemma 8.2 there is a chain map $f: \mathbb{P} \rightarrow F\mathbb{P}$ inducing v_0 on C^0 . By Lemma 8.3, f defines a chain map $V: \mathbb{L}_2\mathbb{P} \rightarrow \mathbb{L}_1\mathbb{P}$. Let $\mathbb{P}_Z$ be its mapping cone:

$$\mathbb{P}_Z^n = \mathbb{L}_1\mathbb{P}^n \oplus \mathbb{L}_2\mathbb{P}^{n+1}, \quad d_{\mathbb{P}_Z} = \begin{pmatrix} \mathbb{L}_1(d) & V \\ 0 & -\mathbb{L}_2(d) \end{pmatrix}.$$

The sequence

$$0 \longrightarrow \mathbb{L}_1\mathbb{P} \longrightarrow \mathbb{P}_Z \longrightarrow \mathbb{L}_2\mathbb{P}[1] \longrightarrow 0$$

is split in each degree, and its outer terms are totally acyclic by Lemma 8.3; the long exact cohomology sequences of this sequence and of its dual show that $\mathbb{P}_Z$ is totally acyclic.

Its cokernel. The first component of $C^0(\mathbb{P}_Z)$ is $C^0(\mathbb{P}) = S$. Its second component is the cokernel of

$$F\mathbb{P}^{-1} \oplus \mathbb{P}^0 \longrightarrow F\mathbb{P}^0 \oplus Q, \quad (x, y) \longmapsto (F(d)x + f^0y, -i\epsilon_0y),$$

which is $(FS \oplus Q)/\{(v_0(s), -i(s)) \mid s \in S\}$, since $F \otimes_E -$ is exact and f induces v_0 ; the structure map is induced by the inclusion of FS . The automorphism $(t, q) \mapsto (t, -q)$ of $FS \oplus Q$ induces an isomorphism $C^0(\mathbb{P}_Z) \cong Z$. Hence, Z is Gorenstein-projective, and $\text{Ext}_\Lambda^a(Z, \Lambda) = 0$ for $a > 0$ by Lemma 8.2. The module Z is finite-dimensional since S , FS and Q are.

The endomorphism complex. Put $\mathcal{H} = \text{Hom}_E(\mathbb{P}, \mathbb{P})$ and $\mathcal{K} = \text{Hom}_E(\mathbb{P}, F\mathbb{P})$, so that $H^a(\mathcal{H}) = \widehat{\text{Ext}}_E^a(S, S)$ and $H^a(\mathcal{K}) = \widehat{\text{Ext}}_E^a(S, FS)$, and let δ be the chain map

$$\delta: \mathcal{H}^{\oplus 2} \longrightarrow \mathcal{K}, \quad (g, j) \longmapsto F(g)f - fj;$$

it induces δ^a in cohomology. Since there are no nonzero maps from $\mathbb{L}_1$ -terms to $\mathbb{L}_2$ -terms, an element Φ of degree n of $\text{Hom}_\Lambda(\mathbb{P}_Z, \mathbb{P}_Z)$ has unique coordinates $(g, j, h) \in \mathcal{H}^n \oplus \mathcal{H}^n \oplus \mathcal{K}^{n-1}$, defined by

$$\Phi = \begin{pmatrix} \mathbb{L}_1(g) & (-1)^n h \\ 0 & (-1)^n \mathbb{L}_2(j) \end{pmatrix},$$

where h stands for the map $\mathbb{L}_2\mathbb{P} \rightarrow \mathbb{L}_1\mathbb{P}$ corresponding to it under Lemma 8.3. Expanding

$$\partial\Phi = d_{\mathbb{P}_Z}\Phi - (-1)^n\Phi d_{\mathbb{P}_Z},$$

the diagonal entries are $\mathbb{L}_1(\partial g)$ and $(-1)^{n+1}\mathbb{L}_2(\partial j)$, and the upper right entry is

$$(-1)^n(dh + (-1)^nhd + fj - F(g)f) = (-1)^{n+1}(\delta(g, j) - \partial h).$$

Hence,

$$\partial(g, j, h) = (\partial g, \partial j, \delta(g, j) - \partial h),$$

and there is a degreewise split short exact sequence of complexes

$$0 \longrightarrow \mathcal{K}[-1] \longrightarrow \text{Hom}_\Lambda(\mathbb{P}_Z, \mathbb{P}_Z) \longrightarrow \mathcal{H}^{\oplus 2} \longrightarrow 0,$$

given by $h \mapsto (0, 0, h)$ and $(g, j, h) \mapsto (g, j)$. A cocycle (g, j) lifts to $(g, j, 0)$, whose boundary is $(0, 0, \delta(g, j))$, so the connecting homomorphism is δ^a , and for every a there is an exact sequence

$$(8.5) \quad 0 \longrightarrow \operatorname{coker} \delta^{a-1} \longrightarrow H^a \operatorname{Hom}_\Lambda(\mathbb{P}_Z, \mathbb{P}_Z) \longrightarrow \ker \delta^a \longrightarrow 0.$$

For $a > 0$ the middle term is $\operatorname{Ext}_\Lambda^a(Z, Z)$ by Lemma 8.2. For $a = 1$ the outer terms vanish since δ^0 is surjective and δ^1 is injective, and for $a \geq 2$ since δ^{a-1} and δ^a are bijective.

Nonprojectivity. Since $\ker \delta^0 \neq 0$, the stable endomorphism ring of S is nonzero, so S is not projective. The first component of Z is S , so Z is not projective by Lemma 8.3. $\square$

Remark 8.6. The proof uses the symmetry of E only to ensure that finitely generated projective modules are injective and that every module embeds into a projective one, and the hypothesis that δ^0 has a nonzero kernel only to ensure that S is not projective; it uses neither Tate duality nor the simplicity of S . The sequence (8.5) holds without the hypotheses on δ . In particular, if δ^1 is injective, then $\operatorname{Ext}_\Lambda^1(Z, Z) \cong \operatorname{coker} \delta^0$. An application of Theorem 8.4 therefore needs a bimodule F for which $\widehat{\operatorname{Ext}}_E^a(S, FS)$ is a copy of $\widehat{\operatorname{Ext}}_E^a(S, S)^{\oplus 2}$ for $a > 0$, compatibly with the two actions of the Tate algebra of S in the definition of δ^a , while in degree zero δ^0 is surjective but not injective; in the construction below, $\widehat{\operatorname{Ext}}_E^0(S, FS)$ is one-dimensional. The bimodule of Section 9 is obtained as a fibre of a map from two twisted copies of E to a bimodule whose evaluation at S has Tate cohomology only in degrees 0 and 3.

9. A COUNTEREXAMPLE TO THE AUSLANDER–REITEN CONJECTURE

Following [Ope26c], we construct a bimodule and a stable morphism satisfying the hypotheses of Theorem 8.4, and deduce a second algebra of infinite little finitistic dimension, with nine simple modules. Throughout,

$$\mathbf{k} = \mathbb{F}_2(q, H_1, H_2),$$

the field of rational functions in three variables over $\mathbb{F}_2$; in characteristic two all signs disappear, and $\delta^a(g, j) = F(g)v + v[a]j$ in Theorem 8.4. The construction uses a ten-dimensional algebra C with a simple module s whose resolution has differentials depending on successive powers of q (Section 9.1); its trivial extension $T = C \ltimes DC$, over which s has polynomial Ext algebra on a generator of degree three (Section 9.2), realised by an explicit Hochschild cocycle (Section 9.3); the tensor square $E = T \otimes_{\mathbf{k}} T$, a bimodule $\mathcal{C}$ built from two cones on that cocycle, and twisted lifts (Sections 9.4 to 9.6); and the fibre F of a map from two twisted copies of E to $\mathcal{C}[3]$ (Section 9.7). Finite identities, such as associativity of C and the cocycle identities, are checked by exact computations recorded in Appendix A; every statement about all degrees is proved in the text.

9.1. The algebra C . Let C be the $\mathbf{k}$ -algebra with basis

$$e, x, y, z \in eCe, \quad u, v \in eCf, \quad t, j \in fCe, \quad f, n \in fCf,$$

where e and f are orthogonal idempotents with $e + f = 1$, multiplication by e and f is prescribed by the corners, and the products of two of the remaining eight basis vectors are zero except for

$$\begin{aligned} xy = qz, \quad yx = z, \quad xu = yu = v, \quad tx = j, \quad ty = q^2j, \quad ut = y + qx, \\ vt = qz, \quad uj = z, \quad tu = (1 + q)n, \quad nt = qj, \quad un = v. \end{aligned}$$

Let N be the span of x, y, z, u, v, t, j, n , let s be the simple left module $C/(N + Ce)$ at f , and let s^r be the simple right module at f .

Lemma 9.1. *The algebra C is associative. It is graded by $\deg u = \deg t = 1$, $\deg x = \deg y = \deg n = 2$, $\deg v = \deg j = 3$, $\deg z = 4$ and $\deg e = \deg f = 0$, its radical is N , and $N^5 = 0$.*

Proof. Every product above is homogeneous for the grading. A product of three basis vectors vanishes under both bracketings if one of them is an idempotent that does not match the corners, or if its total degree exceeds four, since there is no basis vector of degree larger than four. Among the remaining triples of radical basis vectors, the only ones with compatible corners are xut , yut , utx , uty , utu , unt , txu , tyu , tut , tun and ntu , and their products under both bracketings are qz , qz , z , q^2z , $(1+q)v$, qz , 0 , 0 , $q(1+q)j$, 0 and 0 ; for instance $(ut)y = (y+qx)y = q^2z = u(q^2j) = u(ty)$. The associators of all 1000 triples of basis vectors are also checked by an exact computation (Appendix A). The span N of the basis vectors of positive degree is a nilpotent ideal with $N^5 = 0$ and $C/N \cong \mathbf{k} \times \mathbf{k}$, so N is the radical. $\square$

For $i \geq 0$ put $\ell_i = x + q^i y \in eCe$. The left modules Ce and Cf have the bases e, x, y, z, t, j and f, u, v, n , and right multiplication by u and by ℓ_i acts as follows:

$$(9.2) \quad \begin{array}{c|cccccc} a & e & x & y & z & t & j \\ \hline au & u & v & v & 0 & (1+q)n & 0 \\ a\ell_i & \ell_i & q^{i+1}z & z & 0 & (1+q^{i+2})j & 0 \end{array}$$

Proposition 9.3. *The complex*

$$\mathbb{R}: \quad \cdots \longrightarrow Ce \xrightarrow{\cdot \ell_2} Ce \xrightarrow{\cdot \ell_1} Ce \xrightarrow{\cdot \ell_0} Ce \xrightarrow{\cdot u} Cf,$$

with Cf in degree zero and right multiplications as differentials, is a minimal projective resolution of s . Consequently, $\text{Ext}_C^a(s, s) = 0$ for $a > 0$, and $\mathbb{R}\text{Hom}_C(s, C) \cong s^r[-2]$ in the derived category of right C -modules.

Proof. Since q is transcendental over $\mathbb{F}_2$, the elements $1+q$ and $1+q^{i+2}$ are nonzero. By (9.2), the image of $\cdot u$ is the span of u, v and n , which is the radical of Cf , and its kernel is spanned by ℓ_0, z and j . A vector in the kernel of $\cdot \ell_i$ has zero coefficients at e and t , and coefficients at x and y related by $a_y = q^{i+1}a_x$; hence

$$\ker(\cdot \ell_i) = \langle \ell_{i+1}, z, j \rangle, \quad \text{im}(\cdot \ell_i) = \langle \ell_i, z, j \rangle,$$

for every $i \geq 0$, and the complex is exact. All differentials have image in N times their targets, so each surjection onto a syzygy is a projective cover, and the resolution is minimal. Since $\text{Hom}_C(Ce, s) = es = 0$ and $\text{Hom}_C(Cf, s) = \mathbf{k}$, the asserted vanishing of self-extensions follows.

Identifying $\text{Hom}_C(Ce, C) = eC$ and $\text{Hom}_C(Cf, C) = fC$ as right modules, precomposition with right multiplication by b becomes left multiplication by b , and $\text{Hom}_C(\mathbb{R}, C)$ is the complex

$$fC \xrightarrow{\cdot u} eC \xrightarrow{\cdot \ell_0} eC \xrightarrow{\cdot \ell_1} eC \longrightarrow \cdots$$

in degrees $0, 1, 2, \dots$, with $fC = \langle f, t, j, n \rangle$ and $eC = \langle e, x, y, z, u, v \rangle$. Left multiplication by u sends f, t, j, n to the independent vectors $u, y+qx, z, v$, and left multiplication by ℓ_i sends e, x, y, z, u, v to $\ell_i, q^i z, qz, 0, (1+q^i)v, 0$. For $i = 0$ its kernel is $\langle qx + y, z, u, v \rangle$, the image of u , and its image is $\langle \ell_0, z \rangle$; for $i \geq 1$ its kernel is $\langle \ell_{i-1}, z, v \rangle$ and its image is $\langle \ell_i, z, v \rangle$. Hence, the only cohomology is in degree two, spanned by the class of v , on which f acts as the identity and the radical acts as zero, since $vt = qz$ is a boundary. Good truncations identify the complex with $s^r[-2]$. $\square$

The parameter q makes the resolution aperiodic, while the self-extensions of s vanish in positive degrees: on $e\langle \ell_i, z, j \rangle$ the elements x and y act with $x = q^{i+1}y$ and $y \neq 0$, a ratio preserved by isomorphisms of C -modules, so the syzygies $\langle \ell_i, z, j \rangle$ are

pairwise nonisomorphic. The module s is not a counterexample to the Auslander–Reiten conjecture over C , since $\text{Ext}_C^2(s, C) \neq 0$; over $T = C \ltimes DC$, its Ext algebra is polynomial on a generator of degree three (Theorem 9.5).

9.2. The trivial extension T . Let $T = C \ltimes DC$, where the bimodule structure of DC is $(a\phi)(c) = \phi(ca)$ and $(\phi a)(c) = \phi(ac)$, and regard s as a T -module through $T \rightarrow C$. For a basis vector c of C we write c^* for the dual basis vector of DC .

Lemma 9.4. *The algebra T is a twenty-dimensional symmetric algebra, with symmetrising form $\text{tr}(a, \phi) = \phi(1)$. Its radical is $N \oplus DC$, with sixth power zero, and $T/\text{rad } T \cong \mathbf{k} \times \mathbf{k}$.*

Proof. The algebra T is symmetric, with the stated form, by Section 2.3. With $\deg c^* = 5 - \deg c$, the algebra T is graded, with all basis vectors other than e and f in degrees between one and five; their span is a nilpotent ideal with sixth power zero and quotient $\mathbf{k} \times \mathbf{k}$. $\square$

Theorem 9.5. *There is $\tau \in \text{Ext}_T^3(s, s)$ such that $\text{Ext}_T^*(s, s) = \mathbf{k}[\tau]$, a polynomial algebra under the Yoneda product. Every nonzero element of $\text{Ext}_T^3(s, s)$ is such a generator. Moreover, s is not projective and $\underline{\text{End}}_T(s) = \mathbf{k}$.*

Proof. Let $\mathbb{R}$ be the resolution of Proposition 9.3. For a finitely generated projective left C -module V , the map $DC \otimes_C V \rightarrow D \text{Hom}_C(V, C)$, $\phi \otimes v \mapsto (g \mapsto \phi(g(v)))$, is a natural isomorphism of left C -modules, as one sees for $V = C$. Hence, $DC \otimes_C \mathbb{R} \cong D \text{Hom}_C(\mathbb{R}, C)$, which has cohomology $D(s^r) = s$ in degree -2 by Proposition 9.3; since DC acts on it as zero, it is quasi-isomorphic to $s[2]$ as a complex of T -modules. Tensoring the sequence $0 \rightarrow DC \rightarrow T \rightarrow C \rightarrow 0$ of (T, C) -bimodules with the complex $\mathbb{R}$ of projective, hence flat, C -modules gives a short exact sequence of complexes of T -modules, and hence a triangle

$$(9.6) \quad s[2] \longrightarrow \mathcal{J} \longrightarrow s \xrightarrow{\eta} s[3], \quad \mathcal{J} = T \otimes_C \mathbb{R},$$

in $D^-(\text{mod } T)$. The terms $T \otimes_C Ce = Te$ and $T \otimes_C Cf = Tf$ of $\mathcal{J}$ are projective, and $\mathcal{J}$ is bounded above, so adjunction gives $\text{Hom}(\mathcal{J}, s[a]) \cong H^a \text{Hom}_C(\mathbb{R}, s)$, which is $\mathbf{k}$ for $a = 0$ and zero otherwise.

Put $V^a = \text{Hom}(s, s[a])$, which is $\text{Ext}_T^a(s, s)$ for $a \geq 0$ and zero for $a < 0$. Applying $\text{Hom}(-, s[a])$ to (9.6) gives exact sequences

$$\text{Hom}(\mathcal{J}, s[a-1]) \longrightarrow V^{a-3} \xrightarrow{\beta \mapsto \beta[3] \circ \eta} V^a \longrightarrow \text{Hom}(\mathcal{J}, s[a]).$$

For $a = 1$ this gives $V^1 = 0$, and for $a \geq 2$ the outer terms vanish, so that composition with η is an isomorphism $V^{a-3} \rightarrow V^a$. Since $V^0 = \mathbf{k}$ and $V^{-1} = V^{-2} = 0$, we obtain $V^{3m} = \mathbf{k}\eta^m$ for $m \geq 0$ and $V^a = 0$ in all other degrees, and no power of η vanishes. Hence, $\text{Ext}_T^*(s, s) = \mathbf{k}[\eta]$, and every nonzero multiple of η is a generator. Since $\text{Ext}_T^3(s, s) \neq 0$, the module s is not projective, and a nonzero endomorphism of the one-dimensional module s factoring through a projective module would make s a direct summand of a projective module; hence $\underline{\text{End}}_T(s) = \mathbf{k}$. $\square$

9.3. A Hochschild cocycle. The construction of the bimodule F needs a representative of τ by a bimodule map, which the abstract computation of Theorem 9.5 does not provide. Let $I = \mathbf{k}e \oplus \mathbf{k}f \subseteq T$ and $\mathfrak{r} = \text{rad } T$, so that $T = I \oplus \mathfrak{r}$. The bar resolution of T relative to I is the complex $\mathbb{B}$ of T -bimodules with

$$\mathbb{B}^{-n} = T \otimes_I \mathfrak{r}^{\otimes n} \otimes_I T \quad (n \geq 0), \quad \mathbb{B}^n = 0 \quad (n > 0),$$

whose elements we write $a_0[a_1 | \cdots | a_n]a_{n+1}$, with the differential given by the sum of the $n+1$ products of adjacent entries, and augmentation $\mathbb{B}^0 \rightarrow T$ given by multiplication. Its terms are finite direct sums of the projective bimodules $Ti \otimes_{\mathbf{k}} jT$

for $i, j \in \{e, f\}$. The map

$$a_0[a_1 | \cdots | a_n]a_{n+1} \mapsto [\bar{a}_0 | a_1 | \cdots | a_n]a_{n+1},$$

where $\bar{a}_0$ is the component of a_0 in $\mathfrak{r}$, together with $a \mapsto 1[a]$ in augmented degree, is a contraction of the augmented complex, and it is right linear; hence $\mathbb{B}$ is a projective bimodule resolution of T , and $\mathbb{B} \otimes_T s$ is a projective resolution of s .

Let $p: \mathfrak{r}^{\otimes 3} \rightarrow T$ be the I -bilinear map given by the table of 179 coefficients in Appendix A, reproduced from the appendix of [Ope26c]: an entry $abcw$ in the row q^d means that the coefficient of the basis vector w in $p(a, b, c)$ is q^d , where capital letters denote dual basis vectors, so that for example $J = j^*$; all other coefficients are zero. For $\lambda \in \mathbf{k}^\times$ let h_λ be the automorphism $(c, \phi) \mapsto (c, \lambda\phi)$ of T , and let $z_\lambda: \mathfrak{r} \rightarrow \mathfrak{r}$ be the linear map given on basis vectors by

$$z_\lambda(a) = \begin{cases} \lambda qa & \text{if } a \in \{u, v, t, e^*, x^*, y^*, z^*, u^*, v^*\}, \\ 0 & \text{otherwise.} \end{cases}$$

Lemma 9.7. *The map p has the following properties.*

- (1) *It is a Hochschild cocycle: for all $a, b, c, d \in \mathfrak{r}$,*

$$ap(b, c, d) + p(ab, c, d) + p(a, bc, d) + p(a, b, cd) + p(a, b, c)d = 0,$$

where p is extended by zero to inputs with a factor in I .

- (2) *Every nonzero coefficient of $p(a, b, c)$ at a basis vector w , for basis vectors a, b, c , has exactly one more dual basis vector among a, b, c than w has.*
(3) *For all $a, b \in \mathfrak{r}$ and $\lambda \in \mathbf{k}^\times$,*

$$\sum_w p(a, b, h_\lambda(w))w^* = az_\lambda(b) + z_\lambda(ab) + z_\lambda(a)h_\lambda^{-1}(b),$$

where w runs through the radical basis vectors and w^ denotes the dual of w under the trace, so that $(c^*)^* = c$.*

- (4) *$p(t, x, j^*) = 0$ and $p(t, y, j^*) = q^3 f$.*

Proof. These are finitely many polynomial identities in q and λ , and they are checked by an exact computation over $\mathbb{F}_2[q, \lambda]$ on all inputs; see Appendix A. $\square$

Define a map $p: \mathbb{B} \rightarrow \mathbb{B}[3]$ by

$$p(a_0[a_1 | \cdots | a_n]a_{n+1}) = a_0[a_1 | \cdots | a_{n-3}]p(a_{n-2}, a_{n-1}, a_n)a_{n+1} \quad (n \geq 3),$$

and zero on $\mathbb{B}^{-n}$ for $n < 3$. In $dp + pd$, every product of adjacent entries among the first $n - 3$ bars appears twice, and the remaining five terms form the left side of Lemma 9.7(1); hence p is a chain map of bimodule complexes.

Proposition 9.8. *The class τ of the chain map $p \otimes_T s: \mathbb{B} \otimes_T s \rightarrow (\mathbb{B} \otimes_T s)[3]$ in $\text{Ext}_T^3(s, s)$ is nonzero, and therefore generates $\text{Ext}_T^*(s, s)$.*

Proof. The class is represented by the cocycle $c: \mathbb{B}^{-3} \otimes_T s \rightarrow s$ obtained by composing with the augmentation. The element $\zeta = q^2[t|x|j^*] + [t|y|j^*]$ of $s^{\mathfrak{r}} \otimes_T \mathbb{B}^{-3} \otimes_T s$ is a cycle: its outer terms vanish because the radical annihilates simple modules, and since $tx = j$, $ty = q^2 j$, $xj^* = t^*$ and $yj^* = q^2 t^*$, the four inner terms cancel in pairs. A coboundary vanishes on every cycle, whereas $c(\zeta) = q^2 p(t, x, j^*) + p(t, y, j^*)$ acts on s as $q^3 \neq 0$ by Lemma 9.7(4). Hence, $\tau \neq 0$, and Theorem 9.5 applies. $\square$

Lemma 9.9. *For $\lambda \in \mathbf{k}^\times$ the functor $T_{h_\lambda} \otimes_T -$, where T_{h_λ} is T with the right action twisted by h_λ , fixes s up to a canonical isomorphism and acts on $\text{Ext}_T^{3m}(s, s)$ as multiplication by λ^{-m} , for every $m \geq 0$.*

Proof. The map $T_{h_\lambda} \otimes_T M \rightarrow {}_{h_\lambda} M$, $a \otimes m \mapsto h_\lambda^{-1}(a)m$, is an isomorphism, natural in the left module M , onto M with the action restricted along h_λ^{-1} . Since h_λ is the identity modulo DC , which acts as zero on s , the twisted module ${}_{h_\lambda} s$ is s . Applying h_λ^{-1} to all entries of $\mathbb{B} \otimes_T s$ defines an isomorphism from this resolution to its twist which induces the identity on s . Under it, the cocycle c of Proposition 9.8 becomes $c \circ h_\lambda^{-1}$. The only coefficients of p that survive in c are those at f or e , which have no dual factor, and by Lemma 9.7(2) they occur only on inputs with exactly one dual factor; on those inputs $c \circ h_\lambda^{-1} = \lambda^{-1}c$. Hence, the functor sends τ to $\lambda^{-1}\tau$, and, being exact, it preserves Yoneda products, so it sends τ^m to $\lambda^{-m}\tau^m$. $\square$

9.4. The tensor square. Let $E = T \otimes_{\mathbf{k}} T$ and $S = s \otimes_{\mathbf{k}} s$, and put $H^a = \widehat{\text{Ext}}_E^a(S, S)$.

Proposition 9.10. *The algebras E and $E^e = E \otimes_{\mathbf{k}} E^{\text{op}}$ are symmetric, and*

$$\text{Ext}_E^*(S, S) = \mathbf{k}[\tau_1, \tau_2], \quad \tau_1 = \tau \otimes 1, \quad \tau_2 = 1 \otimes \tau,$$

is a polynomial algebra on two generators of degree three. Moreover, S is not projective and $\underline{\text{End}}_E(S) = \mathbf{k}$.

Proof. The tensor product of the symmetrising forms of the factors is a symmetric associative form, nondegenerate because its Gram matrix is the Kronecker product of two invertible matrices; the same applies to E^e . If $\mathbb{Q} \rightarrow s$ is a projective resolution with finitely generated terms, then the total complex $\mathbb{Q} \otimes_{\mathbf{k}} \mathbb{Q}$ is a projective resolution of S by finitely generated E -modules, as one sees by splitting $\mathbb{Q}$, as a complex of vector spaces, into its cohomology and contractible complexes. The isomorphism of complexes

$$\text{Hom}_E(\mathbb{Q} \otimes \mathbb{Q}, S) \cong \text{Hom}_T(\mathbb{Q}, s) \otimes \text{Hom}_T(\mathbb{Q}, s)$$

gives $\text{Ext}_E^{3m}(S, S)$ the basis $\tau_1^i \tau_2^{m-i}$, $0 \leq i \leq m$, and all other positive degrees vanish. The classes τ_1 and τ_2 are represented by lifts of τ acting on different tensor factors, which commute up to a sign, hence commute in characteristic two, so their monomials are the external products $\tau^i \otimes \tau^{m-i}$ and there are no relations. The last assertion follows as in Theorem 9.5. $\square$

For $m \geq 0$ let $P_m \subseteq \mathbf{k}[\tau_1, \tau_2]$ be the space of homogeneous polynomials of degree m in the generators, so that $H^{3m} = P_m$, and put $P_m = 0$ for $m < 0$.

Proposition 9.11. *The Tate groups of S are*

$$H^{3m} = P_m, \quad H^{-3m-1} \cong DP_m \quad (m \geq 0), \quad H^a = 0 \quad \text{otherwise},$$

where the isomorphism is that of Theorem 2.1. For $m \geq 1$, left multiplication by τ_i from H^{-3m-1} to H^{-3m+2} is the transpose $DP_m \rightarrow DP_{m-1}$ of left multiplication by τ_i from P_{m-1} to P_m ; it is surjective with kernel spanned by the dual of τ_{3-i}^m . Multiplication by τ_i on $H^{-1} = DP_0$ is zero.

Proof. The first assertion follows from Proposition 9.10 and Tate duality, Theorem 2.1, with $X = Y = S$: $H^{-1-a} \cong DH^a$. The compatibility of the duality pairing with composition [Lin13, Section 2, (2.8)], recalled after Theorem 2.1, identifies multiplication by τ_i on the negative groups with the transpose of multiplication by τ_i on the positive ones. Multiplication by τ_i from P_{m-1} to P_m is injective with cokernel spanned by the class of τ_{3-i}^m , which gives the description of its transpose. On H^{-1} the product lands in $H^2 = 0$. $\square$

9.5. The bimodule $\mathcal{C}$. Let $\mathcal{L}$ be the mapping cone of $p[-3]: \mathbb{B}[-3] \rightarrow \mathbb{B}$, a complex of projective T -bimodules with $\mathcal{L}^j = \mathbb{B}^{j-2} \oplus \mathbb{B}^j$, and let

$$K = \mathcal{L} \otimes_{\mathbf{k}} \mathcal{L},$$

a complex of projective E -bimodules, that is, of projective E^e -modules, bounded above with finitely generated terms. The complex $\mathcal{L}$ has cohomology T in degrees 0 and 2 and no other cohomology, and K has a filtration with four subquotients $\mathbb{B} \otimes \mathbb{B}$, $(\mathbb{B} \otimes \mathbb{B})[-2]$ twice, and $(\mathbb{B} \otimes \mathbb{B})[-4]$, where $\mathbb{B} \otimes \mathbb{B}$ is a projective bimodule resolution of E ; in particular K is exact in negative degrees. For $N \geq 0$ put

$$C_N = \text{coker}(K^{-N-1} \rightarrow K^{-N}).$$

Lemma 9.12. *For $N \geq 1$, the bimodule C_N is projective as a left and as a right E -module, and $C_{N+1} \cong C_N[-1]$ in $\underline{\text{mod}} E^e$. Let $\mathcal{C}$ be a finitely generated E -bimodule, projective on both sides, representing $C_N[N]$, which does not depend on N . Then $M \mapsto \mathcal{C}M = \mathcal{C} \otimes_E M$ induces an exact functor on $\underline{\text{mod}} E$, and there are triangles*

$$S[-3] \xrightarrow{\tau_1} S \longrightarrow \mathcal{C}_1 S \xrightarrow{\pi_1} S[-2], \quad (\mathcal{C}_1 S)[-3] \xrightarrow{\tau_2} \mathcal{C}_1 S \longrightarrow \mathcal{C} S \xrightarrow{\pi_2} (\mathcal{C}_1 S)[-2]$$

in $\underline{\text{mod}} E$, where $\mathcal{C}_1 S$ is the evaluation at S of a bimodule, projective on both sides, representing

$$\text{coker}((\mathcal{L} \otimes_{\mathbf{k}} \mathbb{B})^{-2} \rightarrow (\mathcal{L} \otimes_{\mathbf{k}} \mathbb{B})^{-1})[1],$$

and the second τ_2 is induced by p acting on the second factor.

Proof. The cokernel functor C_N is exact on degreewise split short exact sequences of complexes that are exact in degree $-N-1$: if b in the subcomplex becomes a boundary dv in the middle, the image of v in the quotient is a cycle, hence a boundary $d\bar{w}$; replacing v by $v - dw$ for a lift w of $\bar{w}$ shows that b is a boundary in the subcomplex. Hence, C_N has a filtration whose factors are the cokernels

$$Z_j = \text{coker}((\mathbb{B} \otimes \mathbb{B})^{-j-1} \rightarrow (\mathbb{B} \otimes \mathbb{B})^{-j})$$

for $j = N, N+2, N+2$ and $N+4$. The augmented resolution $\mathbb{B} \otimes \mathbb{B} \rightarrow E$ splits as a complex of left and as a complex of right E -modules, since E is projective on each side, so the Z_j are projective on each side, and so is C_N , since the filtration splits on each side. Exactness of K in degree $-N$ gives an exact sequence $0 \rightarrow C_{N+1} \rightarrow K^{-N} \rightarrow C_N \rightarrow 0$ with K^{-N} projective over E^e , so $C_{N+1} \cong C_N[-1]$.

A cosyzygy of a bimodule M that is projective on each side, taken in E^e -modules, is again projective on each side: E^e is symmetric, so M embeds into a projective E^e -module, and since modules projective on one side over the symmetric algebra E are injective on that side, the embedding splits on each side. Hence, $\mathcal{C}$ can be chosen projective on both sides. For such a bimodule, $\mathcal{C} \otimes_E -$ is exact and preserves projective modules, and it induces an exact functor on the stable category. Evaluation at S of the degreewise split cone sequences, which split as sequences of right modules, gives the two triangles; the first comes from the cone in the first factor of K , and the second from the action of p on the second factor, as K is the cone of the map induced by $p[-3]$ on $\mathcal{L} \otimes \mathbb{B}$. $\square$

Write $\pi = \pi_1[-2] \circ \pi_2: \mathcal{C}S \rightarrow S[-4]$ for the composite of the two projections, and put $W^a = \widehat{\text{Ext}}_E^a(S, \mathcal{C}S)$.

Proposition 9.13. *For every $a \in \mathbb{Z}$,*

$$W^a \cong \begin{cases} \mathbf{k} & a \in \{0, 3\}, \\ 0 & \text{otherwise,} \end{cases}$$

and $\pi[3]$ induces an isomorphism $W^3 \rightarrow H^{-1}$.

Proof. Put $U^a = \widehat{\text{Ext}}_E^a(S, \mathcal{C}_1 S)$. The first triangle of Lemma 9.12 gives exact sequences

$$0 \longrightarrow \text{coker}(\tau_1: H^{a-3} \rightarrow H^a) \longrightarrow U^a \xrightarrow{\pi_1} \ker(\tau_1: H^{a-2} \rightarrow H^{a+1}) \longrightarrow 0.$$

By Propositions 9.10 and 9.11, multiplication by τ_1 is injective on $\bigoplus_{m \geq 0} P_m$ with cokernel $\mathbf{k}[\tau_2]$, and on the negative part it is surjective with kernel spanned by the duals $\varepsilon_m \in H^{-3m-1}$ of τ_2^m . Since the two possible contributions lie in degrees that are incongruent modulo three,

$$U^{3m} = \mathbf{k}x_m, \quad U^{1-3m} = \mathbf{k}y_m \quad (m \geq 0), \quad U^a = 0 \quad \text{otherwise},$$

where x_m is the image of τ_2^m and $\pi_1(y_m) = \varepsilon_m$. The action of τ_2 on the second tensor factor commutes with all maps of the first triangle, so $\tau_2 x_m = x_{m+1}$, $\tau_2 y_m = y_{m-1}$ for $m \geq 1$ and $\tau_2 y_0 = 0$. The second triangle gives exact sequences

$$0 \longrightarrow \text{coker}(\tau_2: U^{a-3} \rightarrow U^a) \longrightarrow W^a \xrightarrow{\pi_2} \ker(\tau_2: U^{a-2} \rightarrow U^{a+1}) \longrightarrow 0.$$

The cokernel is nonzero only for $a = 0$, where it is $\mathbf{k}x_0$, and the kernel is nonzero only for $a - 2 = 1$, where it is $\mathbf{k}y_0$. Hence, $W^0 \cong \mathbf{k}$, $W^3 \cong \mathbf{k}$ via π_2 onto $U^1 = \mathbf{k}y_0$, and $W^a = 0$ otherwise; since $\pi_1(y_0) = \varepsilon_0$ spans H^{-1} , the composite $\pi[3]$ is an isomorphism $W^3 \rightarrow H^{-1}$. $\square$

9.6. Twisted lifts. For $\lambda \in \mathbf{k}^\times$ let E_λ be the E -bimodule E with the right action twisted by $h_\lambda \otimes h_\lambda$. By Lemma 9.9 and Proposition 9.10, the functor $E_\lambda \otimes_E -$ fixes S and acts on H^{3m} as multiplication by λ^{-m} , for every $m \geq 0$. Let $\beta_0 \in H^{-1}$ be the class of the map $S \rightarrow \Omega S$, $f \otimes f \mapsto f^* \otimes f^*$, where $\Omega S = \text{rad}(E)(f \otimes f)$ is the kernel of the projective cover $E(f \otimes f) \rightarrow S$.

Proposition 9.14. *For every $\lambda \in \mathbf{k}^\times$ there is a morphism $g_\lambda: E_\lambda \rightarrow \mathcal{C}[3]$ in $\text{mod } E^e$ such that $\pi[3] \circ (g_\lambda \otimes_E S) = \lambda^2 \beta_0$, and $\beta_0 \neq 0$.*

Proof. Let $\mathcal{B}$ be the basis of T consisting of the basis of C and its dual basis, and let w^* denote the trace dual of $w \in \mathcal{B}$. The Casimir element $\sum_w w \otimes w^*$ of $T \otimes_{\mathbf{k}} T$ corresponds to the identity under $x \otimes y \mapsto (t \mapsto x \text{tr}(yt))$, so it commutes with the elements of T . Applying h_λ to the first factor and projecting to $T \otimes_I T = \mathbb{B}^0$ gives

$$\xi_\lambda = \sum_{w \in \mathcal{B}} h_\lambda(w) \otimes_I w^*, \quad h_\lambda(a) \xi_\lambda = \xi_\lambda a \quad (a \in T),$$

so $b(x) = x \xi_\lambda$ defines a bimodule map $b: T_{h_\lambda} \rightarrow \mathbb{B}^0$. Its composite with the augmentation is $\sum_w h_\lambda(w) w^*$, which is zero: for a basis vector w of C , the products ww^* and w^*w are the duals of the idempotents at the two ends of w , and each idempotent is the end of an even number of basis vectors of C on each side, namely 6 for e and 4 for f . Separating the same identity according to the left idempotent $r \in \{e, f\}$ of w and removing the idempotent term gives $\sum_w h_\lambda(w) w^* = r^*$, the sum over the radical basis vectors w with left idempotent r ; these identities are also checked in Appendix A.

Let $\mathbb{Q}$ be $\mathbb{B}$ with the right action twisted by h_λ , a resolution of T_{h_λ} , and define maps out of $\mathbb{Q}$ on generators $[a_1 | \cdots | a_n]$, extended by $\psi(a_0 [\cdots] a_{n+1}) = a_0 \psi([\cdots]) h_\lambda^{-1}(a_{n+1})$, which is the rule for bimodule maps from $\mathbb{Q}$. Let $D_b: \mathbb{Q} \rightarrow \mathbb{B}$ be b composed with the augmentation in degree zero and zero elsewhere, and put

$$B_n([a_1 | \cdots | a_n]) = \sum_w [a_1 | \cdots | a_n | h_\lambda(w)] w^*,$$

$$G_n([a_1 | \cdots | a_n]) = [a_1 | \cdots | a_{n-1}] z_\lambda(a_n),$$

with w running through the radical basis vectors and $G_0 = 0$, maps of degrees -1 and $+1$. Projecting $a \xi_\lambda = \xi_\lambda h_\lambda^{-1}(a)$ to the radical in the first factor gives, for a

radical a with right idempotent r ,

$$\sum_w ah_\lambda(w) \otimes_I w^* + \sum_w h_\lambda(w) \otimes_I w^* h_\lambda^{-1}(a) = a \otimes_I r^*.$$

Using this identity and the previous one, all terms of $dB + Bd - D_b$ cancel: the products among the first bars appear twice, the two terms involving the last bar of the source add up to $[a_1 | \cdots | a_n]r^*$, and the product of the appended bar with its coefficient gives the same element. Hence, $dB + Bd = 0$ in negative degrees. In degree zero, for $r \in \{e, f\}$,

$$dB_0(r[]r) = \sum_w h_\lambda(w) \otimes_I w^* + r \otimes_I r^* = r\xi_\lambda = D_b(r[]r),$$

with w running through the radical basis vectors with left idempotent r , so $dB + Bd = D_b$. In $dG + Gd$ the products among the first bars cancel, and what remains is

$$[a_1 | \cdots | a_{n-2}] (a_{n-1} z_\lambda(a_n) + z_\lambda(a_{n-1} a_n) + z_\lambda(a_{n-1}) h_\lambda^{-1}(a_n)),$$

which is pB by Lemma 9.7(3); for $n \leq 1$ both sides vanish. Hence, $pB = dG + Gd$.

The complex $\mathbb{Q} \otimes_{\mathbf{k}} \mathbb{Q}$ resolves E_λ . Define a map $\Phi: \mathbb{Q} \otimes \mathbb{Q} \rightarrow K[3]$ with component $B \otimes D_b$ into the summand $(\mathbb{B} \otimes \mathbb{B})[-4][3]$ of the underlying graded bimodule, component $G \otimes D_b$ into the summand corresponding to the cone in the second tensor factor, and zero components into the other two summands. The component of $d\Phi - \Phi d$ in $(\mathbb{B} \otimes \mathbb{B})[-4][3]$ is $D_b \otimes D_b$; in the summand corresponding to the second tensor factor, the component $(dG + Gd) \otimes D_b$ is cancelled by the term $pB \otimes D_b$ of the cone differential, and the term coming from the first factor vanishes because $pD_b = 0$. Hence, Φ is a chain map in all degrees below zero, where $D_b \otimes D_b$ vanishes. For $n \geq 4$ both complexes are exact in degree $-n$, so Φ induces a map of cokernels $C^{-n}(\mathbb{Q} \otimes \mathbb{Q}) \rightarrow C_{n-3}$, that is, a stable morphism $E_\lambda[-n] \rightarrow \mathcal{C}[3-n]$, and shifting by n gives g_λ . Its composite with the projection onto $(\mathbb{B} \otimes \mathbb{B})[-4][3]$ is the comparison map of the bimodule map

$$E_\lambda \longrightarrow \Omega_{E^e} E \subseteq (\mathbb{B} \otimes \mathbb{B})^0, \quad 1 \longmapsto \xi_\lambda \otimes \xi_\lambda,$$

which takes values in the kernel of the augmentation, since b does; the augmentation is a projective cover, since its source is $\bigoplus_r Er \otimes_{\mathbf{k}} rE$ over the four primitive vertex idempotents r of E and the induced map on tops is an isomorphism.

Evaluating at S , all terms of $\xi_\lambda \otimes_T s$ vanish except the one with $w^* = f$, for which $h_\lambda(w) = \lambda f^*$; hence $\pi[3] \circ (g_\lambda \otimes S)$ is the class of $f \otimes f \mapsto \lambda^2(f^* \otimes f^*)$, which is $\lambda^2 \beta_0$. Finally, $\beta_0 \neq 0$: a map $S \rightarrow \Omega S$ factoring through a projective module factors through a free module E^r , its components $S \rightarrow E$ take values in the left socle of E , and a map $E \rightarrow \Omega S$ is right multiplication by a radical element; for u in the left socle and j in the radical we have $\text{tr}(auj) = \text{tr}(jau) = 0$ for all a , so $uj = 0$, and every such factorisation is zero, whereas $f^* \otimes f^* \neq 0$. $\square$

9.7. The fibre and the counterexample. Let $w \in W^3$ be the element with $\pi[3](w) = \beta_0$, which exists and is nonzero by Propositions 9.13 and 9.14. For a finitely generated E^e -module M , put

$$\Sigma M = \text{coker}(M \rightarrow \text{Hom}_{\mathbf{k}}(E^e, M)),$$

where E^e acts on $\text{Hom}_{\mathbf{k}}(E^e, M)$ by $(r\varphi)(t) = \varphi(tr)$ and $m \mapsto (t \mapsto tm)$ is an injective homomorphism. Its target is projective-injective, since

$$\text{Hom}_{\mathbf{k}}(E^e, M) \cong D(E^e) \otimes_{\mathbf{k}} M \cong E^e \otimes_{\mathbf{k}} M,$$

with M regarded as a vector space. If M is projective on each side, so is ΣM , as in the proof of Lemma 9.12, and ΣM represents $M[1]$. The bimodule $\mathcal{Y} = \Sigma^4 C_1$ represents $\mathcal{C}[3]$. Choose bimodule homomorphisms $g_i: E_{H_i} \rightarrow \mathcal{Y}$ representing $H_i^{-2} g_{H_i}$,

for $i = 1, 2$, so that both evaluate at S to w . Let $Q_{\mathfrak{Y}} = E^e \otimes_{\mathbf{k}} \mathfrak{Y} \rightarrow \mathfrak{Y}$ be the free cover $r \otimes y \mapsto ry$, and define F by the exact sequence

$$(9.15) \quad 0 \longrightarrow F \longrightarrow E_{H_1} \oplus E_{H_2} \oplus Q_{\mathfrak{Y}} \xrightarrow{(g_1, g_2, \cdot)} \mathfrak{Y} \longrightarrow 0,$$

with projections $\rho_i: F \rightarrow E_{H_i}$.

Proposition 9.16. *The bimodule F is finite-dimensional and projective as a left and as a right E -module. There is a unique stable morphism $v: S \rightarrow FS$ with $\rho_1 v = \rho_2 v = \text{id}_S$, and for $V^a = \widehat{\text{Ext}}_E^a(S, FS)$ the map*

$$\delta^a: (H^a)^{\oplus 2} \longrightarrow V^a, \quad (g, j) \longmapsto F(g)v + v[a]j,$$

is bijective for every $a > 0$, while δ^0 is surjective with kernel spanned by $(1, 1)$.

Proof. All terms of (9.15) are finite-dimensional, and $\mathfrak{Y}$ is projective on each side, so the sequence splits on each side and F is projective on each side. Tensoring with S preserves exactness, the free bimodule evaluates to a projective module, and $E_{H_i} \otimes_E S \cong S$; hence there is a triangle

$$FS \xrightarrow{(\rho_1, \rho_2)} S \oplus S \xrightarrow{(w, w)} \mathbb{C}S[3] \longrightarrow FS[1]$$

in $\text{mod } E$, and the ρ_i induce natural transformations $F \otimes_E - \rightarrow E_{H_i} \otimes_E -$ of exact functors on the stable category. Applying $\widehat{\text{Ext}}_E^a(S, -)$ gives exact sequences

$$(H^{a-1})^{\oplus 2} \longrightarrow W^{a+2} \longrightarrow V^a \xrightarrow{(\rho_1, \rho_2)} (H^a)^{\oplus 2} \longrightarrow W^{a+3}.$$

For $a = 0$, since $W^2 = 0$ and $W^3 = \mathbf{k}w$, the group V^0 maps isomorphically onto the kernel of $(c, d) \mapsto (c + d)w$, the diagonal line; this gives the unique v , which spans V^0 . For $a = 1$, the map $(H^0)^{\oplus 2} \rightarrow W^3$ is surjective and $W^4 = 0$, and for $a \geq 2$ both W^{a+2} and W^{a+3} vanish; hence $(\rho_1, \rho_2): V^a \rightarrow (H^a)^{\oplus 2}$ is an isomorphism for all $a > 0$.

By naturality,

$$\rho_i[a] \circ F(g) \circ v = (E_{H_i} \otimes_E g) \circ \rho_i v = E_{H_i} \otimes_E g \quad \text{and} \quad \rho_i[a] \circ v[a] \circ j = j.$$

In degree zero this gives $\delta^0(c, d) = (c + d)v$, which is surjective with kernel $\mathbf{k}(1, 1)$. For $a = 3m$ with $m > 0$, composing δ^a with the isomorphism (ρ_1, ρ_2) gives

$$(g, j) \longmapsto (H_1^{-m}g + j, H_2^{-m}g + j),$$

by Lemma 9.9 and proposition 9.10; its scalar 2×2 matrix has determinant $H_1^{-m} + H_2^{-m} \neq 0$, since H_1^m and H_2^m are distinct monomials. For positive a not divisible by three both sides vanish. $\square$

Theorem 9.17 ([Ope26c, Theorem 1.1]). *Let*

$$\Lambda = \begin{pmatrix} E & 0 \\ F & E \end{pmatrix},$$

and let Z be the Λ -module of Theorem 8.4 for the data of Proposition 9.16. Then Λ is a finite-dimensional algebra over $\mathbf{k} = \mathbb{F}_2(q, H_1, H_2)$ with eight simple modules, all one-dimensional, and Z is a finite-dimensional, nonprojective, Gorenstein-projective Λ -module with $\text{Ext}_{\Lambda}^i(Z, Z \oplus \Lambda) = 0$ for all $i \geq 1$. The same holds after extending scalars to any field containing $\mathbf{k}$.

Proof. The first assertion about Z is Theorem 8.4 applied to Proposition 9.16. The radical of E is $J_E = \text{rad } T \otimes T + T \otimes \text{rad } T$, a sum of two commuting nilpotent ideals with quotient $\mathbf{k}^4$. The ideal of Λ with diagonal entries in J_E and lower left entry in F is nilpotent: a product of $2n$ of its elements has at most one factor contributing through the lower left corner, flanked by $2n - 1$ factors from J_E , and

vanishes if $J_E^n = 0$. Its quotient is $\mathbf{k}^8$, so it is the radical, and Λ has eight one-dimensional simple modules.

For a field extension K of $\mathbf{k}$, a projective resolution of a finitely generated module M by finitely generated projective modules gives

$$K \otimes_{\mathbf{k}} \text{Ext}_{\Lambda}^i(M, N) \cong \text{Ext}_{\Lambda_K}^i(M_K, N_K),$$

since Hom from a finitely generated projective module commutes with extension of scalars. Hence, the vanishing of extensions persists. A complete resolution of Z remains totally acyclic after extension of scalars, and a nonsplit projective cover sequence of Z remains nonsplit, since its class in Ext^1 stays nonzero; so Z_K is Gorenstein-projective and nonprojective. The radical computation is unchanged. $\square$

9.8. An endomorphism algebra of infinite finitistic dimension.

Corollary 9.18. *Let Z' be an indecomposable nonprojective direct summand of the module Z of Theorem 9.17. Then the algebra $\text{End}_{\Lambda}(\Lambda \oplus Z')$ has nine simple modules and infinite little finitistic dimension. The same construction gives such an algebra over every field containing $\mathbf{k}$.*

Proof. The module Z' exists since Z is nonprojective, and it inherits the vanishing of extensions, since Ext is additive. By Theorem 3.7, the algebra $\text{End}_{\Lambda}(\Lambda \oplus Z')$ has infinite little finitistic dimension. Since $\Lambda/\text{rad } \Lambda \cong \mathbf{k}^8$, the algebra Λ is basic with eight indecomposable projective modules, and $\Lambda \oplus Z'$ has nine pairwise nonisomorphic indecomposable summands, which index the simple modules of its endomorphism algebra. For a field K containing $\mathbf{k}$, the same argument applies to Λ_K and an indecomposable nonprojective summand of Z_K , by Theorem 9.17. $\square$

Remark 9.19. The algebra Λ is explicit. With the bimodule representatives used above, $\dim_{\mathbf{k}} E = 400$ and $\dim_{\mathbf{k}} E^e = 160000$, each application of Σ multiplies dimensions by 159999, and

$$\dim_{\mathbf{k}} F = 800 + 159999^5 \dim_{\mathbf{k}} C_1, \quad \dim_{\mathbf{k}} \Lambda = 2 \dim_{\mathbf{k}} E + \dim_{\mathbf{k}} F,$$

where $\dim_{\mathbf{k}} C_1 = 1\,623\,889\,344$ is computed from the dimensions of the terms of K and its cohomology (Appendix A); hence $\dim_{\mathbf{k}} \Lambda \approx 1.7 \cdot 10^{35}$. A computation with minimal bimodule resolutions of the cones over a finite field, recorded in Appendix A, determined that the two cokernels corresponding to C_0 and C_1 would have dimensions 784 704 and 1 377 984, and was stopped before forming them, at a linear system with about $3.6 \cdot 10^6$ unknowns; it gives no lower bound for the dimension of a minimised construction. The algebra in Corollary 9.18 has nine simple modules; the number $3(l+2)$ for Corollary 7.4 is not determined.

10. OBSTRUCTIONS TO SMALLER CONSTRUCTIONS

The algebra of Corollary 9.18 has nine simple modules, and the triangular algebra used to obtain it has eight simple modules and dimension of the order of 10^{35} (Remark 9.19). Most of this size comes from the tensor square $E = T \otimes T$ and from the bimodule cosyzygies over E^e . The analogous construction over T itself, with a single cone, fails in degree zero (Proposition 10.2 and corollary 10.3), and Tate duality also forces the vanishing of the related Toda brackets (Theorem 10.4).

10.1. One cone and two cones. In this subsection A is a finite-dimensional symmetric algebra and s a simple A -module with $\text{End}_A(s) = \mathbf{k}$ and $\text{Ext}_A^*(s, s) = \mathbf{k}[\tau]$, a polynomial algebra on a generator of degree three; for instance $(A, s) = (T, s)$ of Theorem 9.5. Write $H^a = \widehat{\text{Ext}}_A^a(s, s)$. By Theorem 2.1, $H^{3m} = \mathbf{k}\tau^m$ and $H^{-3m-1} \cong D(\mathbf{k}\tau^m)$ for $m \geq 0$, all other H^a vanish, and multiplication by τ is an

isomorphism $H^{-3m-1} \rightarrow H^{-3m+2}$ for $m \geq 1$, by the argument of Proposition 9.11. Let β_0 span H^{-1} .

Proposition 10.1. *Let*

$$s[-3] \xrightarrow{\tau} s \xrightarrow{i} X \xrightarrow{\pi} s[-2]$$

be a triangle in $\underline{\text{mod}} A$, and put $U^a = \widehat{\text{Ext}}_A^a(s, X)$. Then U^0 and U^1 are one-dimensional, i induces an isomorphism $H^0 \rightarrow U^0$, $\pi[1]$ induces an isomorphism $U^1 \rightarrow H^{-1}$, and $U^a = 0$ for $a \notin \{0, 1\}$.

Proof. The triangle gives exact sequences

$$0 \longrightarrow \text{coker}(\tau: H^{a-3} \rightarrow H^a) \xrightarrow{i} U^a \xrightarrow{\pi} \ker(\tau: H^{a-2} \rightarrow H^{a+1}) \longrightarrow 0.$$

Multiplication by τ is an isomorphism $H^{3m-3} \rightarrow H^{3m}$ for $m \geq 1$ and $H^{-3m-1} \rightarrow H^{-3m+2}$ for $m \geq 1$, so the only nonzero cokernel is H^0 , in degree $a = 0$, and the only nonzero kernel is H^{-1} , in degree $a - 2 = -1$. $\square$

In the construction over $T \otimes_{\mathbf{k}} T$ of Section 9, the first cone has one-dimensional groups in the degrees $3m$ and $1 - 3m$ for $m \geq 0$; after the second cone, taken in the other tensor factor, only the groups in degrees zero and three remain (Proposition 9.13). With a single cone the two nonzero degrees are adjacent, and the comparison map of Theorem 8.4 cannot be surjective in degree zero unless at least one of the compositions $w_i[-1] \circ \beta_0$ in Proposition 10.2 is nonzero.

Proposition 10.2. *In the situation of Proposition 10.1, let F be a finite-dimensional A -bimodule, projective on each side, with a triangle*

$$Fs \xrightarrow{(\rho_1, \rho_2)} s \oplus s \xrightarrow{(w_1, w_2)} X[1] \longrightarrow Fs[1]$$

in $\underline{\text{mod}} A$, where $w_1, w_2 \in U^1$ are not both zero. Suppose that $w_1[-1] \circ \beta_0 = 0 = w_2[-1] \circ \beta_0$ in U^0 . Then for every stable morphism $v: s \rightarrow Fs$ the map δ^0 of Theorem 8.4 is not surjective, and the module Z constructed by the formula in Theorem 8.4 satisfies $\text{Ext}_\Lambda^1(Z, Z) \neq 0$.

Proof. Put $V^0 = \widehat{\text{Ext}}_A^0(s, Fs)$. Applying $\widehat{\text{Ext}}_A^0(s, -)$ to the triangle gives an exact sequence

$$(H^{-1})^{\oplus 2} \xrightarrow{(w_1[-1], w_2[-1])} U^0 \longrightarrow V^0 \longrightarrow (H^0)^{\oplus 2} \xrightarrow{(w_1, w_2)} U^1.$$

The first map vanishes by hypothesis, so $U^0 \cong \mathbf{k}$ injects into V^0 . The last map has rank one, since $H^0 = \mathbf{k}$ and the w_i are not both zero, so its kernel is one-dimensional. Hence, $\dim V^0 = 2$. Since $H^0 = \mathbf{k}$ acts by scalars, $\delta^0(c, d) = (c - d)v$ is a multiple of v , so the image of δ^0 has dimension at most one and δ^0 is not surjective. By the exact sequence (8.5) for $a = 1$, the cokernel of δ^0 embeds into $\text{Ext}_\Lambda^1(Z, Z)$. $\square$

Corollary 10.3. *The hypothesis $w_1[-1] \circ \beta_0 = 0 = w_2[-1] \circ \beta_0$ of Proposition 10.2 always holds. Hence, for every bimodule F and triangle as in Proposition 10.2, the formula in Theorem 8.4 never produces a module Z with $\text{Ext}_\Lambda^1(Z, Z) = 0$.*

Proof. Since $\tau \neq 0$, the nondegenerate pairing of Theorem 2.1, which is compatible with composition, makes the map $H^{-4} \rightarrow H^{-1}$, $\gamma \mapsto \gamma[3] \circ \tau$, nonzero, hence surjective onto the one-dimensional space H^{-1} . Choose γ with $\gamma[3] \circ \tau = \beta_0$. For $w \in U^1$,

$$w[-1] \circ \beta_0 = (w[-4] \circ \gamma)[3] \circ \tau,$$

and $w[-4] \circ \gamma \in U^{-3}$, which is zero by Proposition 10.1. $\square$

10.2. A Tate-duality obstruction.

Theorem 10.4. *Let A be a finite-dimensional symmetric algebra and s a nonprojective simple A -module with $\text{End}_A(s) = \mathbf{k}$ and*

$$\text{Ext}_A^1(s, s) = \text{Ext}_A^2(s, s) = \text{Ext}_A^4(s, s) = 0.$$

Then for all $\tau \in \widehat{\text{Ext}}_A^3(s, s)$ and $\beta \in \widehat{\text{Ext}}_A^{-1}(s, s)$ the Toda bracket $\langle \tau, \beta, \beta \rangle$ is defined and equal to $\{0\}$.

Proof. Write $H^a = \widehat{\text{Ext}}_A^a(s, s)$. By Theorem 2.1, $H^{-1-a} \cong DH^a$, so $H^{-2} = H^{-3} = H^{-5} = 0$, and $H^{-1} \cong DH^0$ is one-dimensional, since $H^0 = \underline{\text{End}}_A(s) = \mathbf{k}$ for the nonprojective module s with $\text{End}_A(s) = \mathbf{k}$. The bracket is defined, since $\tau\beta \in H^2 = 0$ and $\beta^2 \in H^{-2} = 0$; it lies in H^0 , and its indeterminacy $\tau H^{-3} + H^1\beta$ is zero. If $\tau = 0$, then 0 is an element of the bracket, obtained with $b = 0$ in the defining system. If $\tau \neq 0$, the nondegenerate pairing of Theorem 2.1, which is compatible with composition, makes the map $H^{-4} \rightarrow H^{-1}$, $\gamma \mapsto \gamma\tau$, nonzero and hence surjective onto the one-dimensional space H^{-1} ; choose $\gamma \in H^{-4}$ with $\gamma\tau = \beta$. The bracket $\langle \tau, \beta, \gamma \rangle$ is defined, since $\beta\gamma \in H^{-5} = 0$, and it lies in $H^{-3} = 0$. By Lemma 2.2,

$$\{0\} = \langle \tau, \beta, \gamma \rangle \circ \tau \subseteq \langle \tau, \beta, \gamma\tau \rangle = \langle \tau, \beta, \beta \rangle.$$

Since the indeterminacy vanishes, $\langle \tau, \beta, \beta \rangle = \{0\}$. $\square$

Corollary 10.5. *Let A be a finite-dimensional symmetric algebra and s a simple A -module. If $\text{End}_A(s) = \mathbf{k}$ and $\text{Ext}_A^*(s, s) = \mathbf{k}[\tau]$ with τ of degree $p \geq 3$, then $\langle \tau, \beta, \beta \rangle = \{0\}$ for every $\beta \in \widehat{\text{Ext}}_A^{-1}(s, s)$.*

Proof. For $p = 3$ this is Theorem 10.4. Let $p > 3$. The bracket is defined: β^2 lies in $\widehat{\text{Ext}}_A^{-2}(s, s) \cong D\text{Ext}_A^1(s, s) = 0$ by Theorem 2.1, and the composite of τ and β lies in $\text{Ext}_A^{p-1}(s, s) = 0$, since $0 < p-1 < p$. It lies in $\widehat{\text{Ext}}_A^{p-3}(s, s) = \text{Ext}_A^{p-3}(s, s)$, which vanishes since $0 < p-3 < p$. $\square$

The tensor square $S = s \otimes s$ over $E = T \otimes T$ of Section 9 also satisfies the hypotheses of Theorem 10.4, since $\text{Ext}_E^*(S, S) = \mathbf{k}[\tau_1, \tau_2]$ vanishes in degrees 1, 2 and 4; so the corresponding brackets vanish there as well. The construction over $T \otimes_{\mathbf{k}} T$ does not depend on such a bracket: it uses a second cone, after which the only nonzero groups W^a are in degrees 0 and 3, as in Proposition 9.13, so that the term $W^2 = 0$ replaces the term U^0 of Proposition 10.2. Theorem 10.4 does not exclude other constructions, for instance with simple modules whose stable endomorphism ring is larger than $\mathbf{k}$, with other brackets, or with a different construction from stable morphisms to modules.

Remark 10.6. For trivial extensions $T = C \ltimes DC$, graded with C in degree zero and DC in degree one, there is a second explanation, by weights. The twists h_λ of Section 9.3 act on a homogeneous stable class of internal degree δ by λ^δ , under the convention of Lemma 9.9; a bracket of homogeneous classes with zero indeterminacy and target concentrated in internal degree zero vanishes unless the internal degrees add up to zero. For the pair (T, s) of Section 9, let β_0 span $\widehat{\text{Ext}}_T^{-1}(s, s)$. Then $\delta(\tau) = -1$ by Lemma 9.9 and $\delta(\beta_0) = 1$, since β_0 is dual to the identity under a form of internal degree one, so the weights do not cancel. This argument, and a six-dimensional example in which the weights cancel and a bracket of the form $\langle \tau, \beta, \beta \rangle$ is nonzero, are recorded with their verification in the repository of the project; see Appendix B. The six-dimensional example has periodic, not polynomial, Ext algebra, in accordance with Theorem 10.4.

10.3. Computational evidence. The following computations concern constructions with a single cone, or without cones. Their scripts and parameters are described in Appendix A; the results agree with Corollary 10.3 in the stated cases and degrees. They are finite computations with the stated scope, not proofs in all degrees.

- (1) An example without cones:

$$\Lambda_0 = \begin{pmatrix} T & 0 \\ T_{H_1} \oplus T_{H_2} & T \end{pmatrix},$$

of dimension 80 with four simple modules, and a nonprojective module Z_0 of dimension ten. Over finite fields $\mathbb{F}_{2^{16}}$, in three runs and up to degree six, $\text{Ext}_{\Lambda_0}^a(Z_0, Z_0)$ has dimensions $1, 0, 0, 0, 0, 0$ for $a = 1, \dots, 6$ and $\text{Ext}_{\Lambda_0}^a(Z_0, \Lambda_0) = 0$; the same holds exactly over $\mathbb{F}_2(q, H_1, H_2)$ up to degree two. The surviving class is the cokernel of δ^0 , as predicted by (8.5).

- (2) The candidate constructed over T from the cone of τ , lifts of the twists and a fibre F_1 , of dimension 352, giving an algebra Λ_1 of dimension 392 with four simple modules and a module Z_1 of dimension 16. Over $\mathbb{F}_{2^{16}}$, in two samples, $\text{Ext}_{\Lambda_1}^a(Z_1, Z_1)$ has dimensions $1, 0, 0$ and $\text{Ext}_{\Lambda_1}^a(Z_1, \Lambda_1) = 0$ for $a = 1, 2, 3$, and δ^0 has rank one, as in Proposition 10.2.
- (3) Three further attempts to use a single cone over algebras of dimensions 6, 12 and 40 fail because the relevant simple modules do not have polynomial Ext algebras on one generator of degree three; their Ext dimensions were computed over $\mathbb{F}_{2^{16}}$ up to degree twelve, using a tensor-product formula in degrees 2 to 12 for the forty-dimensional example.

APPENDIX A. COMPUTATIONS

Certificates check finitely many polynomial identities exactly, over polynomial rings such as $\mathbb{F}_2[q]$ or $\mathbb{F}_2[q, \lambda]$; they are part of the proofs of the statements that cite them, and multilinearity extends them to all inputs. *Experiments* compute homological invariants in finitely many degrees, over finite fields or over $\mathbb{F}_2(q, H_1, H_2)$; they are evidence with the stated scope and prove nothing about other degrees or other fields. All scripts and their saved outputs are in the directory `computations/` of the repository described in Appendix D. A rerun of the scripts, recorded in the repository, executed 45 commands with Python 3.14 and SageMath 10.10, and their mathematical output agreed with the saved results. Three finite-field runs for Λ_0 were not repeated, and two stages for Λ_1 and two stages of the attempt using minimal bimodule resolutions over $T \otimes_{\mathbf{k}} T$ were rerun from saved intermediate results. The finite fields $\mathbb{F}_{2^{16}}$ below are defined by the modulus $x^{16} + x^5 + x^3 + x^2 + 1$, and the parameters q, H_1 and H_2 are powers of a generator, with exponents recorded in the saved outputs.

The cochain. The following table, reproduced from the appendix of [Ope26c], defines the cochain p of Section 9.3. Lower-case letters denote the basis of C of Section 9.1 and capital letters the dual basis of DC ; an entry $abcw$ in the row q^d means that the coefficient of the basis vector w in $p(a, b, c)$ is q^d , and all other coefficients are zero. A script confirms that the table below and the source table have the same 179 entries in the same order.

coefficient	input and output letters										
q^2	EuNT	EyZX	FNtU	JtXZ	NtEU	NtXV	XzJT	ZxEY	tEun	tXuf	
q^2	tXvn	tXzj	tuFn	tuNf	tvNn	uFNT	vNNT	vtJu	xEyz	zJtx	
q	EuUE	EuVX	EvVE	ExXE	EyYE	EzZE	FNnF	FVvF	FVxU	FtTF	
q	FtYU	Ftun	Ftxj	JFtX	JNtZ	JUyX	JVyZ	JjEE	JjTT	JjXX	
q	NFtU	NNtV	NUyU	NVyV	NjJN	NjZV	NnNN	NnVV	Ntuf	Ntnv	
q	Ntzj	TtEE	TtXX	UuFF	UuUU	UvNF	UvVU	UxXU	Uyun	Uyxj	
q	UzZU	VvNN	VvVV	VxEU	VxXV	Vyuf	Vyvn	Vyzj	VzZV	XxEE	
q	XxXX	YuFT	YuUY	YvNT	YvVY	YxXY	YyEE	YyJJ	YyTT	YyXX	
q	YyYY	YyZZ	YzZY	ZxXZ	ZzEE	ZzJJ	ZzTT	ZzXX	jJFF	jJUJ	
q	jJjj	jJnn	jYyj	jZvn	jZzj	nNFF	nNUU	ntTn	tJjt	tTNN	
q	tTVV	tTtt	tYyt	tZzt	tzJn	uFVX	uFty	uNFT	uNUY	uNte	
q	uUyy	uVye	uJJu	vNUE	vNVX	vNty	vVyy	vtTv	xEuv	xJFT	
q	xJUy	xJjx	xJte	xTtx	xXuU	xXvv	xXyy	xXzz	xYyx	xZye	
q	xZzx	xuFv	xuNu	xvNv	yxJu	yzJv	zJNT	zJvY	zYyz	zZuu	
q	zZvv	zZyy	zZzz								
q^3	EyJT	JtEY	tEyj	tyJf							
1	JNjX	JVzX	NNjU	NVzU	NjYU	Njun	Njxj	Ntxt	TNtX	TVyX	
1	Vyxt	VzYU	Vzun	Vzxj	YuNJ	YuVZ	jYun	juNn	jxJn	txTn	
1	uNNJ	uNVZ	uNjy	uVzy	xJNJ	xJVZ	xTNT	xTVY	yxTv	zYuv	
1	zuNv	zxJv									

Certificates for Section 9.

- (1) *The algebras C and T* (Lemmas 9.1 and 9.4; 08-D-E/foundations_certificate.py, over $\mathbb{F}_2[q]$). All 1000 associators of basis vectors of C and all 8000 of T vanish; the grading is respected by all products; the trace form of T is symmetric, associative and has the stated dual bases; $(ut)^2 = q(1+q)z$. An earlier, independently written script (02-ar-finite-data/) checks the same identities over $\mathbb{F}_2(q)$ and $\mathbb{F}_{2^{16}}$ and computes the resolution of s in degrees up to eight, in agreement with Proposition 9.3.
- (2) *The cochain* (Lemma 9.7; 08-D-E/finite-cochain_certificate.py, over $\mathbb{F}_2[q]$ with the coefficients of λ^0 and λ^1 compared separately, hence over $\mathbb{F}_2[q, \lambda]$). Every entry has radical inputs, matching corners and weight -1 ; the cocycle identity holds on all $18^4 = 104976$ words of four radical basis vectors; the boundary identity holds on all 324 pairs; $p(t, x, j^*) = 0$, $p(t, y, j^*) = q^3 f$, and the bar chain ζ of Proposition 9.8 is a cycle. The script 08-D-E/certificate_receipt.py confirms that its table equals the source table.
- (3) *Casimir identities* (Proposition 9.14; the same script). The sum

$$\sum_w h_\lambda(w) w^*$$

over the twenty basis vectors vanishes, and the sum over the radical basis vectors with left idempotent r equals r^* , for $r \in \{e, f\}$. The other identities used in that proof are proved in the text.

Dimensions. The dimension of C_1 in Remark 9.19 is computed exactly from the corner dimensions of T and $\text{rad } T$, which give the dimensions of the terms of the bar resolution, and from the cohomology E , E^2 and E of K in degrees 0, 2 and 4 (09-sizes/dimensions.py). The terms $K^0, \dots, K^4$ have dimensions 1 761 405 952, 148 453 376, 11 713 792, 818 688 and 43 264, which gives $\dim C_1 = 1\,623\,889\,344$ and

$$\begin{aligned} \dim_{\mathbf{k}} \Lambda &= 1600 + 159999^5 \cdot 1\,623\,889\,344 \\ &= 170\,271\,818\,183\,326\,072\,615\,867\,045\,851\,312\,256. \end{aligned}$$

The computation (06-two-factor/, over $\mathbb{F}_{2^{16}}$) used minimal bimodule resolutions of the cones and determined, from the dimensions of the terms and of the homology

of the complex, cokernels of dimensions 784704 and 1377984; it stopped, before forming the bimodule matrices, at a linear system with 3591168 unknowns per right-hand side, beyond the bound of $2 \cdot 10^6$ set in advance. The same computation gives the dimensions of the Tate cohomology groups in Proposition 9.13 in degrees $-4 \leq a \leq 7$ for its sample parameters.

Experiments for Section 10.

- (1) *The algebra Λ_0 (03-testbed-lambda0/)*. Three runs over $\mathbb{F}_{2^{16}}$ in degrees up to six and one exact run over $\mathbb{F}_2(q, H_1, H_2)$ in degrees up to two compute minimal projective resolutions of Z_0 and the groups $\text{Ext}^a(Z_0, Z_0)$ and $\text{Ext}^a(Z_0, \Lambda_0)$. A separate script compares the surviving class with the cokernel of δ^0 ; this map has rank one and kernel spanned by $(1, 1)$.
- (2) *The algebra Λ_1 constructed over T (05-candidate1/)*. Two runs over $\mathbb{F}_{2^{16}}$ construct minimal bimodule covers, the cocycle, the cone of dimension 312, normalised lifts, the fibre of dimension 352, the algebra Λ_1 and the module Z_1 , and compute the dimensions of the Tate cohomology groups of the cone in degrees $-4 \leq a \leq 4$ (nonzero only in degrees 0 and 1, as in Proposition 10.1), the matrix of δ^0 , and Ext^a for $a = 1, 2, 3$.
- (3) *Further attempts using a single cone (07-one-factor-search/, over $\mathbb{F}_{2^{16}}$)*. For the six-dimensional trivial extension of $\mathbf{k}(1 \rightarrow 2)$, its tensor product with $\mathbf{k}[\epsilon]/(\epsilon^2)$, and the trivial extension of T , minimal resolutions of the simple modules give Ext dimensions up to degree twelve, directly for the first two and, for the third, directly in degrees 0 and 1 and by a tensor-product formula in degrees 2 to 12. None has polynomial Ext algebra on a generator of degree three. The same directory contains the six-dimensional example mentioned in Remark 10.6.
- (4) *Brackets for T (04-toda-bracket/, over $\mathbb{F}_2(q)$ and over $\mathbb{F}_{2^n}$ for $n = 8, 12, 16$)*. An explicit defining system shows $\langle \tau, \beta_0, \beta_0 \rangle = \{0\}$, in agreement with Theorem 10.4.

Checks of signs and finite cases. The remaining scripts check signs and finite cases of general arguments; they are not needed for the proofs. They cover the finite quotients of G for $m \leq 10$ (D-C-selection-checks.py); the signs of the bar decomposition, of the simulation and two examples, including Example 4.2 (D-A-checks.py); the rectification for an algebra with a nonzero null-homotopic relation (D-B-rectification-check.py); and the signs of the cone in the proof of Theorem 8.4 (D-D-cone-signs.py).

APPENDIX B. VERIFICATION AND FORMALISATION RECORD

All checks were carried out by AI models; none is a human verification, and the formalised statements are the only results checked by a proof assistant. The complete reports are in the repository described in Appendix D.

Proof notes and fresh-context verification. Before any section was written, every result received a written proof note, either compiled from the first part of the work or drafted from scratch, and the note was then checked by a fresh session of GPT-6 Astra that had not seen the note being written, was given the statements and proofs with the earlier verdicts and statuses removed as far as possible, and was instructed not to use any remaining assessment as evidence. Each drafted section was checked again, by another fresh session, against its note and its sources. Table 1 lists, for each part of the article, the author of the proof note, the verification of the note and the verification of the section; the effort settings are those of the Codex command-line interface, where *ultra* is the highest.

TABLE 1. Authors of the proof notes and verifications, by part of the article.

Part	Proof note	Verification of the note	Verification of the section
Section 3	Claude Opus 5.5	GPT-6 Astra (max): errors and gaps in five blocks, repaired	GPT-6 Astra (ultra): two local slips and an overstated sentence repaired
Sections 4 and 7	GPT-6 Astra (max)	GPT-6 Astra (ultra): no error; two wording repairs	GPT-6 Astra (ultra): wording repairs
Section 5	GPT-6 Astra (max); Claude Opus 5.5 for Section 5.1	GPT-6 Astra (max): no error	GPT-6 Astra (ultra): one range and wording repairs
Section 6	GPT-6 Astra (ultra)	GPT-6 Astra (ultra): no error; one wording repair	GPT-6 Astra (ultra): wording and scope repairs
Section 8	GPT-6 Astra (ultra)	GPT-6 Astra (ultra): no error	GPT-6 Astra (ultra): no error; one remark narrowed
Section 9	GPT-6 Astra (ultra)	GPT-6 Astra (ultra): no error; one missing input supplied, one statement narrowed	GPT-6 Astra (ultra): two steps restored from the note, one definition completed, wording repairs
Section 10	Claude Opus 5.5, from proofs by GPT-6 Astra (medium to max)	GPT-6 Astra (max): theorems confirmed, two readings corrected; the search computations not rechecked	GPT-6 Astra (ultra): a gap in the argument for a single cone closed by a direct proof (Corollary 10.3)

The errors found and repaired in this way include an invalid inference from dimensions alone in an earlier version of Proposition 3.5, an unstated convention on left and right modules in Proposition 3.8, a wrong lower bound on the size of a construction of a counterexample to the Auslander–Reiten conjecture using minimal bimodule resolutions, a misreading of the scope of Theorem 10.4, and several statements whose wording claimed more than their proofs. Two broader claims of the first part of the work were withdrawn.

Second checks. After the article was completed, Sections 3 to 10 were checked by fresh sessions of Claude Opus 5.5, in two phases. GPT-6 Astra had checked all proof notes and drafted most of those for Sections 4 to 9 (Table 1); the notes for Sections 3 and 10 had been compiled by Claude Opus 5.5. In the first phase, the checker received only the definitions and statements, with all proofs, remarks and examples removed, and tried to prove each statement; in the second, it checked the proofs of the article step by step against its own attempts. In the first phase the checker gave its own proof of every statement except the existence of the maps g_λ in Proposition 9.14, which the proof in the article supplies, and the third case of Corollary 5.5, which rests on a cited result of Schofield, read in the source. In the second phase no error was found; the proof of Corollary 10.5 omitted the verification that the bracket is defined for $p > 3$, which has been added, and several statements were reworded. The separation of the first phase rested on instructions

to the checker, and the statements of the lemmas outline the proofs, so the first phase was not fully blind.

To estimate how reliable the verification of the proof notes is, eight realistic errors were inserted by Claude Opus 5.5 into copies of the proof notes for Sections 4 and 6 to 9, and the original verification was repeated on the copies by fresh sessions of GPT-6 Astra (ultra). All eight errors were found, and no error was claimed at an unaltered place. Eight errors are too few to estimate a detection rate, and errors of a non-local kind are not measured. Finally, the citations of the Stacks Project were compared with the tagged statements; one wrong tag was corrected.

Formalisation. The following results are formally verified in Lean 4 (version 4.33.1) against Mathlib (commit 0df444a3), without additional axioms beyond `propext`, `Classical.choice` and `Quot.sound`. Items 1–5 are unconditional. Item 6 is conditional on explicit hypotheses, listed below.

- (1) Proposition 3.1, for an arbitrary abelian category with enough projective objects, stated as the conjunction of the projective dimension bound and its sharpness.
- (2) Theorem 3.3 over an arbitrary ring, in two stages. For a resolution of a nonzero right module E by finitely generated projective modules, the vanishing of the groups $\text{Ext}_{A^{\text{op}}}^i(E, A)$ of Mathlib for all $i \geq 0$ is equivalent to the exactness of the dual complex, and the cokernels C_n of the dual complex are then finitely generated left modules of projective dimension exactly n , for every $n \geq 1$; moreover C_n is the transpose of the presentation $P_n \rightarrow P_{n-1}$ of $\Omega^{n-1}E$. Minimality of this presentation, which the report includes in the definition of the transpose, is not formalised, since Mathlib has no projective covers.
- (3) The group of Section 5: Definition 5.6 as a presented group with exactly the stated relations, its finite presentation (Proposition 5.9, with Mathlib’s notion of a finitely presented group), Proposition 5.11, Proposition 5.13, Proposition 5.14 for every $m \geq 1$, and Corollary 5.15.
- (4) The argument of Proposition 5.4, as a statement of linear algebra: if a linear form on a finite-dimensional vector space $\mathcal{V}$ vanishes on $\mathbb{T}^t v$ for all large t , then it vanishes on $\mathbb{T}^t v$ for $t \geq \dim \mathcal{V}$; this holds over any field and without assuming that the vectors $\mathbb{T}^t v$ span $\mathcal{V}$. The passage from Grothendieck groups and rank functions to this statement is not formalised.
- (5) Lemma 6.1, for an arbitrary localisation with a calculus of right fractions; Lemma 6.6, in Mathlib’s idempotent completion of a pretriangulated category; and Proposition 6.7 without the assertion about K_0 , with the objects C and V and their decompositions constructed from the two cones of the proof.
- (6) The results of Section 10, in a $\mathbf{k}$ -linear pretriangulated category with finite-dimensional morphism spaces, under two further hypotheses: a perfect pairing between the spaces of morphisms $X \rightarrow Y[a]$ and $Y \rightarrow X[-1-a]$, natural and compatible with composition as in Theorem 2.1 and the paragraph after it; and, for Propositions 10.1 and 10.2 and corollary 10.3, an object s whose morphisms $s \rightarrow s[n]$, $n \geq 0$, form the graded algebra $\mathbf{k}[\tau]$ with τ of degree 3, as in Theorem 9.5. Toda brackets are defined from triangles as in Section 2.2, and their properties used in the proofs, including Lemma 2.2, are proved. The formal statements cover Proposition 10.1, Theorem 10.4, Corollary 10.5, and the parts of Proposition 10.2 and Corollary 10.3 that concern the triangle: the vanishing of the composites, the dimension of

$\widehat{\text{Ext}}_A^0(s, Fs)$ and the failure of surjectivity of $(c, d) \mapsto (c - d)v$. The assertions about $\text{Ext}_\Lambda^1(Z, Z)$ and the identification of this map with δ^0 require Theorem 8.4 and are not formalised.

The formalisation of Proposition 3.1, statements and proof, was written by Claude Opus 5.5. All other items were implemented by GPT-6 Astra from specifications by Claude Opus 5.5, who then reviewed their formal statements and, for item 6, the formal hypotheses. Acceptance required five checks: a scan of the sources for placeholders and new axioms, a build with warnings treated as errors, an axiom report, a listing of the formal statements, and an independent replay of the proof terms by the Lean kernel checker. The formalisation is included in the repository described in Appendix D (directory `lean/formalisation`), with its check logs (`lean/gates`). OpenAI's formalisations. The repository that accompanies the OpenAI preprints, `openai/math` (commit `fd4aeeb2`, Mathlib `d13f23b7`, Lean 4.34.1), poses existence statements of the counterexamples as challenges for the checking tool Comparator and provides Lean proofs of them, written independently of the work recorded here. The three challenges on little finitistic dimension, the Auslander–Reiten conjecture and Tachikawa's second conjecture were read by two AI models against the classical definitions and against this article, and found to be faithful encodings. The first states the numerical content of Corollary 7.4. The second states, over the field $\mathbb{F}_2(q, H_1, H_2)$ and over every field containing it, the existence of an algebra and a module as in Theorem 9.17, without identifying them with the algebra Λ and the module Z constructed here. The third states the existence asserted in the preprint on Tachikawa's second conjecture, which this article does not use. None of the three covers Theorem 3.7, so Corollary 9.18 is not among the formalised statements. A fourth challenge concerns the left–right asymmetry of finitistic dimensions stated in the main preprint, which this article does not treat; it was not examined. OpenAI's formalisation overlaps only in part with the one described above: it has counterparts of item 3 except Corollary 5.15, with an independent presentation of the group, and of Lemma 6.1 and Proposition 6.7 from item 5, and of two auxiliary lemmas of item 2, but nothing corresponding to items 1, 4 and 6 or to Theorem 3.3 itself. Both comparisons were made by reading the sources and are recorded in the files `audit/openai-lean-audit.md` and `audit/lean-comparison-openai.md` of the repository. OpenAI's proofs themselves have not been checked for this article: neither Comparator nor an independent replay by the Lean kernel has been run.

Scope of the formalisation here. The search of Mathlib made for the feasibility review found that Mathlib lacked, at the time, theories needed to formalise the proofs of this article in full, in particular derived tensor products over noncommutative algebras and dg resolutions for the main construction, and stable categories of symmetric algebras with Tate duality for the route through the Auslander–Reiten conjecture. This estimates the cost of formalising the present proofs; it does not show that a complete formalisation is infeasible, and OpenAI's repository provides Lean proofs of the existence statements. Of the smaller parts identified by the feasibility review, only the finite computations of Section 9 have not been formalised here. The evaluation is recorded in the repository.

APPENDIX C. THE PREPRINT ON TACHIKAWA'S SECOND CONJECTURE

The two preprints discussed in this article were released together with a third, *A counterexample to Tachikawa's second conjecture* [Ope26a]. Its main theorem [Ope26a, Theorem 1.1] asserts that over $\mathbf{k} = \mathbb{F}_2(q, H_1, H_2)$ there are a finite-dimensional symmetric algebra A and a nonprojective module M with

$\text{Ext}_A^i(M, M) = 0$ for all $i > 0$. Since A is self-injective, M is then also a counterexample to the Auslander–Reiten conjecture, and its consequences for the endomorphism algebra $\Gamma = \text{End}_A(A \oplus M)^{\text{op}}$ include infinite dominant dimension and infinite little left finitistic dimension. We have not verified that preprint, and nothing in this article depends on it. The existence of A and M is one of the Lean challenges of OpenAI’s repository, for which the repository provides a Lean proof, not checked here (Appendix B); according to the repository’s description of its scope, the consequences after extension of scalars and those for the endomorphism algebra are not part of that formalisation.

Given A and M , Theorem 3.7 applies directly and gives infinite little finitistic dimension of $\text{End}_A(A \oplus M)$, with one more simple module than A if A is basic and M indecomposable. The preprint also states the existence [Ope26a, Corollary 1.2(2)] of a simple Γ -module S with $\text{Ext}_\Gamma^i(S, \Gamma) = 0$ for all $i \geq 0$; Theorem 3.7 establishes this property for a module that need not be simple, and Theorem 3.3, applied to S regarded as a right module over Γ^{op} , gives explicit modules of every finite projective dimension over Γ^{op} . The preprint’s proof of infinite left finitistic dimension of Γ proceeds through dominant dimension instead. In Section 9, a bimodule and a morphism in the stable category of a symmetric algebra instead give a counterexample to the Auslander–Reiten conjecture over a triangular algebra, without requiring a module with vanishing self-extensions over the symmetric algebra itself.

APPENDIX D. DECLARATION OF THE USE OF AI AND OUTLINE OF THE PROCESS

This article was prepared autonomously by AI agents, as instructed by Gustavo Jasso, between 7 and 8 October 2026. All its text, proofs, computations and formal proofs were produced by AI models; no part of it was written by a human. Gustavo Jasso designed the task and took the decisions recorded below. No human has checked the mathematics of this article, and it is not peer reviewed.

Models. The models and their roles were as follows; the names are those of the project’s records.

- Claude Opus 5.5 (Anthropic), run in Claude Code: coordination of the work, the analysis of the preprints, the general criteria and obstructions of Section 3 and Section 10 (part of them jointly with the other models, as recorded in the repository), the proof notes compiled from earlier results, the checking of verification reports, the first stage of the Lean formalisation, and the first draft of the prose of this article.
- GPT-6 Astra (OpenAI), run in Codex (command-line interface, version 0.161), with reasoning effort between medium and its highest setting: independent verification of every proof in fresh sessions, all heavy computations, the proof notes for the constructions of the two preprints, the Tate-duality obstruction of Theorem 10.4, the later stages of the Lean formalisation, and the block-by-block review of the article.
- Claude Fable 5.1 (Anthropic): two consultations on possible simpler constructions and on the obstructions to them; the second was incomplete.
- Claude Sonnet (Anthropic), as subagents: retrieval of literature and bibliographic data, summaries of instructions, and one computation; their output was checked before use. In the second checks described in Appendix B, Claude Sonnet 5.5 coordinated the work and drafted several edits, which were reviewed by Claude Opus 5.5.

A ChatGPT Pro 100 subscription and a premium seat in the Claude Team plan for scientists (at a discounted rate) were available for the work.

Process. The first part (7–8 October 2026) analysed the main preprint [Ope26b] and searched for a simpler counterexample to the little finitistic dimension conjecture; it produced the criteria of Section 3 and the obstructions of Section 10, but found no counterexample verifiable by hand. The second part produced this article from that record.

Every mathematical claim carried a status (heuristic, plausible, supported by computation, proved by an AI model, or verified by a second, independent check of a different kind) and every change to a stronger status was accompanied by a record of the supporting evidence. Proofs were checked step by step in fresh sessions, sometimes of the same model that wrote them. For the checks of the proof notes, earlier verdicts were removed as far as possible, and the reviewers were instructed not to use any remaining assessment as evidence; several errors and gaps were found and corrected in this way, and two broader claims were withdrawn. Computations used exact arithmetic over finite fields and rational function fields, and their scope is stated wherever they are used (Appendix A). Citations were checked against the sources, whose statements and locators were read before being cited. Parts of Sections 3, 5 and 6 are formally verified in Lean against Mathlib, and the arguments of Section 10 are formally verified under explicitly stated hypotheses (Appendix B). The completed article was reviewed block by block by fresh sessions of GPT-6 Astra without waiting for human approval of each block, in three parts. The reviewers checked 337 blocks, corrected punctuation, typesetting and a few slips directly, and proposed 15 further changes, concerning notation and the precision of statements about scope, attribution and evidence; each proposal was examined by Claude Opus 5.5, and all were adopted. No error in a proof was found at this stage.

Repository. The repository <https://github.com/gjassoah/findim-report> contains the complete record of the work: the plan, the notes, the verification reports, the scripts and their outputs, the Lean sources and the evidence of their checks, the sources of this article, and a log of all interactions between Gustavo Jasso and the agents, in which his messages are reproduced verbatim and the agents' replies are summarised.

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(G. Jasso) MATHEMATISCHES INSTITUT, UNIVERSITÄT ZU KÖLN, WEYERTAL 86–90, 50931 KÖLN, GERMANY

Email address: gjasso@math.uni-koeln.de

URL: <https://gustavo.jasso.info>